

Supplementary Material of Hidden Human-Like Nature of Machine-Generated Texts: Theory and Detection Enhancement

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I. DETAILED DISCUSSION OF OUR CONTRIBUTIONS

Our contributions lie in revealing the hidden human-like nature of machine-generated texts, theoretically analyzing how such hidden human-like spans affect detection, and showing how these theoretical insights can guide the design of more effective detection enhancement frameworks.

- **We reveal the hidden human-like nature of machine-generated texts (Section (III-A) of the main text).** Existing works [1], [2], [3] mainly focus on explicit mixed-text scenarios, where a document contains both human-written and machine-generated content. For example, RoFT [3] studies cases where machines continue writing human text. In contrast, we highlight a more subtle and overlooked phenomenon: even when a text is fully generated by an LLM, some of its spans may still be highly consistent with human writing. Owing to the strong generative capabilities of modern LLMs and their exposure to large-scale human-written corpora, such hidden human-like spans may appear in fully machine-generated texts and provide weak or even misleading evidence for machine authorship. Considering the difficulty of detecting mixed texts [1], this finding suggests that the real-world detection challenge may be more severe than previously recognized, even when the input text is entirely machine-generated.
- **We theoretically demonstrate the impact of hidden human-like spans on detection (Section (III-C) of the main text).** Although hidden human-like spans are generated by machines, they are distributionally close to human writing and can therefore be modeled as human-distribution components from the perspective of detection. Existing studies [1], [2] have empirically shown that mixed texts are difficult to detect, but theoretical understanding remains limited. We take a step toward filling this gap by deriving sentence-complexity bounds that characterize how the proportion of hidden human-like spans affects the best possible detector. The analysis systematically explains why such spans hinder detection and further motivates detection strategies that explicitly reduce their influence.
- **We develop a theory-motivated detection enhancement framework (Section (IV) of the main text).** Guided by the theoretical observation that fewer hidden human-like spans lead to lower detection complexity, we establish a filtering-based enhancement principle that reduces the influence of hidden human-like spans, and then instantiate it as a practical stacked detection enhancement framework. Specifically, we formulate span retention as a latent-variable problem, revisit the optimization objective based on retained machine-indicative evidence, and adopt a hard-EM-inspired procedure to balance feasibility and efficiency. Notably, the main technical contribution is not limited to a specific hard-EM implementation. Rather, it lies in the general conceptual framework, where other inference and optimization strategies, such as EM variants [4], [5] and variational inference [6], can also be incorporated. This framework design may inspire future research on detection enhancement under hidden human-like components.
- **We demonstrate the effectiveness of the proposed framework from both theoretical and experimental aspects (Sections (IV-A) and (V) of the main text, and Section VII in the Supplementary Material).** Theorem (2) of the main text provides a theoretical bound for the proposed filtering-based enhancement framework, showing a smaller sentence complexity than Theorem (1). This theoretically supports the benefit of reducing the influence of hidden human-like spans. Extensive experiments under cross-LLM, cross-domain, paraphrasing-attack, shorter-text, and different human-like-span-ratio settings further demonstrate that the proposed framework consistently enhances existing detectors. Remarkably, the framework can also improve detection in a training-free manner by only applying the stacked inference structure without additional optimization. This plug-and-play property makes it flexible and scalable for practical deployment.

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II. MORE DISCUSSION ON VALIDATION OF IMPLICIT MIXED TEXT

Due to the lack of a complete human-written corpus, it is difficult to accurately estimate the true proportion of hidden human-like spans in MGTs. Therefore, our goal is not to precisely calculate this proportion, but to verify the existence of such spans by showing that a conservative lower bound is greater than zero.

Specifically, let $\mathcal{H}$ denote the set of observable human-written sentences in a given dataset, and let $\mathcal{M}$ denote the set of LLM-generated sentences. Based on exact sentence consistency, we define the observed proportion of hidden human-like spans as

$$P = \frac{|\mathcal{H} \cap \mathcal{M}|}{|\mathcal{M}|}. \quad (1)$$

This metric only counts generated sentences that exactly match observable human-written sentences. In real-world scenarios, the complete set of possible human-written sentences, denoted as $\mathcal{H}_{all}$, is far larger than the observable set $\mathcal{H}$, i.e., $\mathcal{H} \subseteq \mathcal{H}_{all}$. Therefore, the true proportion satisfies

$$P_{all} = \frac{|\mathcal{H}_{all} \cap \mathcal{M}|}{|\mathcal{M}|} \geq P. \quad (2)$$

Thus, once $P > 0$, we can confirm that at least some hidden human-like spans exist in MGTs. Moreover, exact matching is a strict criterion; many generated spans may be highly consistent with human writing in semantics or style without being exactly identical. Hence, the actual prevalence of hidden human-like spans in real-world MGTs may be larger than the observed lower bound.

III. MORE THEORETICAL RESULTS

A. Sentence Complexity of Mixed Text Detection under IID Setting

Theorem 1 (Sentence Complexity of Mixed Text Detection under IID Setting): Assume the total variation distance between the human and machine distributions is $TV(m, h) = \delta > 0$. Let the text contain n sentences, with α representing the proportion of hidden human-like spans within the MGT. To achieve an AUROC of ϵ , the required sentences n for the best possible detector is:

$$n = \Omega \left(\frac{1}{\delta^2(1-\alpha)^2} \ln \left(\frac{1}{1-\epsilon} \right) \right).$$

B. Sentence Complexity of Label-Agnostic Filtering-based Method under IID Setting

Theorem 2 (Sentence Complexity of Filtering-based Method under IID Setting): Consider the MGT detection under the assumption of Theorem 1. If we filter an $\alpha_s (< \alpha)$ proportion of hidden human-like spans from all texts, then to achieve an AUROC of ϵ , the required sentences n for the best possible detector is

$$n = \Omega \left(\frac{1-\alpha_s}{\delta^2(1-\alpha)} \ln \left(\frac{1}{1-\epsilon} \right) \right).$$

IV. PROOFS

A. Proof of Theorem 1 of the Supplementary Material

Our proof follows the proof in existing work [7]. First, according to our assumption, the Total Variance Distance $TV(m, h)$ between machine sentence distribution and human sentence distribution is δ , where $\delta > 0$. According to the definition of TV distance, there exists some set $A \in \mathcal{S}$ such that given the sentences $s^m \sim m(s)$ and $s^h \sim h(s)$, it holds

$$p(s^m \in A) - p(s^h \in A) = \delta$$

If we set $p(s^h \in A) = q$, it implies that $p(s^m \in A) = q + \delta$. For convenience, we define $\beta = 1 - \alpha$, which denotes the proportion of machine-indicative sentences in the MGT. Then, in the MGT with hidden human-like spans of n sentences $\{s_i\}_{i=1}^n$, βn machine-indicative sentences $\{s_i\}_{i=1}^{\beta n}$ are from $m(s)$ and αn hidden human-like sentences $\{s_i\}_{i=\beta n+1}^n$ are modeled as being from $h(s)$. Given $p(s^h \in A) = q$ and $p(s^m \in A) = q + \delta$, we can deduce that, on average, $(q + \beta\delta)n$ sentences will be in A . Similarly, for human-written text of n sentences $\{s_i\}_{i=1}^n$ from $h(s)$, on average, qn sentences will be in A . Therefore, applying the Chernoff bound, we have

$$\mathbb{P} \left(\text{at least } \left(q + \frac{\beta\delta}{2} \right) n \text{ sentences of the MGT with hidden human-like spans are in } A \right) \leq \exp^{-\frac{\beta^2\delta^2n}{2}} \quad (3)$$

and

$$\mathbb{P} \left(\text{at most } \left(q + \frac{\beta\delta}{2} \right) n \text{ sentences of human-written text are in } A \right) \leq \exp^{-\frac{\beta^2\delta^2n}{2}} \quad (4)$$

Let A' denote the set of n -tuples containing more than $\left(q + \frac{\beta\delta}{2}\right)n$ sentences of A . If the distribution of MGTs with hidden human-like spans is defined as $m^{\otimes\beta n}h^{\otimes\alpha n}$ and the human-written text distribution is $h^{\otimes n}$, we can bound

$$\begin{aligned} TV(m^{\otimes\beta n}h^{\otimes\alpha n}, h^{\otimes n}) &\geq p\left(\{s_i^m\}_{i=1}^{\beta n} \cup \{s_i^h\}_{i=1}^{\alpha n} \in A'\right) - p\left(\{s_i^h\}_{i=1}^n \in A'\right) \\ &\geq \left(1 - \exp^{-\frac{\beta^2\delta^2n}{2}}\right) - \exp^{-\frac{\beta^2\delta^2n}{2}} \\ &= 1 - 2\exp^{-\frac{\beta^2\delta^2n}{2}}. \end{aligned} \quad (5)$$

Invoking existing theoretical result (Proposition 1 in [7]), we have

$$AUROC \leq \frac{1}{2} + TV(m^{\otimes\beta n}h^{\otimes\alpha n}, h^{\otimes n}) - \frac{TV(m^{\otimes\beta n}h^{\otimes\alpha n}, h^{\otimes n})^2}{2} \quad (6)$$

If we need the AUROC of the best possible detector to satisfy $AUROC \geq \epsilon$, then we need

$$\frac{1}{2} + TV(m^{\otimes\beta n}h^{\otimes\alpha n}, h^{\otimes n}) - \frac{TV(m^{\otimes\beta n}h^{\otimes\alpha n}, h^{\otimes n})^2}{2} \geq \epsilon \quad (7)$$

Since the left-hand side is the monotonically increasing function of $TV(m^{\otimes\beta n}h^{\otimes\alpha n}, h^{\otimes n})$, combined with Eq. (5), it holds from the minimum value that

$$\frac{1}{2} + \left(1 - 2\exp^{-\frac{\beta^2\delta^2n}{2}}\right) - \frac{\left(1 - 2\exp^{-\frac{\beta^2\delta^2n}{2}}\right)^2}{2} \geq \epsilon \quad (8)$$

After expanding the squares and rearranging the terms, we get

$$\frac{1 - \epsilon}{2} \geq \exp^{-n\beta^2\delta^2} \quad (9)$$

Taking the logarithm of both sides and rearranging the terms, we can get

$$n \geq \frac{1}{\beta^2\delta^2} \ln\left(\frac{2}{1 - \epsilon}\right) = \frac{1}{(1 - \alpha)^2\delta^2} \ln\left(\frac{2}{1 - \epsilon}\right) \quad (10)$$

The theorem is proved.

B. Proof of Theorem 1 of the main text

This proof follows the existing work [7]. First, we need to introduce a lemma from existing work [8] to support our proof.

Lemma 1: Let n be the number of samples drawn sequentially from $\mathbb{P}(S_1, S_2 \cdots S_n) = \prod_{j=1}^L \tau_j$, where τ_j are independent subsets consisting of c_j dependent sequences $(s_1, s_2 \cdots s_{c_j})$ such that $\sum_{j=1}^L c_j = n$. Under the dependence structure in the Non-IID setting of the main text, for any $\delta > \frac{\sum_{j=1}^L (c_j - 1)\rho_j}{n}$, it holds that

$$\begin{aligned} \mathbb{P}(\bar{S} - \mathbb{E}[\bar{S}] \geq \delta) &\leq \exp \frac{-2\left(n\delta - \sum_{j=1}^L (c_j - 1)\rho_j\right)^2}{n} \\ \mathbb{P}(\mathbb{E}[\bar{S}] - \bar{S} \geq \delta) &\leq \exp \frac{-2\left(n\delta - \sum_{j=1}^L (c_j - 1)\rho_j\right)^2}{n} \end{aligned}$$

where $\bar{S} = \frac{1}{n} \sum_{i=1}^n s_i$ and $\mathbb{E}[S_i | S_{i-1} = s_{i-1}, \dots, S_1 = s_1] = \frac{\rho}{i-1} \sum_{k=1}^{i-1} s_k + (1 - \rho)\mathbb{E}[S_i]$.

This lemma provides a theoretical upper bound in the non-IID setting. When the dependence strength $\rho = 0$, it degenerates to the Chernoff bound.

Based on this lemma, we can follow a similar proof in the IID setting. If we set $p(s^h \in A) = q$, it implies that $p(s^m \in A) = q + \delta$. For convenience, we define $\beta = 1 - \alpha$, which denotes the proportion of machine-indicative sentences in the MGT with hidden human-like spans. Then, in the MGT with hidden human-like spans of n sentences $\{s_i\}_{i=1}^n$, βn machine-indicative sentences $\{s_i\}_{i=1}^{\beta n}$ are from $m(s)$ and αn hidden human-like sentences $\{s_i\}_{i=\beta n+1}^n$ are modeled as being from $h(s)$. Given $p(s^h \in A) = q$ and $p(s^m \in A) = q + \delta$, we can deduce that, on average, $(q + \beta\delta)n$ sentences will be in A . Similarly, for human-written text of n sentences $\{s_i\}_{i=1}^n$ from $h(s)$, on average, qn sentences will be in A . Therefore, applying the above lemma, we have

$$\mathbb{P}\left(\text{at least } \left(q + \frac{\beta\delta}{2}\right)n \text{ sentences of MGT with hidden human-like spans are in } A\right) \leq \exp \frac{-2\left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1)\rho_j\right)^2}{n} \quad (11)$$

and

$$\mathbb{P} \left(\text{at most } \left(q + \frac{\beta\delta}{2} \right) n \text{ sentences of human-written text are in } A \right) \leq \exp \frac{-2 \left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1) \rho_j \right)^2}{n} \quad (12)$$

Let A' denote the set of n -tuples containing more than $\left(q + \frac{\beta\delta}{2} \right) n$ sentences of A . If the distribution of MGTs with hidden human-like spans is defined as $m^{\otimes \beta n} h^{\otimes \alpha n}$ and the human-written text distribution is $h^{\otimes n}$, we have

$$\begin{aligned} TV(m^{\otimes \beta n} h^{\otimes \alpha n}, h^{\otimes n}) &\geq p \left(\{s_i^m\}_{i=1}^{\beta n} \cup \{s_i^h\}_{i=1}^{\alpha n} \in A' \right) - p \left(\{s_i^h\}_{i=1}^n \in A' \right) \\ &\geq \left(1 - \exp \frac{-2 \left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1) \rho_j \right)^2}{n} \right) - \exp \frac{-2 \left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1) \rho_j \right)^2}{n} \\ &= 1 - 2 \exp \frac{-2 \left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1) \rho_j \right)^2}{n}. \end{aligned} \quad (13)$$

If we need the AUROC of the best possible detector to satisfy $AUROC \geq \epsilon$, then we need

$$\frac{1}{2} + TV(m^{\otimes \beta n} h^{\otimes \alpha n}, h^{\otimes n}) - \frac{TV(m^{\otimes \beta n} h^{\otimes \alpha n}, h^{\otimes n})^2}{2} \geq \epsilon \quad (14)$$

Since the left-hand side is the monotonically increasing function of $TV(m^{\otimes \beta n} h^{\otimes \alpha n}, h^{\otimes n})$, combined with Eq. (13), it holds from the minimum value that

$$\frac{1}{2} + \left(1 - 2 \exp \frac{-2 \left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1) \rho_j \right)^2}{n} \right) - \frac{\left(1 - 2 \exp \frac{-2 \left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1) \rho_j \right)^2}{n} \right)^2}{2} \geq \epsilon \quad (15)$$

After expanding the squares and rearranging the terms, we get

$$\frac{1 - \epsilon}{2} \geq \exp \frac{-4 \left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1) \rho_j \right)^2}{n} \quad (16)$$

Let $\Lambda = \sum_{j=1}^L (c_j - 1) \rho_j$ and $\gamma(\epsilon) = \ln \left(\frac{2}{1 - \epsilon} \right)$. Further rearranging, we get

$$\ln \gamma(\epsilon) \leq \frac{4}{n} \left(\beta n \frac{\delta}{2} - \Lambda \right)^2 = n\beta^2\delta^2 - 4\Lambda\beta\delta + \frac{4}{n}\Lambda^2 \quad (17)$$

Further rearranging the terms, we can get

$$\beta^2\delta^2 n^2 - n(4\beta\Lambda\delta + \gamma(\epsilon)) + 4\Lambda^2 \geq 0 \quad (18)$$

The solution of the quadratic equation is given by

$$\begin{aligned} n &\geq \frac{\gamma(\epsilon)}{2\beta^2\delta^2} + 2\frac{\Lambda}{\beta\delta} + \frac{1}{2\beta^2\delta^2} \sqrt{(4\Lambda\beta\delta + \gamma(\epsilon))^2 - 16\Lambda^2\beta^2\delta^2} \\ &= \frac{\gamma(\epsilon)}{2\beta^2\delta^2} + 2\frac{\Lambda}{\beta\delta} + \frac{1}{2\beta^2\delta^2} \sqrt{\gamma(\epsilon)^2 + 8\Lambda\beta\delta\gamma(\epsilon)} \\ &= \frac{\gamma(\epsilon)}{2\beta^2\delta^2} + 2\frac{\Lambda}{\beta\delta} + \frac{1}{2\beta^2\delta^2} \sqrt{\frac{2\gamma(\epsilon)^2 + 16\Lambda\beta\delta\gamma(\epsilon)}{2}} \\ &\geq \frac{\gamma(\epsilon)}{2\beta^2\delta^2} + 2\frac{\Lambda}{\beta\delta} + \frac{1}{2\beta^2\delta^2} \left(\frac{\sqrt{2}\gamma(\epsilon) + 4\sqrt{\Lambda\beta\delta\gamma(\epsilon)}}{2} \right) \\ &= \frac{(2 + \sqrt{2})\gamma(\epsilon)}{4\beta^2\delta^2} + 2\frac{\Lambda}{\beta\delta} + \sqrt{\frac{\Lambda\gamma(\epsilon)}{\beta^3\delta^3}} \end{aligned} \quad (19)$$

Using the order notation, we obtain

$$\begin{aligned} n &= \Omega \left(\frac{1}{(1 - \alpha)^2\delta^2} \ln \left(\frac{1}{1 - \epsilon} \right) + \frac{1}{(1 - \alpha)\delta} \sum_{j=1}^L (c_j - 1) \rho_j \right. \\ &\quad \left. + \left(\frac{1}{(1 - \alpha)^3\delta^3} \cdot \left(\sum_{j=1}^L (c_j - 1) \rho_j \right) \cdot \ln \left(\frac{1}{1 - \epsilon} \right) \right)^{1/2} \right). \end{aligned} \quad (20)$$

Then the theorem is proved.

C. Proof of Theorem 2 of the Supplementary Material

Similar to the proof for Theorem 1, if we define $p(s^h \in A) = q$, it implies that $p(s^m \in A) = q + \delta$. Then, in the MGT with hidden human-like spans of $(1 - \alpha_s)n$ sentences $\{s_i\}_{i=1}^{(1-\alpha_s)n}$, where βn machine-indicative sentences $\{s_i\}_{i=1}^{\beta n}$ are from $m(s)$ and $(\alpha - \alpha_s)n$ hidden human-like sentences $\{s_i\}_{i=\beta n+1}^{(1-\alpha_s)n}$ are modeled as being from $h(s)$. Given $p(s^h \in A) = q$ and $p(s^m \in A) = q + \delta$, we can deduce that, on average, $(1 - \alpha_s)qn + \beta\delta n$ sentences will be in A . Similarly, for human-written text, we collect $(1 - \alpha_s)n$ sentences $\{s_i\}_{i=1}^{(1-\alpha_s)n}$ from $h(s)$ and on average $(1 - \alpha_s)qn$ sentences will be in A . Therefore, applying the Chernoff bound, we have

$$\mathbb{P}\left(\text{at least } \left((1 - \alpha_s)q + \frac{\beta\delta}{2}\right)n \text{ sentences of MGT with hidden human-like spans are in } A\right) \leq \exp^{-\frac{\beta^2\delta^2n}{2(1-\alpha_s)}} \quad (21)$$

and

$$\mathbb{P}\left(\text{at most } \left((1 - \alpha_s)q + \frac{\beta\delta}{2}\right)n \text{ sentences of human-written text are in } A\right) \leq \exp^{-\frac{\beta^2\delta^2n}{2(1-\alpha_s)}} \quad (22)$$

Let A' denote the set of $(1 - \alpha_s)n$ -tuples containing more than $\left((1 - \alpha_s)q + \frac{\beta\delta}{2}\right)n$ sentences of A . If the distribution of MGTs with hidden human-like spans is defined as $m^{\otimes\beta n}h^{\otimes(\alpha-\alpha_s)n}$, and the human-written distribution is $h^{\otimes(1-\alpha_s)n}$, we can bound

$$\begin{aligned} \text{TV}\left(m^{\otimes\beta n}h^{\otimes(\alpha-\alpha_s)n}, h^{\otimes(1-\alpha_s)n}\right) &\geq p\left(\{s_i^m\}_{i=1}^{\beta n} \cup \{s_i^h\}_{i=1}^{(\alpha-\alpha_s)n} \in A'\right) - p\left(\{s_i^h\}_{i=1}^{(1-\alpha_s)n} \in A'\right) \\ &\geq \left(1 - \exp^{-\frac{\beta^2\delta^2n}{2(1-\alpha_s)}}\right) - \exp^{-\frac{\beta^2\delta^2n}{2(1-\alpha_s)}} \\ &= 1 - 2\exp^{-\frac{\beta^2\delta^2n}{2(1-\alpha_s)}}. \end{aligned}$$

If we define $\beta_s = \frac{\beta}{\sqrt{1-\alpha_s}}$, then $\text{TV}\left(m^{\otimes\beta n}h^{\otimes(\alpha-\alpha_s)n}, h^{\otimes(1-\alpha_s)n}\right) \geq 1 - 2\exp^{-\frac{\beta_s^2\delta^2n}{2}}$. Similar to the proof of Theorem 1, we have

$$n \geq \frac{1}{\beta_s^2\delta^2} \ln\left(\frac{2}{1-\epsilon}\right) = \frac{1-\alpha_s}{(1-\alpha)^2\delta^2} \ln\left(\frac{2}{1-\epsilon}\right) \quad (23)$$

The theorem is proved.

D. Proof of Theorem 2 of the main text

Similar to the proof for Theorem 1, if we define $p(s^h \in A) = q$, it implies that $p(s^m \in A) = q + \delta$. Then, in the MGT with hidden human-like spans of $(1 - \alpha_s)n$ sentences $\{s_i\}_{i=1}^{(1-\alpha_s)n}$, where βn machine-indicative sentences $\{s_i\}_{i=1}^{\beta n}$ are from $m(s)$ and $(\alpha - \alpha_s)n$ hidden human-like sentences $\{s_i\}_{i=\beta n+1}^{(1-\alpha_s)n}$ are modeled as being from $h(s)$. Given $p(s^h \in A) = q$ and $p(s^m \in A) = q + \delta$, we can deduce that, on average, $(1 - \alpha_s)qn + \beta\delta n$ sentences will be in A . Similarly, for human-written text, we collect $(1 - \alpha_s)n$ sentences $\{s_i\}_{i=1}^{(1-\alpha_s)n}$ from $h(s)$ and on average $(1 - \alpha_s)qn$ sentences will be in A . Therefore, applying Lemma 1, we have

$$\mathbb{P}\left(\text{at least } \left((1 - \alpha_s)q + \frac{\beta\delta}{2}\right)n \text{ sentences of MGT with hidden human-like spans are in } A\right) \leq \exp^{\frac{-2\left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1)\rho_j\right)^2}{(1-\alpha_s)n}} \quad (24)$$

and

$$\mathbb{P}\left(\text{at most } \left((1 - \alpha_s)q + \frac{\beta\delta}{2}\right)n \text{ sentences of human-written text are in } A\right) \leq \exp^{\frac{-2\left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1)\rho_j\right)^2}{(1-\alpha_s)n}} \quad (25)$$

Let A' denote the set of $(1 - \alpha_s)n$ -tuples containing more than $\left((1 - \alpha_s)q + \frac{\beta\delta}{2}\right)n$ sentences of A . If the distribution of MGTs with hidden human-like spans is defined as $m^{\otimes\beta n}h^{\otimes(\alpha-\alpha_s)n}$, and the human-written distribution is $h^{\otimes(1-\alpha_s)n}$, we can bound

$$\begin{aligned} \text{TV}\left(m^{\otimes\beta n}h^{\otimes(\alpha-\alpha_s)n}, h^{\otimes(1-\alpha_s)n}\right) &\geq p\left(\{s_i^m\}_{i=1}^{\beta n} \cup \{s_i^h\}_{i=1}^{(\alpha-\alpha_s)n} \in A'\right) - p\left(\{s_i^h\}_{i=1}^{(1-\alpha_s)n} \in A'\right) \\ &\geq \left(1 - \exp^{\frac{-2\left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1)\rho_j\right)^2}{(1-\alpha_s)n}}\right) - \exp^{\frac{-2\left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1)\rho_j\right)^2}{(1-\alpha_s)n}} \\ &= 1 - 2\exp^{\frac{-2\left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1)\rho_j\right)^2}{(1-\alpha_s)n}}. \end{aligned} \quad (26)$$

If we need the AUROC of the best possible detector to satisfy $AUROC \geq \epsilon$, we need

$$\frac{1}{2} + \text{TV} \left(m^{\otimes \beta n} h^{\otimes (\alpha - \alpha_s)n}, h^{\otimes (1 - \alpha_s)n} \right) - \frac{\text{TV} \left(m^{\otimes \beta n} h^{\otimes (\alpha - \alpha_s)n}, h^{\otimes (1 - \alpha_s)n} \right)^2}{2} \geq \epsilon \quad (27)$$

Since the left-hand side is the monotonically increasing function of $\text{TV} \left(m^{\otimes \beta n} h^{\otimes (\alpha - \alpha_s)n}, h^{\otimes (1 - \alpha_s)n} \right)$, combined with Eq. (26), it holds from the minimum value that

$$\frac{1}{2} + \left(1 - 2 \exp \frac{-2 \left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1) \rho_j \right)^2}{(1 - \alpha_s)n} \right) - \frac{\left(1 - 2 \exp \frac{-2 \left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1) \rho_j \right)^2}{(1 - \alpha_s)n} \right)^2}{2} \geq \epsilon \quad (28)$$

After expanding the squares and rearranging the terms, we get

$$\frac{1 - \epsilon}{2} \geq \exp \frac{-4 \left(\beta n \frac{\delta}{2} - \sum_{j=1}^L (c_j - 1) \rho_j \right)^2}{(1 - \alpha_s)n} \quad (29)$$

Let $\Lambda = \sum_{j=1}^L (c_j - 1) \rho_j$ and $\gamma(\epsilon) = (1 - \alpha_s) \ln \left(\frac{2}{1 - \epsilon} \right)$. Further rearranging, we get

$$\gamma(\epsilon) \leq \frac{4}{n} \left(\beta n \frac{\delta}{2} - \Lambda \right)^2 = n \beta^2 \delta^2 - 4 \Lambda \beta \delta + \frac{4}{n} \Lambda^2 \quad (30)$$

Then similar to the proof of Theorem 1 and using $(1 - \alpha_s) \ln \left(\frac{2}{1 - \epsilon} \right)$ to replace $\ln \left(\frac{2}{1 - \epsilon} \right)$, we have

$$\begin{aligned} n = \Omega & \left(\frac{1 - \alpha_s}{(1 - \alpha)^2 \delta^2} \ln \left(\frac{1}{1 - \epsilon} \right) + \frac{1}{(1 - \alpha) \delta} \sum_{j=1}^L (c_j - 1) \rho_j \right. \\ & \left. + \left(\frac{1 - \alpha_s}{(1 - \alpha)^3 \delta^3} \cdot \left(\sum_{j=1}^L (c_j - 1) \rho_j \right) \cdot \ln \left(\frac{1}{1 - \epsilon} \right) \right)^{1/2} \right). \end{aligned}$$

The theorem is proved.

E. Proof of Theorem 3 of the main text

For the unfiltered MGT in Theorem 1, the proportion of machine-indicative spans is

$$\beta = 1 - \alpha.$$

Following the proof of Theorem 1, the corresponding TV lower bound is

$$\text{TV}_1 \geq 1 - 2 \exp \left(- \frac{2 \left(\beta n \frac{\delta}{2} - \Lambda \right)^2}{n} \right), \text{ where } \Lambda = \sum_{j=1}^L (c_j - 1) \rho_j.$$

Now consider imperfect filtering. The filter correctly removes an α_s proportion of hidden human-like spans and mistakenly removes an α_h proportion of machine-indicative spans. Therefore, the retained MGT contains $(1 - \alpha - \alpha_h)n = \tilde{\beta}n$ machine-indicative spans and $(\alpha - \alpha_s)n$ hidden human-like spans. The retained text length is $(1 - \alpha_s - \alpha_h)n = \ell n$. For the retained MGT, the expected number of spans falling in A is $\tilde{\beta}n(q + \delta) + (\alpha - \alpha_s)nq$. Since $\tilde{\beta} + \alpha - \alpha_s = 1 - \alpha - \alpha_h + \alpha - \alpha_s = 1 - \alpha_s - \alpha_h = \ell$, we have $\tilde{\beta}n(q + \delta) + (\alpha - \alpha_s)nq = \ell q n + \tilde{\beta} \delta n$. For the retained human-written text, we collect ℓn spans from $h(s)$, and the expected number of spans falling in A is $\ell q n$. Applying the same non-IID concentration lemma used in Theorem 2, we have

$$P \left(\text{at least } \left(\ell q + \frac{\tilde{\beta} \delta}{2} \right) n \text{ spans of the retained MGT are in } A \right) \leq \exp \left(- \frac{2 \left(\tilde{\beta} n \frac{\delta}{2} - \Lambda \right)^2}{\ell n} \right),$$

and

$$P \left(\text{at most } \left(\ell q + \frac{\tilde{\beta} \delta}{2} \right) n \text{ spans of the retained human-written text are in } A \right) \leq \exp \left(- \frac{2 \left(\tilde{\beta} n \frac{\delta}{2} - \Lambda \right)^2}{\ell n} \right).$$

Let A' denote the set of ℓn -tuples containing more than $\left(\ell q + \frac{\tilde{\beta}\delta}{2}\right)n$ spans of A . The retained MGT distribution is $m^{\otimes \tilde{\beta}n} h^{\otimes (\alpha - \alpha_s)n}$, and the retained human-written distribution is $h^{\otimes \ell n}$. Therefore,

$$TV_{\text{imp}} \geq 1 - 2 \exp\left(-\frac{2\left(\tilde{\beta}n\frac{\delta}{2} - \Lambda\right)^2}{\ell n}\right).$$

We now compare this lower bound with the unfiltered lower bound. Since $1 - 2\exp(-x)$ is monotonically increasing in x , TV_{imp} has a larger lower bound than TV_1 if

$$\frac{\left(\tilde{\beta}n\frac{\delta}{2} - \Lambda\right)^2}{\ell n} > \frac{\left(\beta n\frac{\delta}{2} - \Lambda\right)^2}{n}.$$

Equivalently,

$$\frac{\left(\tilde{\beta}n\frac{\delta}{2} - \Lambda\right)^2}{\ell} > \left(\beta n\frac{\delta}{2} - \Lambda\right)^2.$$

When the effective margins are positive, we can take square roots on both sides and obtain

$$\frac{\tilde{\beta}n\frac{\delta}{2} - \Lambda}{\sqrt{\ell}} > \beta n\frac{\delta}{2} - \Lambda.$$

Rearranging gives

$$n\frac{\delta}{2}(\tilde{\beta} - \beta\sqrt{\ell}) > \Lambda(1 - \sqrt{\ell}).$$

Dividing both sides by n gives

$$\frac{\delta}{2}(\tilde{\beta} - \beta\sqrt{\ell}) > \frac{\Lambda}{n}(1 - \sqrt{\ell}).$$

Thus, the imperfectly filtered text has a larger TV lower bound if

$$\tilde{\beta} > \beta\sqrt{\ell}$$

and

$$\frac{\Lambda}{n} < \frac{\delta(\tilde{\beta} - \beta\sqrt{\ell})}{2(1 - \sqrt{\ell})}.$$

Substituting $\beta = 1 - \alpha$, $\tilde{\beta} = 1 - \alpha - \alpha_h$, $\ell = 1 - \alpha_s - \alpha_h$ yields the proposition. The proof is completed.

V. MORE DISCUSSION OF THE PROPOSED FRAMEWORK

A. Rationality of the approximation of z_i

In the main text, we argue that the effect of different z_i on $p(z_i|S_i; \theta^t)$ is usually much greater than $p(y_i|S_i, z_i; \theta^t)$, which allows us to reasonably approximate $z_i \approx \arg \max_{z_i} p(z_i|S_i; \theta^t)$. We use proof by contradiction to show that this approximation is mild.

Specifically, let $z_1 = \arg \max_{z_i} p(z_i|S_i; \theta^t)$, $z_2 = \arg \max_{z_i} p(y_i|S_i, z_i; \theta^t)p(z_i|S_i; \theta^t)$. If the approximation is unreasonable, then $z_1 \neq z_2$, let us assume that there are k positions between z_1 and z_2 that are different. For a well-trained detector, the detection results are usually close to 0 or 1. For example, it is reasonable to assume that $p(z_{i,j}|S_i; \theta^t) > 0.95$ or < 0.05 , which is often more serious in reality. Then, we know $p(z_1|S_i; \theta^t) > 20^k p(z_2|S_i; \theta^t)$. If $z_2 = \arg \max_{z_i} p(y_i|S_i, z_i; \theta^t)p(z_i|S_i; \theta^t) \neq z_1$, then at least $p(y_i|S_i, z_1; \theta^t) > 20^k p(y_i|S_i, z_2; \theta^t)$ must be satisfied. Since $p(y_i|S_i, z_1; \theta^t)$ represents the detection probability of paragraph $S_i \odot z_1$, this means that modifying two subsequences of a paragraph can change the detection probability by a factor of 400, which is generally difficult. Therefore, our approximation is mild.

B. Differences from Existing Mixed Text Detection Works

Despite recent attention to human-machine mixed text detection [1], [2], our work differs from existing studies in the following key aspects:

- **Scenario.** Existing studies mainly focus on explicit mixed-text scenarios, where human-written and machine-generated contents are directly combined in the same document. In contrast, our work studies a more subtle and overlooked phenomenon: even when a text is fully generated by an LLM, some of its spans may still be highly consistent with human writing. Owing to the strong generative capability of modern LLMs, such hidden human-like spans may provide weak or even misleading evidence for machine authorship. This finding suggests that the real-world detection challenge may be more severe than previously recognized, even when the input text is entirely machine-generated.

- **Evaluation.** Previous studies mainly demonstrate the difficulty of mixed-text detection through empirical evaluations on specific datasets, while the theoretical understanding of why such texts hinder detection remains limited. Our work takes a step toward filling this gap by theoretically analyzing how hidden human-like spans affect MGT detection. The derived sentence-complexity bounds reveal the factors that influence detection difficulty and provide a principled basis for designing more effective detection enhancement methods.
- **Methodology.** Different from existing works that primarily evaluate or detect mixed texts, we propose a model-agnostic enhancement framework for existing detectors. Guided by the theoretical analysis, our framework reduces the influence of hidden human-like spans by formulating span retention as a latent-variable problem and optimizing it through a hard-EM-inspired procedure. Extensive experiments across cross-LLM, cross-domain, and adversarial paraphrasing settings validate its effectiveness and generality.

C. Formal Description of the Proposed Framework’s Validity

The effectiveness of the proposed framework can be understood from three aspects: information bottleneck theory, attention mechanism, and the granularity of text processing:

- **Bottleneck theory.** Information bottleneck theory suggests that, during forward propagation, neural networks tend to suppress information irrelevant to the prediction and gradually focus on the most task-relevant parts of the input [9]. In our setting, hidden human-like spans in MGTs can be regarded as weakly discriminative or even misleading components, because they are distributionally close to human writing and provide limited evidence for machine authorship. The proposed stacked framework explicitly implements this bottleneck intuition by filtering confidently human-like subsequences before final detection, thereby encouraging the detector to rely more on machine-indicative evidence. Under this view, the proposed framework achieves iterative self-enhancement: in the E-step, the detector filters confidently human-like subsequences to obtain more machine-indicative retained texts; in the M-step, the detector is optimized on these retained texts, which further improves its ability to estimate reliable retention masks in the next E-step.
- **Attention mechanism.** Existing theoretical results (Theorem 3.1 in [10]) establish the temporal consistency of attention weights: if the attention weight $\alpha_{t,l}$ of the l -th token is large at step t , then $\alpha_{t+1,l}$ is likely to remain large at step $t+1$. This indicates that once important tokens are identified, they can continue to influence subsequent representations and predictions. In the proposed framework, the retained text preserves subsequences that are more likely to contain machine-indicative evidence. Therefore, even though filtering may remove some local context, important tokens and subsequences can still exert persistent influence through the attention mechanism. This helps explain why the framework can remain effective even when some confidently human-like subsequences are removed.
- **Granularity of text processing.** Most models default to categorizing a text S as either machine or human (the format S , where $S \sim h(S)$ or $m(S)$), while we refine this to the sentence level (the format $S = [s_1; s_2; \dots; s_n]$ where each $s_i \sim h(s)$ or $m(s)$). This finer granularity allows the detector to reduce the influence of hidden human-like spans and focus on local evidence that is more useful for MGT detection. Our work takes the first step from the paragraph level to the sentence level, but undeniably, an even finer granularity is to further understand from sentence level to clause level (the format $s_i = "a_1, a_2, \dots, \text{and } a_n"$ where each $a_i \sim h(s)$ or $a_i \sim m(s)$), which may provide valuable insights for future detection enhancements.

D. Limitation

The proposed framework has the following potential limitations:

- To maintain the framework’s flexibility and efficiency, we adopt a simple fixed-length sequence splitting strategy, which may separate semantically dependent contexts. Encouragingly, extensive experimental results (Table I of the main text, and Tables I-XV of the Supplementary Material) show that this straightforward approximation already achieves satisfactory enhancement effects. Nevertheless, developing more adaptive segmentation strategies remains a promising direction for further improving detection.
- The proposed framework is currently applied to model-based detectors. However, its core idea of reducing the influence of hidden human-like spans is general and may also inspire future extensions to feature-based or training-free detectors.
- The effectiveness of the proposed stacked framework depends on the base detector’s ability to estimate whether a subsequence should be retained as machine-indicative evidence. If the base detector is too weak to provide meaningful subsequence-level signals, the enhancement effect may be limited. However, enhancing such weak detectors is less practically valuable, as their original detection capability is already insufficient for deployment. We therefore focus on improving detectors with reasonable baseline performance.
- The proposed framework is generally more suitable for paragraph-level or longer-text detection, where sufficient contextual evidence is available for filtering hidden human-like spans. For very short texts, filtering may further reduce the already limited detection evidence. Although our experiments show encouraging improvements under shorter-text settings, designing more specialized strategies for very short texts remains an important future direction.

Algorithm 1 Soft Stacked Detection Enhancement Framework

```

1: Input: Train data  $\mathcal{D} = \{(S_i, y_i)\}_{i=1}^N$ , the detection model  $f(S, \theta^0)$ , training epochs  $T$ , filtering ratio  $\tau$ , E-step detection threshold  $r_e$ , and learning rate  $\eta$ .
2: function INFERENCE( $S_i, f(S, \theta^t)$ ) ▷ Detector Inference
3:   Split  $S_i$  into a set of sequences  $\{S_{i,j}\}$ , where each  $S_{i,j}$  contains at most  $k$  sentences.
4:   Calculate  $z_i$  according to Eq. (6).
5:    $\hat{S}_i = S_i \odot z_i$ .
6:   Return  $f(\hat{S}_i, \theta)$ .
7: end function
8: function TRAIN( $\mathcal{D}$ ) ▷ Detector Training
9:   for  $t = 0$  to  $T - 1$  do
10:    for each batch of samples  $\mathcal{D}_B \sim \mathcal{D}$  do
11:      Calculate  $\mathcal{Q}(\theta, \theta^t)$  according to Eq. (31). ▷ E-step
12:       $\theta^{t+1} = \theta + \eta \nabla_{\theta} \mathcal{Q}(\theta, \theta^t)$ . ▷ M-step
13:    end for
14:  end for
15:  Return the trained model  $f(S, \theta^T)$ .
16: end function

```

- The theoretical analysis relies on a fixed proportion α of hidden human-like spans. This design helps isolate the core mechanism by which human-like components affect detection difficulty and keeps the analysis tractable. As a first step, our theory provides useful foundations and intuitions for understanding this phenomenon. Future theoretical work may consider more complex settings, such as variable span proportions, heterogeneous span distributions, or adaptive filtering ratios.

VI. SOFT STACKED DETECTION ENHANCEMENT FRAMEWORK

In addition to the default stacked detection framework optimized by the hard EM algorithm, we provide a soft (traditional) EM version. For the soft E-step, we calculate the expected value of the log-likelihood function of θ , with respect to the current conditional distribution of z_i given (S_i, y_i) and the current estimates of the parameters θ^t :

$$Q(\theta; \theta^t) := \sum_{(S_i, y_i) \in \mathcal{D}} \sum_{z_i} p(z_i | S_i, y_i; \theta^t) \log p(y_i, S_i, z_i; \theta).$$

For the posterior distribution $p(z_i | S_i, y_i; \theta^t)$, similar to the approximation of the hard-EM version, we approximate $p(z_i | S_i, y_i; \theta^t) \approx p(z_i | S_i; \theta^t)$. This can be predicted by the detection model $f(S, \theta)$, i.e.,

$$p(z_i | S_i; \theta^t) := \prod_{j=1}^{n_i} f(S_{i,j}, \theta^t)^{z_i} (1 - f(S_{i,j}, \theta^t))^{1-z_i}$$

For each z_i , we can model $p(y_i | S_i, z_i; \theta)$ as:

$$p(y_i | S_i, z_i; \theta) = f(S_i \odot z_i, \theta)$$

Therefore, we have

$$Q(\theta, \theta^t) = \sum_{(S_i, y_i)} \sum_{z_i} \prod_{j=1}^{n_i} f(S_{i,j}, \theta^t)^{z_i} (1 - f(S_{i,j}, \theta^t))^{1-z_i} (y_i \log(f(S_i \odot z_i, \theta)) + (1 - y_i) \log(1 - f(S_i \odot z_i, \theta))) \quad (31)$$

In the M-step, we need to maximize $Q(\theta, \theta^t)$ to update the model parameters:

$$\theta^{t+1} = \arg \max_{\theta} Q(\theta, \theta^t), \quad (32)$$

which can be updated by the gradient descent algorithm.

For the inference phase, we use the same processing as the hard version described in the main text and calculate z_i according to Eq. (6), and the model's prediction is $f(S_i \odot z_i, \theta)$. The detailed algorithm flow is shown in Alg. 1. Besides, an experimental evaluation of this version is given in Section VII-G of the Supplementary Material.

VII. ADDITIONAL EXPERIMENTS

A. Implementation Details

For fair comparison, both the model-based baselines (i.e., ChatGPT-D, OpenAI-D, MPU, RADAR) and the enhanced version are fine-tuned on the training set and use the same hyperparameters. Specifically, all detection models are fine-tuned for 5 epochs, and the Adam optimizer is used for training, with a learning rate of $5e-5$. The hyperparameter setting is tuned once for each (Base Detector + Dataset) combination on the validation set. Specifically, for ChatGPT-STK, the E-step detection threshold r_e is set to 0.01 in Essay, Reuters, and SQuAD1 datasets, the length k is set to 3, 3, 1, and the maximum filter ratio τ is set to 0.2, 0.2, and 0.8. For OpenAI-STK, the E-step detection threshold r_e is set to 0.01, 0.005, 0.005 in Essay, Reuters, and SQuAD1 datasets, the length k is set to 4, 4, 1, and the maximum filter ratio τ is set to 0.5. For MPU-STK and RADAR-STK, the E-step detection threshold r_e is set to 0.005 in Essay, Reuters, and SQuAD1 datasets, the length k is set to 7, 3, 1, and the maximum filter ratio τ is set to 0.5. **For sensitivity analysis of the parameters and guidelines for the hyperparameter selection in practical operation, please refer to Appendix VII-H.**

Notably, the default settings are obtained through grid search, aiming to show the maximum potential of detection enhancement. Our sensitivity analysis (Section VII-H of the Supplementary Material) shows that compared to the original detection models (with the x-axis value of 0), the enhanced models generally outperform them across various hyperparameter settings, reducing the need for performance tuning and having wide practical applicability. For the sentence split, we use the `sent_tokenize` function from the `nlTK` tool library to divide texts into sentences. For feature-based methods, we use the default implementation of MGTBench¹. The experiments are conducted on a PC with an Inter(R) Xeon(R) Gold 6230 CPU @ 2.10GHz, 60GB memory, and a NVIDIA Tesla V100.

B. More Results for Performance Comparison

First, in addition to the performance comparison of detectors trained on ChatGPT texts shown in Table I of the main text, we also extensively evaluated the effectiveness of the detectors trained on different LLM texts to demonstrate the versatility of the proposed framework. The results of these evaluations are presented in Table I-XII of the Supplementary Material. From these tables, it is evident that the proposed framework effectively enhances original detectors (i.e., ChatGPT-D, OpenAI-D, and MPU) in most settings. For example, the proposed framework demonstrates a significant improvement for ChatGPT-D in approximately 98% of the settings (194 out of 198), and it offers a competitive improvement for OpenAI-D in approximately 86% of the settings (169 out of 198). The widespread enhancement on texts from unseen LLMs underscores the practical applicability of the proposed framework, as it aligns more closely with real-world scenarios. Besides, we find that all detectors exhibit low performance on SQuAD1. The reason is that each SQuAD1 sample contains a shared human-written question concatenated with either a human-written or machine-generated answer. Such human-like contextual spans weaken the discriminative evidence for machine authorship, which echoes our theoretical finding that hidden human-like spans hinder detection. Finally, our enhancement is less pronounced on SQuAD1. This is because our framework’s effectiveness relies on the base detector’s ability to confidently identify and filter human-like subsequences in the E-step. If a detector cannot provide reliable subsequence-level signals, i.e., it is very weak, the enhancement effect may be limited. Nevertheless, our enhancement effect is encouraging and remains positive.

Second, in addition to the enhancements shown in Fig. 3 of the main text for ChatGPT-D, Fig. 1-3 of the Supplementary Material demonstrate the improved performance for three other model-based detectors used in the experiments, OpenAI-D, MPU, and RADAR. We observe similar findings as in Section V-B of the main text, where the original detectors are enhanced in most settings, with a more pronounced improvement for weaker detectors.

Third, we include ROC curves in Fig. 4 of the Supplementary Material. As observed, when focusing on extremely low FPR (0%–5%), the enhanced detectors show significant advantages, which is highly valuable for MGT detection tasks that prioritize low false positives.

Finally, we further demonstrate the broad effectiveness of our enhancement framework on the Chinese dataset [11] and the code dataset [12]. Table XIII of the Supplementary Material presents the performance of both the original detectors and their enhanced versions (-E) for detecting Chinese MGTs and code MGTs. We observe that the proposed enhancement strategy still exhibits promising potential for both Chinese and code texts.

C. Evaluation of Expected Cost under Plausible Base Rates

In this section, we provide a detailed analysis of the total expected cost as a function of the base rate. Concretely, we define a formal expected cost function:

$$E(\text{Cost}) = P(\text{Human}) \cdot P(\text{FP}) \cdot C_{FP} + P(\text{Machine}) \cdot P(\text{FN}) \cdot C_{FN}.$$

where $P(\text{Human})$ and $P(\text{Machine})$ are the base rates. Besides, we model a plausible cost asymmetry, $C_{FP} = 10$ and $C_{FN} = 1$, to reflect the significant repercussions of misclassifying human-generated text. Table XIV of the Supplementary

¹<https://github.com/xinleihe/MGTBench>



Fig. 1: Average detection performance (x-axis) of detectors (OpenAI-D and our boosting strategy OpenAI-STK) tested across various LLMs, where these detectors are trained on texts generated by specific LLM (y-axis).

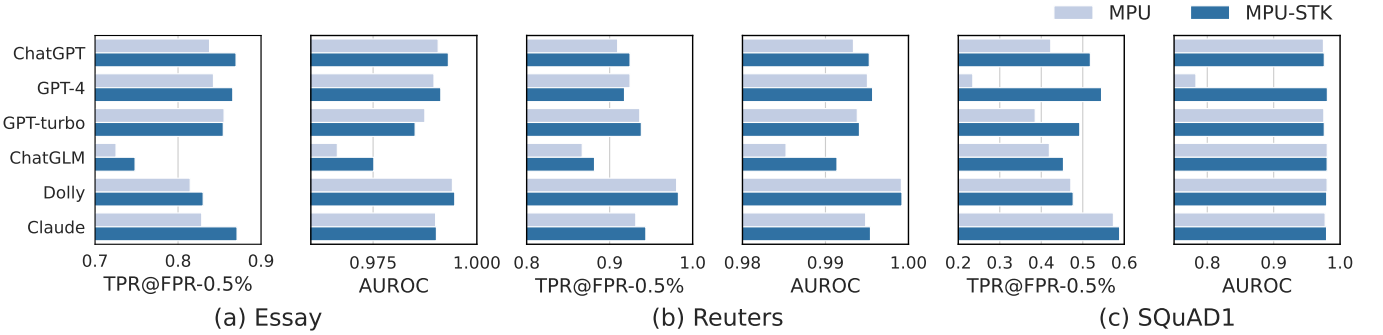


Fig. 2: Average detection performance (x-axis) of detectors (MPU and our boosting strategy MPU-STK) tested across various LLMs, where these detectors are trained on texts generated by specific LLM (y-axis).

Material reports the total expected cost across a range of plausible MGT base rates (10% to 50%), demonstrating that our method achieves lower total expected cost in imbalanced real-world scenarios. Notably, this setup corresponds to the ChatGPT column for the Essay task in Table I, where MPU and MPU-STK both display consistent, superior performance (e.g., 99.92%), achieving the same expected cost.

D. More Results for Robustness Comparison

In addition to the robustness results shown in the main text, Fig. 5 of the Supplementary Material shows the robust performance on OpenAI-D. It can be seen that the proposed strategy can also improve the robustness of the detection model in adversarial environments, which is very valuable.

E. Performance Comparison w.r.t. Accuracy

In addition to demonstrating the performance in terms of TPR@FPR-0.5% and AUROC, we also present a performance comparison in terms of accuracy to compare with GPTZero, as shown in Table XV of the Supplementary Material. Consistent with the previous results, the proposed enhancement framework significantly improves detection accuracy in most settings. For example, on the Essay dataset, the accuracy of OpenAI-D increases from 91.86 to 92.64. Moreover, the zero-shot performance of GPTZero is inferior to that of fine-tuned model-based detectors. This is particularly evident on the SQuAD1 dataset, where each sample contains shared human-written contextual spans, indicating GPTZero’s limitation in detecting texts with hidden human-like components.

F. Visualization

To more intuitively demonstrate the effectiveness of the proposed enhancement framework, we use t-SNE to visualize the last hidden states of ChatGPT-D (first row) and ChatGPT-STK (second row) when detecting texts generated by different LLMs (different columns), as shown in Fig. 6 of the Supplementary Material. First, compared with the original detector, the enhanced detector produces more separable representations between human-written and machine-generated texts. This indicates that the proposed framework helps the detector focus more on machine-indicative evidence after reducing the influence of hidden human-like spans. Second, MGTs exhibit lower diversity than human-written texts, which is consistent with existing findings [13].

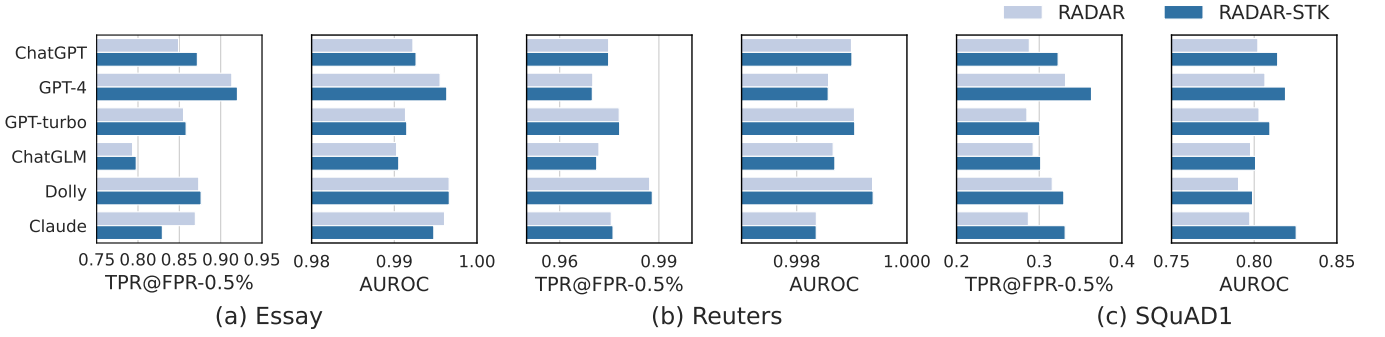


Fig. 3: Average detection performance (x-axis) of detectors (RADAR and our boosting strategy RADAR-STK) tested across various LLMs, where these detectors are trained on texts generated by specific LLM (y-axis).

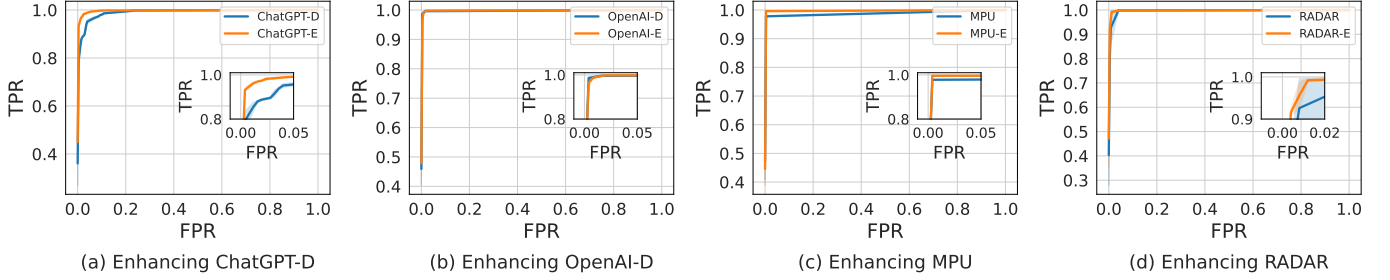


Fig. 4: ROC curves when detecting ChatGPT texts on Essay dataset. Detectors are trained on ChatGPT texts.

G. Comparison with Soft Stacked Detection Enhancement Framework

This section compares two EM-inspired detection enhancement frameworks to highlight the rationality of the hard EM-based approach. Due to the significant memory and computational costs of the soft version, we only experimented on the SQuAD1 dataset with fewer sequence numbers, setting the sequence number to 2. The results are shown in Fig. 7 of the Supplementary Material, where variants with the "soft" suffix represent the soft version. As can be seen, the overall performance of the soft version is inferior to the hard version, which aligns with many existing findings [14], [15], [16] that the soft EM algorithm tends to allocate too much probability mass to the tail, wasting probability mass on unimportant hidden variables, thereby resulting in poorer performance. Additionally, the soft version is competitive on the poorer-performing ChatGPT-D but underperforms on the stronger detectors (OpenAI-D and MPU). This observation is consistent with existing findings [17] that for better parameter models, EM tends to drift further than hard EM (thus losing accuracy). In summary, the experimental results demonstrate the rationality of optimization based on the hard EM algorithm.

H. Sensitivity Analysis of Hyperparameters

Sensitivity of Sequence Length k . Fig. 8 illustrates the detection performance under different sequence lengths, where k denotes the number of sentences in each subsequence. Notably, on the x-axis, 0 indicates the original detector without the proposed framework. From the figure, we observe that within a certain range, detection performance improves as k increases. This is because overly short subsequences provide limited contextual information, making it difficult for the detector to reliably estimate whether they should be filtered as human-like or retained as machine-indicative evidence. A moderately larger k provides richer context and thus enables more reliable latent-mask estimation. However, when the sequence length becomes excessively large, performance starts to decline. We suspect that overly long subsequences may entangle hidden human-like spans with machine-indicative evidence, making it harder to filter only the confidently human-like parts. Therefore, for most detectors, $k = 3$ or 4 serves as a good compromise between contextual sufficiency and filtering granularity. Furthermore, when the base detector performs poorly, such as on SQuAD1, a smaller k is more suitable for finer-grained exploration. In such cases, we recommend $k = 1$ or 2.

Sensitivity of Filter Ratio τ . Fig. 9 presents the detection performance under different filter ratios τ . The results show that as τ increases, detection performance first improves and then declines, which supports the necessity of introducing this hyperparameter. Specifically, a moderate filtering ratio helps remove confidently human-like subsequences and reduces their interference with detection. However, an excessively large τ may remove useful machine-indicative evidence and leave too few retained subsequences for reliable prediction, thereby leading to performance degradation. Consequently, $\tau = 0.5$ serves as a suitable compromise in most settings.

TABLE I: Performance concerning TPR@FPR-0.5%. Detectors are trained on ChatGPT text.

Dataset	Method	ChatGPT	GPT-4	GPT-turbo	ChatGLM	Dolly	Claude/StableLM	Avg.
SQuAD1	Log-Likelihood	1.04 $\pm$ 0.57	5.00 $\pm$ 2.93	2.11 $\pm$ 0.29	1.85 $\pm$ 1.12	5.93 $\pm$ 3.44	21.85 $\pm$ 10.14	6.30
	Log-Rank	1.97 $\pm$ 1.19	4.88 $\pm$ 3.25	2.34 $\pm$ 0.83	1.97 $\pm$ 1.35	4.88 $\pm$ 2.72	19.77 $\pm$ 11.36	5.97
	DetectGPT	0.35 $\pm$ 0.69	0.36 $\pm$ 0.48	0.47 $\pm$ 0.44	0.58 $\pm$ 0.73	0.58 $\pm$ 0.37	0.81 $\pm$ 0.28	0.52
	F-DetectGPT	4.28 $\pm$ 1.35	6.90 $\pm$ 1.53	2.69 $\pm$ 1.41	2.77 $\pm$ 1.12	5.35 $\pm$ 2.19	28.44 $\pm$ 3.93	8.41
	RepreGuard	0.58 $\pm$ 0.37	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.35 $\pm$ 0.46	0.12 $\pm$ 0.23	0.12 $\pm$ 0.23	0.19
	Lastde	1.16 $\pm$ 0.82	0.83 $\pm$ 0.48	2.22 $\pm$ 0.57	1.50 $\pm$ 0.78	1.86 $\pm$ 0.44	2.31 $\pm$ 1.21	1.65
	Binoculars	1.39 $\pm$ 0.46	2.02 $\pm$ 1.04	2.46 $\pm$ 0.86	1.39 $\pm$ 0.28	2.67 $\pm$ 1.63	11.56 $\pm$ 2.64	3.58
	ChatGPT-D	1.39 $\pm$ 1.07	1.19 $\pm$ 0.84	1.64 $\pm$ 1.07	0.81 $\pm$ 0.46	1.28 $\pm$ 1.13	5.43 $\pm$ 1.85	1.96
	ChatGPT-STK	1.16 $\pm$ 0.73	4.52 $\pm$ 1.67	1.17 $\pm$ 0.64	0.81 $\pm$ 0.46	1.51 $\pm$ 1.36	6.01 $\pm$ 3.61	2.53
	OpenAI-D	41.85 $\pm$ 16.81	45.00 $\pm$ 18.20	37.54 $\pm$ 16.76	29.48 $\pm$ 11.36	58.02 $\pm$ 24.87	72.83 $\pm$ 14.76	47.45
	OpenAI-STK	44.86 $\pm$ 17.81	50.48 $\pm$ 23.10	41.64 $\pm$ 21.26	29.83 $\pm$ 13.42	66.40 $\pm$ 22.81	73.41 $\pm$ 12.49	51.10
	MPU	30.64 $\pm$ 27.70	29.40 $\pm$ 29.38	51.23 $\pm$ 29.07	23.01 $\pm$ 18.60	61.05 $\pm$ 31.23	58.38 $\pm$ 13.31	42.28
	MPU-STK	37.80 $\pm$ 17.68	67.26 $\pm$ 22.09	59.88 $\pm$ 22.36	28.44 $\pm$ 12.41	59.65 $\pm$ 30.22	57.92 $\pm$ 11.15	51.83
	RADAR	27.05 $\pm$ 14.62	30.71 $\pm$ 10.01	34.27 $\pm$ 10.12	19.54 $\pm$ 13.00	31.74 $\pm$ 4.38	29.48 $\pm$ 11.57	28.80
	RADAR-STK	33.18 $\pm$ 11.40	32.26 $\pm$ 8.40	38.25 $\pm$ 5.54	24.16 $\pm$ 9.88	32.33 $\pm$ 1.93	33.53 $\pm$ 11.31	32.28

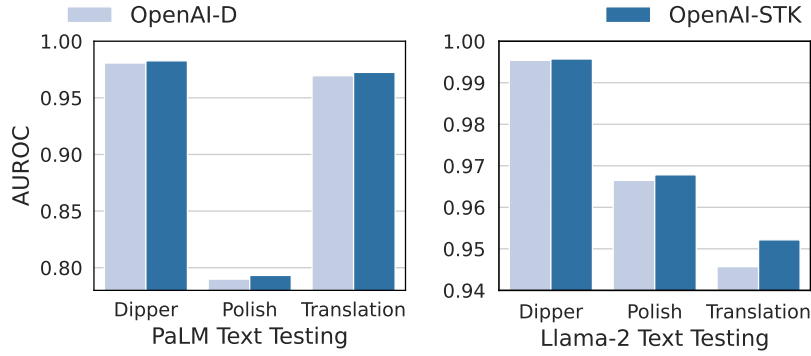


Fig. 5: Enhance the robustness of OpenAI-D. Here we use three attacks: Dipper, Polish, and Translation.

Sensitivity of Confidence Threshold r_e . Fig. 10 illustrates the detection performance under different confidence thresholds r_e . In standard classification tasks, a threshold of 0.5 is commonly used. However, as discussed in Section IV-B of the main text, estimating subsequence-level retention masks is more challenging than text-level detection, especially when the proportion of hidden human-like spans is unknown. Therefore, a stricter threshold is needed to filter only confidently human-like subsequences. This conservative design can mitigate mis-filtering, prevent useful machine-indicative evidence from being removed, and thus avoid performance degradation. The sensitivity results support this observation: using 0.5 is generally not optimal, while smaller thresholds lead to more stable improvements. Moreover, the proposed framework is relatively insensitive within a small-threshold range, suggesting that $r_e = 0.005\text{--}0.05$ is a practical choice in most settings.

I. Case Study

We illustrate examples that can be correctly detected by the proposed enhancement framework but not by the original detector, as shown in Figs. 11-15. Instead of directly using the complete text, the proposed framework identifies and filters confidently human-like subsequences (highlighted in green), allowing the detector to focus more on the remaining machine-indicative evidence. This makes the final prediction less affected by weak or misleading human-like spans. It is worth noting that, due to the lack of sentence-level ground-truth labels, we employ a voting mechanism with five detectors (ChatGPT-D, OpenAI-D, MPU, GPTZero, and ZeroGPT) to obtain pseudo labels. The filtered subsequences are consistently identified as human-written by these detectors, indicating that the proposed framework can capture high-confidence human-like subsequences.

Furthermore, our case analysis reveals several distinctions between filtered human-like subsequences and retained machine-indicative texts:

- **Sentence structure.** Human-like subsequences tend to exhibit relatively short and succinct sentence structures (e.g., “A randomized control trial found no difference...”). In contrast, retained machine-indicative texts often contain more complex structures, including frequent subordinate clauses and detailed arguments. For instance: “The use of time is fundamental... However, ... For example, in The Winter’s Tale... These visual transitions... In contrast, the Novel conveys... While the reader experiences... This gradual progression... Overall, time in the Novel... The use of time also impacts... For instance, in The Winter’s Tale... Similarly, Hermione is reanimated... In the Novel, character development... For example, Emma Woodhouse...”.

TABLE II: Performance concerning AUROC. Detectors are trained on ChatGPT text.

Dataset	Method	ChatGPT	GPT-4	GPT-turbo	ChatGLM	Dolly	Claude/StableLM	Avg.
Essay	Log-Likelihood	98.48 $\pm$ 0.43	96.46 $\pm$ 1.05	98.85 $\pm$ 0.42	98.78 $\pm$ 0.15	90.64 $\pm$ 0.35	92.16 $\pm$ 0.32	95.90
	Log-Rank	98.64 $\pm$ 0.43	96.85 $\pm$ 0.89	98.92 $\pm$ 0.40	98.94 $\pm$ 0.16	90.41 $\pm$ 0.50	91.47 $\pm$ 0.45	95.87
	DetectGPT	96.86 $\pm$ 0.97	95.64 $\pm$ 0.45	44.40 $\pm$ 1.73	95.60 $\pm$ 0.62	92.55 $\pm$ 0.58	46.36 $\pm$ 0.71	78.57
	F-DetectGPT	83.86 $\pm$ 1.52	84.45 $\pm$ 1.58	94.12 $\pm$ 0.72	93.62 $\pm$ 0.73	55.86 $\pm$ 2.54	58.96 $\pm$ 1.52	78.48
	RepreGuard	98.81 $\pm$ 0.31	97.52 $\pm$ 0.38	41.79 $\pm$ 1.84	94.46 $\pm$ 1.02	94.78 $\pm$ 1.25	44.62 $\pm$ 2.02	78.66
	Lastde	99.68 $\pm$ 0.23	99.51 $\pm$ 0.14	93.99 $\pm$ 0.64	99.93 $\pm$ 0.04	97.99 $\pm$ 0.48	83.94 $\pm$ 0.81	95.84
	Binoculars	99.29 $\pm$ 0.50	98.73 $\pm$ 0.47	99.57 $\pm$ 0.17	99.87 $\pm$ 0.07	95.74 $\pm$ 0.56	92.18 $\pm$ 0.57	97.56
	ChatGPT-D	98.71 $\pm$ 0.98	98.14 $\pm$ 1.63	91.31 $\pm$ 2.24	99.73 $\pm$ 0.16	86.39 $\pm$ 7.11	48.70 $\pm$ 11.69	87.16
	ChatGPT-STK	98.98 $\pm$ 1.26	98.41 $\pm$ 1.91	92.78 $\pm$ 1.29	99.80 $\pm$ 0.25	89.96 $\pm$ 6.56	61.59 $\pm$ 13.52	90.25
	OpenAI-D	99.55 $\pm$ 0.78	99.73 $\pm$ 0.34	99.25 $\pm$ 0.53	99.87 $\pm$ 0.20	98.59 $\pm$ 0.86	87.63 $\pm$ 2.72	97.44
	OpenAI-STK	99.87 $\pm$ 0.14	99.82 $\pm$ 0.13	99.57 $\pm$ 0.10	99.95 $\pm$ 0.03	98.75 $\pm$ 0.44	90.64 $\pm$ 1.32	98.10
	MPU	100.00 $\pm$ 0.01	99.91 $\pm$ 0.07	98.63 $\pm$ 0.21	99.99 $\pm$ 0.02	97.74 $\pm$ 0.46	98.14 $\pm$ 0.55	99.07
	MPU-STK	100.00 $\pm$ 0.00	99.91 $\pm$ 0.07	98.50 $\pm$ 0.27	99.99 $\pm$ 0.01	98.07 $\pm$ 0.39	99.41 $\pm$ 0.29	99.31
	RADAR	99.93 $\pm$ 0.02	99.78 $\pm$ 0.09	99.85 $\pm$ 0.05	99.97 $\pm$ 0.02	97.61 $\pm$ 0.59	98.22 $\pm$ 0.26	99.22
	RADAR-STK	99.94 $\pm$ 0.03	99.72 $\pm$ 0.13	99.86 $\pm$ 0.06	99.97 $\pm$ 0.01	97.88 $\pm$ 0.62	98.20 $\pm$ 0.42	99.26
Reuters	Log-Likelihood	97.59 $\pm$ 0.37	74.85 $\pm$ 0.50	98.54 $\pm$ 0.38	99.54 $\pm$ 0.22	60.09 $\pm$ 2.13	85.66 $\pm$ 1.04	86.05
	Log-Rank	97.86 $\pm$ 0.35	79.95 $\pm$ 0.29	98.82 $\pm$ 0.37	99.65 $\pm$ 0.20	61.24 $\pm$ 2.15	84.88 $\pm$ 1.13	87.07
	DetectGPT	92.78 $\pm$ 1.34	85.71 $\pm$ 2.02	49.24 $\pm$ 1.65	91.59 $\pm$ 1.67	83.41 $\pm$ 1.73	66.59 $\pm$ 2.53	78.22
	F-DetectGPT	96.23 $\pm$ 0.19	66.02 $\pm$ 1.53	98.85 $\pm$ 0.30	98.58 $\pm$ 0.35	40.38 $\pm$ 1.31	66.16 $\pm$ 2.58	77.70
	RepreGuard	98.71 $\pm$ 0.49	96.86 $\pm$ 0.83	47.69 $\pm$ 1.79	96.99 $\pm$ 0.68	6.54 $\pm$ 1.05	21.22 $\pm$ 2.42	61.33
	Lastde	99.68 $\pm$ 0.09	99.28 $\pm$ 0.30	99.36 $\pm$ 0.19	99.48 $\pm$ 0.29	95.45 $\pm$ 0.97	86.48 $\pm$ 1.29	96.62
	Binoculars	99.37 $\pm$ 0.14	93.10 $\pm$ 0.78	99.43 $\pm$ 0.28	99.71 $\pm$ 0.19	75.70 $\pm$ 1.89	87.10 $\pm$ 1.28	92.4
	ChatGPT-D	99.79 $\pm$ 0.23	99.14 $\pm$ 0.65	99.64 $\pm$ 0.37	99.46 $\pm$ 0.49	89.21 $\pm$ 5.80	68.66 $\pm$ 12.94	92.65
	ChatGPT-STK	99.89 $\pm$ 0.17	99.29 $\pm$ 0.48	99.86 $\pm$ 0.20	99.54 $\pm$ 0.17	91.13 $\pm$ 3.36	84.61 $\pm$ 6.87	95.72
	OpenAI-D	99.80 $\pm$ 0.20	99.46 $\pm$ 0.31	99.82 $\pm$ 0.20	99.68 $\pm$ 0.23	96.95 $\pm$ 0.51	87.26 $\pm$ 1.08	97.16
	OpenAI-STK	99.99 $\pm$ 0.01	99.83 $\pm$ 0.07	100.00 $\pm$ 0.00	99.83 $\pm$ 0.12	98.00 $\pm$ 0.37	91.69 $\pm$ 1.10	98.22
	MPU	100.00 $\pm$ 0.00	99.81 $\pm$ 0.08	99.99 $\pm$ 0.02	99.99 $\pm$ 0.01	97.55 $\pm$ 0.63	98.69 $\pm$ 0.35	99.34
	MPU-STK	100.00 $\pm$ 0.00	99.80 $\pm$ 0.14	100.00 $\pm$ 0.00	99.98 $\pm$ 0.01	98.02 $\pm$ 0.58	99.36 $\pm$ 0.36	99.53
	RADAR	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.99 $\pm$ 0.01	99.99 $\pm$ 0.02	99.60 $\pm$ 0.20	99.82 $\pm$ 0.11	99.90
	RADAR-STK	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.99 $\pm$ 0.01	99.99 $\pm$ 0.01	99.60 $\pm$ 0.20	99.82 $\pm$ 0.11	99.90
SQuAD1	Log-Likelihood	69.62 $\pm$ 0.28	69.61 $\pm$ 1.06	67.18 $\pm$ 0.67	65.67 $\pm$ 0.56	67.24 $\pm$ 1.10	85.82 $\pm$ 1.96	70.86
	Log-Rank	68.78 $\pm$ 0.34	69.52 $\pm$ 0.79	66.33 $\pm$ 0.59	64.99 $\pm$ 0.65	66.83 $\pm$ 1.04	85.50 $\pm$ 1.97	70.33
	DetectGPT	52.60 $\pm$ 2.74	47.47 $\pm$ 1.42	52.05 $\pm$ 1.87	50.88 $\pm$ 2.47	50.24 $\pm$ 2.20	51.12 $\pm$ 2.44	50.73
	F-DetectGPT	67.91 $\pm$ 0.23	63.84 $\pm$ 1.73	64.67 $\pm$ 0.20	63.36 $\pm$ 0.42	66.32 $\pm$ 1.36	79.62 $\pm$ 0.63	67.62
	RepreGuard	48.70 $\pm$ 1.48	50.14 $\pm$ 3.33	48.74 $\pm$ 0.82	49.15 $\pm$ 1.28	50.78 $\pm$ 3.83	50.82 $\pm$ 10.13	49.72
	Lastde	58.18 $\pm$ 0.59	56.58 $\pm$ 1.03	58.46 $\pm$ 0.35	57.33 $\pm$ 0.39	58.30 $\pm$ 0.68	67.65 $\pm$ 0.76	59.42
	Binoculars	66.34 $\pm$ 0.21	62.58 $\pm$ 1.33	63.55 $\pm$ 0.36	62.49 $\pm$ 0.50	64.22 $\pm$ 1.56	77.28 $\pm$ 1.06	66.08
	ChatGPT-D	56.27 $\pm$ 1.38	43.03 $\pm$ 2.98	54.69 $\pm$ 1.29	53.76 $\pm$ 1.57	54.41 $\pm$ 1.32	63.75 $\pm$ 2.58	54.32
	ChatGPT-STK	56.99 $\pm$ 0.85	58.55 $\pm$ 1.43	55.14 $\pm$ 1.48	55.03 $\pm$ 1.41	55.28 $\pm$ 0.59	64.33 $\pm$ 1.40	57.55
	OpenAI-D	99.19 $\pm$ 0.28	99.34 $\pm$ 0.27	98.63 $\pm$ 0.75	94.95 $\pm$ 1.21	97.74 $\pm$ 0.28	99.53 $\pm$ 0.22	98.23
	OpenAI-STK	99.30 $\pm$ 0.28	99.31 $\pm$ 0.28	98.73 $\pm$ 0.75	94.90 $\pm$ 0.98	97.84 $\pm$ 0.39	99.62 $\pm$ 0.15	98.28
	MPU	98.61 $\pm$ 0.54	98.79 $\pm$ 0.71	98.71 $\pm$ 0.83	93.53 $\pm$ 1.36	96.45 $\pm$ 0.94	98.85 $\pm$ 0.48	97.49
	MPU-STK	98.70 $\pm$ 0.35	99.21 $\pm$ 0.59	98.74 $\pm$ 0.73	93.71 $\pm$ 1.47	96.51 $\pm$ 0.77	98.84 $\pm$ 0.45	97.62
	RADAR	82.07 $\pm$ 1.92	80.63 $\pm$ 1.26	81.19 $\pm$ 1.82	78.14 $\pm$ 1.21	77.76 $\pm$ 2.07	81.53 $\pm$ 2.07	80.22
	RADAR-STK	83.45 $\pm$ 1.85	81.46 $\pm$ 1.00	83.17 $\pm$ 0.73	79.05 $\pm$ 0.96	79.00 $\pm$ 1.69	82.37 $\pm$ 2.33	81.42

- **Sentence length.** Human-like subsequences generally favor short to medium-length sentences, often around 12–15 words. By contrast, retained machine-indicative texts usually contain longer sentences, often around 20–25 words, while their shorter sentences typically serve as transitions or summaries (e.g., “Overall, time in the Novel is more mimetic than in Drama.”).
- **Linguistic style.** Human-like subsequences typically use concise analytical expressions, whereas retained machine-indicative texts tend to adopt a more formal style and frequently include specialized terms such as “mimetic”, “interiority”, “staging”, and “pacing”.

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TABLE III: Performance concerning AUROC. Detectors are trained on GPT-4 texts.

Dataset	Method	ChatGPT	GPT-4	GPT-turbo	ChatGLM	Dolly	Claude/StableLM	Avg.
Essay	Log-Likelihood	98.48 $\pm$ 0.43	96.46 $\pm$ 1.05	98.85 $\pm$ 0.42	98.78 $\pm$ 0.15	90.64 $\pm$ 0.35	92.16 $\pm$ 0.32	95.90
	Log-Rank	98.64 $\pm$ 0.43	96.85 $\pm$ 0.89	98.92 $\pm$ 0.40	98.94 $\pm$ 0.16	90.41 $\pm$ 0.50	91.47 $\pm$ 0.45	95.87
	DetectGPT	96.86 $\pm$ 0.97	95.64 $\pm$ 0.45	44.40 $\pm$ 1.73	95.60 $\pm$ 0.62	92.55 $\pm$ 0.58	46.36 $\pm$ 0.71	78.57
	F-DetectGPT	83.86 $\pm$ 1.52	84.45 $\pm$ 1.58	94.12 $\pm$ 0.72	93.62 $\pm$ 0.73	55.86 $\pm$ 2.54	58.96 $\pm$ 1.52	78.48
	RepreGuard	99.81 $\pm$ 0.04	95.90 $\pm$ 0.40	41.79 $\pm$ 1.84	94.46 $\pm$ 1.02	94.78 $\pm$ 1.25	44.62 $\pm$ 2.02	78.56
	Lastde	99.68 $\pm$ 0.23	99.51 $\pm$ 0.14	93.99 $\pm$ 0.64	99.93 $\pm$ 0.04	97.99 $\pm$ 0.48	83.94 $\pm$ 0.81	95.84
	Binoculars	99.29 $\pm$ 0.50	98.73 $\pm$ 0.47	99.57 $\pm$ 0.17	99.87 $\pm$ 0.07	95.74 $\pm$ 0.56	92.18 $\pm$ 0.57	97.56
	ChatGPT-D	99.38 $\pm$ 0.45	99.43 $\pm$ 0.65	91.96 $\pm$ 3.77	99.96 $\pm$ 0.03	94.81 $\pm$ 2.35	68.44 $\pm$ 12.85	92.33
	ChatGPT-STK	99.66 $\pm$ 0.12	99.66 $\pm$ 0.06	93.04 $\pm$ 1.65	99.92 $\pm$ 0.07	94.63 $\pm$ 1.05	72.64 $\pm$ 2.02	93.26
	OpenAI-D	99.72 $\pm$ 0.34	99.66 $\pm$ 0.26	98.51 $\pm$ 0.73	99.89 $\pm$ 0.14	98.39 $\pm$ 0.83	81.29 $\pm$ 4.23	96.24
	OpenAI-STK	99.84 $\pm$ 0.22	99.81 $\pm$ 0.16	99.15 $\pm$ 0.58	99.95 $\pm$ 0.04	98.44 $\pm$ 1.14	86.54 $\pm$ 5.25	97.29
	MPU	99.99 $\pm$ 0.01	99.95 $\pm$ 0.03	98.77 $\pm$ 0.34	99.99 $\pm$ 0.01	98.02 $\pm$ 0.37	97.09 $\pm$ 0.60	98.97
	MPU-STK	99.99 $\pm$ 0.00	99.91 $\pm$ 0.09	98.83 $\pm$ 0.21	99.99 $\pm$ 0.01	98.15 $\pm$ 0.29	97.92 $\pm$ 0.57	99.13
	RADAR	99.92 $\pm$ 0.02	99.89 $\pm$ 0.05	99.89 $\pm$ 0.10	99.98 $\pm$ 0.01	98.66 $\pm$ 0.24	98.98 $\pm$ 0.26	99.55
	RADAR-STK	99.93 $\pm$ 0.01	99.91 $\pm$ 0.04	99.90 $\pm$ 0.08	99.98 $\pm$ 0.01	98.99 $\pm$ 0.23	99.09 $\pm$ 0.23	99.63
Reuters	Log-Likelihood	97.59 $\pm$ 0.37	74.85 $\pm$ 0.50	98.54 $\pm$ 0.38	99.54 $\pm$ 0.22	60.09 $\pm$ 2.13	85.66 $\pm$ 1.04	86.05
	Log-Rank	97.86 $\pm$ 0.35	79.95 $\pm$ 0.29	98.82 $\pm$ 0.37	99.65 $\pm$ 0.20	61.24 $\pm$ 2.15	84.88 $\pm$ 1.13	87.07
	DetectGPT	92.78 $\pm$ 1.34	85.71 $\pm$ 2.02	49.24 $\pm$ 1.65	91.59 $\pm$ 1.67	83.41 $\pm$ 1.73	66.59 $\pm$ 2.53	78.22
	F-DetectGPT	96.23 $\pm$ 0.19	66.02 $\pm$ 1.53	98.85 $\pm$ 0.30	98.58 $\pm$ 0.35	40.38 $\pm$ 1.31	66.16 $\pm$ 2.58	77.70
	RepreGuard	98.71 $\pm$ 0.49	96.86 $\pm$ 0.83	47.69 $\pm$ 1.79	96.99 $\pm$ 0.68	6.54 $\pm$ 1.05	21.22 $\pm$ 2.40	61.34
	Lastde	99.68 $\pm$ 0.09	99.28 $\pm$ 0.30	99.36 $\pm$ 0.19	99.48 $\pm$ 0.29	95.45 $\pm$ 0.97	86.48 $\pm$ 1.29	96.62
	Binoculars	99.37 $\pm$ 0.14	93.10 $\pm$ 0.78	99.43 $\pm$ 0.28	99.71 $\pm$ 0.19	75.70 $\pm$ 1.89	87.10 $\pm$ 1.28	92.4
	ChatGPT-D	99.59 $\pm$ 0.53	99.29 $\pm$ 0.78	99.37 $\pm$ 0.57	99.07 $\pm$ 0.76	90.45 $\pm$ 3.16	67.02 $\pm$ 7.87	92.47
	ChatGPT-STK	99.73 $\pm$ 0.17	99.56 $\pm$ 0.36	99.39 $\pm$ 0.51	99.65 $\pm$ 0.24	92.08 $\pm$ 0.68	74.16 $\pm$ 7.11	94.10
	OpenAI-D	99.65 $\pm$ 0.44	99.63 $\pm$ 0.44	99.63 $\pm$ 0.42	99.58 $\pm$ 0.48	98.51 $\pm$ 0.78	80.91 $\pm$ 1.35	96.32
	OpenAI-STK	99.76 $\pm$ 0.25	99.82 $\pm$ 0.20	99.78 $\pm$ 0.25	99.75 $\pm$ 0.24	99.09 $\pm$ 0.43	83.85 $\pm$ 2.12	97.01
	MPU	100.00 $\pm$ 0.00	99.94 $\pm$ 0.05	100.00 $\pm$ 0.00	99.99 $\pm$ 0.01	99.48 $\pm$ 0.33	97.62 $\pm$ 0.97	99.51
	MPU-STK	100.00 $\pm$ 0.00	99.90 $\pm$ 0.09	100.00 $\pm$ 0.00	99.99 $\pm$ 0.01	99.39 $\pm$ 0.27	98.13 $\pm$ 0.77	99.57
	RADAR	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.96 $\pm$ 0.02	100.00 $\pm$ 0.01	99.53 $\pm$ 0.16	99.66 $\pm$ 0.14	99.86
	RADAR-STK	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.96 $\pm$ 0.02	100.00 $\pm$ 0.01	99.53 $\pm$ 0.16	99.66 $\pm$ 0.14	99.86
SQuAD1	Log-Likelihood	69.62 $\pm$ 0.28	69.61 $\pm$ 1.06	67.18 $\pm$ 0.67	65.67 $\pm$ 0.56	67.24 $\pm$ 1.10	85.82 $\pm$ 1.96	70.86
	Log-Rank	68.78 $\pm$ 0.34	69.52 $\pm$ 0.79	66.33 $\pm$ 0.59	64.99 $\pm$ 0.65	66.83 $\pm$ 1.04	85.50 $\pm$ 1.97	70.33
	DetectGPT	49.18 $\pm$ 3.68	52.08 $\pm$ 2.03	48.92 $\pm$ 2.56	50.32 $\pm$ 2.60	48.93 $\pm$ 1.94	49.87 $\pm$ 2.69	49.88
	F-DetectGPT	67.91 $\pm$ 0.23	63.84 $\pm$ 1.73	64.67 $\pm$ 0.20	63.36 $\pm$ 0.42	66.32 $\pm$ 1.36	79.62 $\pm$ 0.63	67.62
	RepreGuard	50.42 $\pm$ 1.93	51.82 $\pm$ 2.79	49.93 $\pm$ 1.50	50.09 $\pm$ 1.53	51.49 $\pm$ 3.61	56.06 $\pm$ 8.17	51.63
	Lastde	58.18 $\pm$ 0.59	56.58 $\pm$ 1.03	58.46 $\pm$ 0.35	57.33 $\pm$ 0.39	58.30 $\pm$ 0.68	67.65 $\pm$ 0.76	59.42
	Binoculars	66.34 $\pm$ 0.21	62.58 $\pm$ 1.33	63.55 $\pm$ 0.36	62.49 $\pm$ 0.50	64.22 $\pm$ 1.56	77.28 $\pm$ 1.06	66.08
	ChatGPT-D	57.52 $\pm$ 0.82	61.68 $\pm$ 3.26	55.34 $\pm$ 0.48	54.90 $\pm$ 0.69	55.55 $\pm$ 1.19	66.07 $\pm$ 2.47	58.51
	ChatGPT-STK	57.38 $\pm$ 0.50	59.11 $\pm$ 0.81	55.74 $\pm$ 0.67	55.09 $\pm$ 0.41	55.50 $\pm$ 0.82	66.01 $\pm$ 1.32	58.14
	OpenAI-D	91.19 $\pm$ 1.69	99.72 $\pm$ 0.16	89.44 $\pm$ 1.71	86.38 $\pm$ 1.76	88.06 $\pm$ 1.10	96.31 $\pm$ 1.27	91.85
	OpenAI-STK	99.45 $\pm$ 0.23	99.05 $\pm$ 0.46	98.85 $\pm$ 0.73	95.83 $\pm$ 0.83	98.09 $\pm$ 0.25	99.71 $\pm$ 0.17	98.50
	MPU	73.14 $\pm$ 5.54	99.11 $\pm$ 0.50	69.43 $\pm$ 6.64	68.98 $\pm$ 5.69	71.34 $\pm$ 5.03	87.73 $\pm$ 2.94	78.29
	MPU-STK	99.11 $\pm$ 0.19	98.96 $\pm$ 0.74	98.91 $\pm$ 0.56	94.88 $\pm$ 1.20	97.53 $\pm$ 0.47	99.10 $\pm$ 0.23	98.08
	RADAR	82.32 $\pm$ 2.14	80.70 $\pm$ 0.82	81.86 $\pm$ 1.18	78.54 $\pm$ 0.60	78.42 $\pm$ 2.00	82.04 $\pm$ 2.07	80.65
	RADAR-STK	83.62 $\pm$ 1.57	82.09 $\pm$ 1.48	83.29 $\pm$ 1.39	79.50 $\pm$ 0.99	80.05 $\pm$ 1.83	82.79 $\pm$ 1.96	81.89

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TABLE IV: Performance concerning FPR@TPR-0.5%. Detectors are trained on GPT-4 texts.

Dataset	Method	ChatGPT	GPT-4	GPT-turbo	ChatGLM	Dolly	Claude/StableLM	Avg.
Essay	Log-Likelihood	24.08 $\pm$ 22.67	37.70 $\pm$ 30.63	23.12 $\pm$ 24.38	5.86 $\pm$ 7.02	12.45 $\pm$ 12.04	2.48 $\pm$ 3.06	17.62
	Log-Rank	28.72 $\pm$ 26.90	46.48 $\pm$ 36.20	25.04 $\pm$ 26.83	28.19 $\pm$ 27.90	17.34 $\pm$ 15.65	2.96 $\pm$ 3.73	24.79
	DetectGPT	37.04 $\pm$ 10.21	24.75 $\pm$ 10.03	5.52 $\pm$ 1.93	21.45 $\pm$ 14.25	15.45 $\pm$ 7.05	4.48 $\pm$ 2.57	18.12
	F-DetectGPT	4.24 $\pm$ 1.51	3.85 $\pm$ 2.05	31.28 $\pm$ 3.18	35.74 $\pm$ 3.97	0.00 $\pm$ 0.00	0.16 $\pm$ 0.20	12.55
	RepreGuard	92.08 $\pm$ 2.28	16.72 $\pm$ 8.60	0.32 $\pm$ 0.16	31.89 $\pm$ 2.22	20.86 $\pm$ 3.22	0.16 $\pm$ 0.20	27
	Lastde	94.24 $\pm$ 1.83	88.77 $\pm$ 3.22	48.96 $\pm$ 3.15	98.39 $\pm$ 1.08	79.57 $\pm$ 5.36	15.52 $\pm$ 3.60	70.91
	Binoculars	91.60 $\pm$ 7.82	94.26 $\pm$ 2.49	85.36 $\pm$ 18.99	98.55 $\pm$ 1.03	62.83 $\pm$ 17.81	25.20 $\pm$ 12.61	76.3
	ChatGPT-D	89.92 $\pm$ 5.69	89.75 $\pm$ 7.23	43.52 $\pm$ 19.38	99.04 $\pm$ 0.70	55.54 $\pm$ 10.95	8.24 $\pm$ 5.98	64.33
	ChatGPT-STK	90.88 $\pm$ 1.39	93.61 $\pm$ 1.77	48.80 $\pm$ 3.50	99.04 $\pm$ 1.03	56.65 $\pm$ 3.69	8.40 $\pm$ 1.29	66.23
	OpenAI-D	82.08 $\pm$ 30.45	86.56 $\pm$ 12.64	71.04 $\pm$ 27.40	97.75 $\pm$ 2.94	70.64 $\pm$ 26.78	18.48 $\pm$ 12.66	71.09
	OpenAI-STK	95.68 $\pm$ 5.07	95.16 $\pm$ 3.43	82.48 $\pm$ 16.97	97.75 $\pm$ 1.86	78.28 $\pm$ 14.75	30.80 $\pm$ 16.05	80.03
	MPU	99.44 $\pm$ 0.32	99.51 $\pm$ 0.40	73.04 $\pm$ 6.10	99.44 $\pm$ 0.41	78.80 $\pm$ 2.72	55.36 $\pm$ 5.44	84.26
	MPU-STK	99.52 $\pm$ 0.39	99.43 $\pm$ 0.56	74.08 $\pm$ 6.42	99.52 $\pm$ 0.30	79.66 $\pm$ 4.67	67.36 $\pm$ 12.94	86.59
	RADAR	97.28 $\pm$ 1.32	95.98 $\pm$ 2.54	96.80 $\pm$ 2.01	99.20 $\pm$ 0.51	81.37 $\pm$ 7.64	77.44 $\pm$ 9.59	91.35
	RADAR-STK	97.92 $\pm$ 0.69	96.23 $\pm$ 2.62	97.36 $\pm$ 1.30	99.36 $\pm$ 0.32	82.49 $\pm$ 6.18	78.80 $\pm$ 8.18	92.03
Reuters	Log-Likelihood	77.84 $\pm$ 5.19	14.88 $\pm$ 5.98	86.08 $\pm$ 3.38	93.76 $\pm$ 2.03	11.20 $\pm$ 4.45	15.04 $\pm$ 6.86	49.80
	Log-Rank	82.40 $\pm$ 5.24	25.92 $\pm$ 7.08	90.96 $\pm$ 4.12	96.80 $\pm$ 0.88	14.00 $\pm$ 4.82	17.60 $\pm$ 8.29	54.61
	DetectGPT	4.40 $\pm$ 2.62	0.64 $\pm$ 0.54	2.32 $\pm$ 1.87	2.56 $\pm$ 2.80	0.48 $\pm$ 0.47	3.04 $\pm$ 1.61	2.24
	F-DetectGPT	48.00 $\pm$ 9.48	6.80 $\pm$ 1.88	92.96 $\pm$ 1.65	88.96 $\pm$ 4.80	0.00 $\pm$ 0.00	0.48 $\pm$ 0.39	39.53
	RepreGuard	26.88 $\pm$ 19.31	9.04 $\pm$ 6.02	0.08 $\pm$ 0.16	0.80 $\pm$ 1.01	0.00 $\pm$ 0.00	0.16 $\pm$ 0.20	6.16
	Lastde	92.64 $\pm$ 1.06	88.96 $\pm$ 1.71	87.36 $\pm$ 3.51	98.56 $\pm$ 0.65	66.00 $\pm$ 3.27	9.68 $\pm$ 3.04	73.87
	Binoculars	95.84 $\pm$ 0.93	70.88 $\pm$ 3.93	96.08 $\pm$ 0.93	99.28 $\pm$ 0.30	40.88 $\pm$ 4.94	26.16 $\pm$ 4.18	71.52
	ChatGPT-D	85.76 $\pm$ 23.50	82.08 $\pm$ 30.65	81.92 $\pm$ 29.00	83.04 $\pm$ 31.12	54.96 $\pm$ 20.38	9.28 $\pm$ 6.55	66.17
	ChatGPT-STK	94.48 $\pm$ 3.66	93.60 $\pm$ 6.28	90.00 $\pm$ 5.58	97.92 $\pm$ 1.11	56.08 $\pm$ 14.30	8.88 $\pm$ 8.47	73.49
	OpenAI-D	79.84 $\pm$ 38.52	83.84 $\pm$ 27.78	79.68 $\pm$ 37.85	79.36 $\pm$ 38.68	61.92 $\pm$ 23.84	16.96 $\pm$ 7.37	66.93
	OpenAI-STK	86.48 $\pm$ 25.04	94.00 $\pm$ 9.82	87.04 $\pm$ 24.12	96.08 $\pm$ 5.45	72.24 $\pm$ 13.89	26.16 $\pm$ 6.47	77.00
	MPU	100.00 $\pm$ 0.00	99.12 $\pm$ 0.64	99.92 $\pm$ 0.16	99.84 $\pm$ 0.20	88.40 $\pm$ 6.73	67.44 $\pm$ 16.46	92.45
	MPU-STK	100.00 $\pm$ 0.00	99.12 $\pm$ 0.64	100.00 $\pm$ 0.00	99.68 $\pm$ 0.30	86.64 $\pm$ 4.03	65.28 $\pm$ 13.06	91.79
	RADAR	99.92 $\pm$ 0.16	100.00 $\pm$ 0.00	99.04 $\pm$ 1.12	99.92 $\pm$ 0.16	92.16 $\pm$ 2.66	90.96 $\pm$ 3.28	97.00
	RADAR-STK	99.92 $\pm$ 0.16	100.00 $\pm$ 0.00	99.04 $\pm$ 1.12	99.92 $\pm$ 0.16	92.16 $\pm$ 2.66	90.88 $\pm$ 3.33	96.99
SQuAD1	Log-Likelihood	1.04 $\pm$ 0.57	5.00 $\pm$ 2.93	2.11 $\pm$ 0.29	1.85 $\pm$ 1.12	5.93 $\pm$ 3.44	21.85 $\pm$ 10.14	6.30
	Log-Rank	1.97 $\pm$ 1.19	4.88 $\pm$ 3.25	2.34 $\pm$ 0.83	1.97 $\pm$ 1.35	4.88 $\pm$ 2.72	19.77 $\pm$ 11.36	5.97
	DetectGPT	0.58 $\pm$ 0.52	1.31 $\pm$ 1.02	0.12 $\pm$ 0.23	0.58 $\pm$ 0.63	0.58 $\pm$ 0.64	0.46 $\pm$ 0.43	0.60
	F-DetectGPT	4.28 $\pm$ 1.35	6.90 $\pm$ 1.53	2.69 $\pm$ 1.41	2.77 $\pm$ 1.12	5.35 $\pm$ 2.19	28.44 $\pm$ 3.93	8.41
	RepreGuard	0.46 $\pm$ 0.43	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.35 $\pm$ 0.46	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.13
	Lastde	1.16 $\pm$ 0.82	0.83 $\pm$ 0.48	2.22 $\pm$ 0.57	1.50 $\pm$ 0.78	1.86 $\pm$ 0.44	2.31 $\pm$ 1.21	1.65
	Binoculars	1.39 $\pm$ 0.46	2.02 $\pm$ 1.04	2.46 $\pm$ 0.86	1.39 $\pm$ 0.28	2.67 $\pm$ 1.63	11.56 $\pm$ 2.64	3.58
	ChatGPT-D	1.27 $\pm$ 0.92	4.40 $\pm$ 1.17	0.94 $\pm$ 0.70	0.69 $\pm$ 0.23	1.63 $\pm$ 0.85	5.09 $\pm$ 1.34	2.34
	ChatGPT-STK	1.04 $\pm$ 0.67	4.05 $\pm$ 1.38	1.17 $\pm$ 0.52	1.04 $\pm$ 0.57	1.86 $\pm$ 1.19	7.75 $\pm$ 4.24	2.82
	OpenAI-D	9.60 $\pm$ 6.35	88.93 $\pm$ 13.08	10.88 $\pm$ 8.84	7.63 $\pm$ 4.41	20.58 $\pm$ 7.94	52.83 $\pm$ 23.86	31.74
	OpenAI-STK	49.02 $\pm$ 11.18	38.57 $\pm$ 21.65	46.08 $\pm$ 16.91	35.26 $\pm$ 8.20	69.53 $\pm$ 16.99	83.58 $\pm$ 7.79	53.67
	MPU	5.20 $\pm$ 6.26	92.14 $\pm$ 9.53	2.34 $\pm$ 1.57	4.51 $\pm$ 5.26	9.77 $\pm$ 2.79	27.05 $\pm$ 13.30	23.50
	MPU-STK	54.22 $\pm$ 18.76	46.31 $\pm$ 31.59	54.85 $\pm$ 15.39	43.70 $\pm$ 14.10	69.30 $\pm$ 22.98	58.73 $\pm$ 25.58	54.52
	RADAR	35.61 $\pm$ 9.70	29.88 $\pm$ 10.88	38.48 $\pm$ 3.40	28.09 $\pm$ 8.28	33.14 $\pm$ 3.23	33.87 $\pm$ 8.37	33.18
	RADAR-STK	39.77 $\pm$ 6.43	31.55 $\pm$ 11.45	42.22 $\pm$ 2.32	31.79 $\pm$ 5.32	36.28 $\pm$ 2.46	36.30 $\pm$ 7.33	36.32

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TABLE V: Performance concerning AUROC. Detectors are trained on GPT-turbo texts.

Dataset	Method	ChatGPT	GPT-4	GPT-turbo	ChatGLM	Dolly	Claude/StableLM	Avg.
Essay	Log-Likelihood	98.48 $\pm$ 0.43	96.46 $\pm$ 1.05	98.85 $\pm$ 0.42	98.78 $\pm$ 0.15	90.64 $\pm$ 0.35	92.16 $\pm$ 0.32	95.90
	Log-Rank	98.64 $\pm$ 0.43	96.85 $\pm$ 0.89	98.92 $\pm$ 0.40	98.94 $\pm$ 0.16	90.41 $\pm$ 0.50	91.47 $\pm$ 0.45	95.87
	DetectGPT	96.86 $\pm$ 0.97	95.64 $\pm$ 0.45	44.40 $\pm$ 1.73	95.60 $\pm$ 0.62	92.55 $\pm$ 0.58	46.36 $\pm$ 0.71	78.57
	F-DetectGPT	83.86 $\pm$ 1.52	84.45 $\pm$ 1.58	94.12 $\pm$ 0.72	93.62 $\pm$ 0.73	55.86 $\pm$ 2.54	58.96 $\pm$ 1.52	78.48
	RepreGuard	0.19 $\pm$ 0.04	2.48 $\pm$ 0.38	53.65 $\pm$ 1.76	5.54 $\pm$ 1.02	5.22 $\pm$ 1.25	55.38 $\pm$ 2.02	20.41
	Lastde	99.68 $\pm$ 0.23	99.51 $\pm$ 0.14	93.99 $\pm$ 0.64	99.93 $\pm$ 0.04	97.99 $\pm$ 0.48	83.94 $\pm$ 0.81	95.84
	Binoculars	99.29 $\pm$ 0.50	98.73 $\pm$ 0.47	99.57 $\pm$ 0.17	99.87 $\pm$ 0.07	95.74 $\pm$ 0.56	92.18 $\pm$ 0.57	97.56
	ChatGPT-D	97.11 $\pm$ 0.39	96.26 $\pm$ 2.23	96.66 $\pm$ 1.76	99.62 $\pm$ 0.15	80.88 $\pm$ 7.54	66.73 $\pm$ 17.76	89.54
	ChatGPT-STK	96.86 $\pm$ 1.34	96.37 $\pm$ 0.90	96.53 $\pm$ 1.33	99.48 $\pm$ 0.31	83.37 $\pm$ 5.26	67.94 $\pm$ 15.24	90.09
	OpenAI-D	99.77 $\pm$ 0.31	99.87 $\pm$ 0.08	99.83 $\pm$ 0.32	99.88 $\pm$ 0.16	98.51 $\pm$ 0.81	97.62 $\pm$ 0.92	99.25
	OpenAI-STK	99.73 $\pm$ 0.15	99.75 $\pm$ 0.14	99.91 $\pm$ 0.15	99.89 $\pm$ 0.11	<u>97.93</u> $\pm$ 0.66	97.82 $\pm$ 0.85	99.17
	MPU	99.86 $\pm$ 0.05	99.79 $\pm$ 0.10	99.97 $\pm$ 0.02	99.93 $\pm$ 0.04	94.38 $\pm$ 0.53	98.55 $\pm$ 0.19	98.75
	MPU-STK	99.85 $\pm$ 0.04	99.74 $\pm$ 0.13	<u>99.96</u> $\pm$ 0.03	<u>99.91</u> $\pm$ 0.05	93.57 $\pm$ 0.80	98.03 $\pm$ 0.33	98.51
	RADAR	99.88 $\pm$ 0.05	<u>99.82</u> $\pm$ 0.05	99.97 $\pm$ 0.02	99.98 $\pm$ 0.00	97.03 $\pm$ 0.64	<u>98.13</u> $\pm$ 0.26	99.14
	RADAR-STK	<u>99.87</u> $\pm$ 0.07	99.81 $\pm$ 0.06	99.97 $\pm$ 0.02	99.98 $\pm$ 0.00	97.15 $\pm$ 0.95	98.13 $\pm$ 0.65	99.15
Reuters	Log-Likelihood	97.59 $\pm$ 0.37	74.85 $\pm$ 0.50	98.54 $\pm$ 0.38	99.54 $\pm$ 0.22	60.09 $\pm$ 2.13	85.66 $\pm$ 1.04	86.05
	Log-Rank	97.86 $\pm$ 0.35	79.95 $\pm$ 0.29	98.82 $\pm$ 0.37	99.65 $\pm$ 0.20	61.24 $\pm$ 2.15	84.88 $\pm$ 1.13	87.07
	DetectGPT	74.86 $\pm$ 34.84	69.99 $\pm$ 29.66	48.45 $\pm$ 0.95	74.21 $\pm$ 33.87	68.96 $\pm$ 27.57	58.45 $\pm$ 14.49	65.82
	F-DetectGPT	96.23 $\pm$ 0.19	66.02 $\pm$ 1.53	98.85 $\pm$ 0.30	98.58 $\pm$ 0.35	40.38 $\pm$ 1.31	66.16 $\pm$ 2.58	77.70
	RepreGuard	79.19 $\pm$ 39.00	77.94 $\pm$ 37.62	47.96 $\pm$ 2.09	78.01 $\pm$ 37.74	24.22 $\pm$ 35.00	33.91 $\pm$ 23.98	56.87
	Lastde	99.68 $\pm$ 0.09	99.28 $\pm$ 0.30	99.36 $\pm$ 0.19	99.48 $\pm$ 0.29	95.45 $\pm$ 0.97	86.48 $\pm$ 1.29	96.62
	Binoculars	99.37 $\pm$ 0.14	93.10 $\pm$ 0.78	99.43 $\pm$ 0.28	99.71 $\pm$ 0.19	75.70 $\pm$ 1.89	87.10 $\pm$ 1.28	92.4
	ChatGPT-D	99.12 $\pm$ 0.82	98.01 $\pm$ 0.98	99.52 $\pm$ 0.48	99.02 $\pm$ 0.50	81.10 $\pm$ 6.36	70.01 $\pm$ 19.60	91.13
	ChatGPT-STK	99.19 $\pm$ 0.78	98.11 $\pm$ 0.72	99.65 $\pm$ 0.42	99.11 $\pm$ 0.47	82.17 $\pm$ 6.04	75.57 $\pm$ 17.69	92.30
	OpenAI-D	99.78 $\pm$ 0.37	99.49 $\pm$ 0.45	99.79 $\pm$ 0.36	99.72 $\pm$ 0.30	96.88 $\pm$ 0.80	86.58 $\pm$ 3.28	97.04
	OpenAI-STK	<u>99.96</u> $\pm$ 0.01	99.58 $\pm$ 0.11	<u>99.99</u> $\pm$ 0.00	99.78 $\pm$ 0.14	95.84 $\pm$ 0.91	92.77 $\pm$ 1.47	97.99
	MPU	100.00 $\pm$ 0.00	99.74 $\pm$ 0.17	100.00 $\pm$ 0.00	99.97 $\pm$ 0.01	<u>97.30</u> $\pm$ 0.68	99.29 $\pm$ 0.29	99.38
	MPU-STK	100.00 $\pm$ 0.00	99.71 $\pm$ 0.22	100.00 $\pm$ 0.00	99.95 $\pm$ 0.03	97.11 $\pm$ 0.57	<u>99.68</u> $\pm$ 0.21	99.41
	RADAR	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	<u>99.99</u> $\pm$ 0.01	99.99 $\pm$ 0.01	99.61 $\pm$ 0.16	99.84 $\pm$ 0.09	99.90
	RADAR-STK	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	<u>99.99</u> $\pm$ 0.01	<u>99.99</u> $\pm$ 0.01	99.61 $\pm$ 0.16	99.84 $\pm$ 0.09	99.91
SQuAD1	Log-Likelihood	69.62 $\pm$ 0.28	69.61 $\pm$ 1.06	67.18 $\pm$ 0.67	65.67 $\pm$ 0.56	67.24 $\pm$ 1.10	85.82 $\pm$ 1.96	70.86
	Log-Rank	68.78 $\pm$ 0.34	69.52 $\pm$ 0.79	66.33 $\pm$ 0.59	64.99 $\pm$ 0.65	66.83 $\pm$ 1.04	85.50 $\pm$ 1.97	70.33
	DetectGPT	49.94 $\pm$ 3.77	48.22 $\pm$ 2.29	50.12 $\pm$ 2.77	51.15 $\pm$ 2.35	50.92 $\pm$ 2.02	49.83 $\pm$ 2.68	50.03
	F-DetectGPT	67.91 $\pm$ 0.23	63.84 $\pm$ 1.73	64.67 $\pm$ 0.20	63.36 $\pm$ 0.42	66.32 $\pm$ 1.36	79.62 $\pm$ 0.63	67.62
	RepreGuard	50.92 $\pm$ 1.75	46.75 $\pm$ 0.72	50.74 $\pm$ 1.31	51.03 $\pm$ 1.14	46.51 $\pm$ 1.75	40.00 $\pm$ 1.83	47.66
	Lastde	58.18 $\pm$ 0.59	56.58 $\pm$ 1.03	58.46 $\pm$ 0.35	57.33 $\pm$ 0.39	58.30 $\pm$ 0.68	67.65 $\pm$ 0.76	59.42
	Binoculars	66.34 $\pm$ 0.21	62.58 $\pm$ 1.33	63.55 $\pm$ 0.36	62.49 $\pm$ 0.50	64.22 $\pm$ 1.56	77.28 $\pm$ 1.06	66.08
	ChatGPT-D	58.69 $\pm$ 0.60	41.91 $\pm$ 2.79	57.44 $\pm$ 0.65	55.98 $\pm$ 0.35	55.61 $\pm$ 0.43	65.41 $\pm$ 1.86	55.84
	ChatGPT-STK	57.78 $\pm$ 1.56	59.87 $\pm$ 1.17	56.56 $\pm$ 1.09	56.06 $\pm$ 1.24	56.23 $\pm$ 0.85	66.53 $\pm$ 1.93	58.84
	OpenAI-D	99.43 $\pm$ 0.33	99.30 $\pm$ 0.38	98.51 $\pm$ 0.81	<u>95.88</u> $\pm$ 1.18	98.06 $\pm$ 0.40	99.58 $\pm$ 0.23	98.46
	OpenAI-STK	<u>99.43</u> $\pm$ 0.42	99.32 $\pm$ 0.32	<u>98.45</u> $\pm$ 0.86	95.98 $\pm$ 1.13	<u>98.04</u> $\pm$ 0.41	99.60 $\pm$ 0.24	98.47
	MPU	98.59 $\pm$ 0.82	98.80 $\pm$ 0.68	98.43 $\pm$ 0.94	94.27 $\pm$ 1.94	96.61 $\pm$ 0.68	98.60 $\pm$ 0.87	97.55
	MPU-STK	98.63 $\pm$ 0.81	99.23 $\pm$ 0.68	98.38 $\pm$ 0.94	94.28 $\pm$ 1.92	96.53 $\pm$ 0.73	98.65 $\pm$ 0.91	97.62
	RADAR	82.20 $\pm$ 2.31	80.49 $\pm$ 1.02	81.25 $\pm$ 1.39	78.44 $\pm$ 0.74	77.80 $\pm$ 1.91	81.56 $\pm$ 2.58	80.29
	RADAR-STK	82.95 $\pm$ 2.70	80.98 $\pm$ 1.16	82.09 $\pm$ 1.42	79.03 $\pm$ 0.53	78.67 $\pm$ 2.06	81.97 $\pm$ 2.97	80.95

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TABLE VI: Performance concerning TPR@FPR-0.5%. Detectors are trained on GPT-turbo texts.

Dataset	Method	ChatGPT	GPT-4	GPT-turbo	ChatGLM	Dolly	Claude/StableLM	Avg.
Essay	Log-Likelihood	24.08 $\pm$ 22.67	37.70 $\pm$ 30.63	23.12 $\pm$ 24.38	5.86 $\pm$ 7.02	12.45 $\pm$ 12.04	2.48 $\pm$ 3.06	17.62
	Log-Rank	28.72 $\pm$ 26.90	46.48 $\pm$ 36.20	25.04 $\pm$ 26.83	28.19 $\pm$ 27.90	17.34 $\pm$ 15.65	2.96 $\pm$ 3.73	24.79
	DetectGPT	37.04 $\pm$ 10.21	24.75 $\pm$ 10.03	5.52 $\pm$ 1.93	21.45 $\pm$ 14.25	15.45 $\pm$ 7.05	4.48 $\pm$ 2.57	18.12
	F-DetectGPT	4.24 $\pm$ 1.51	3.85 $\pm$ 2.05	31.28 $\pm$ 3.18	35.74 $\pm$ 3.97	0.00 $\pm$ 0.00	0.16 $\pm$ 0.20	12.55
	RepreGuard	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0
	Lastde	94.24 $\pm$ 1.83	88.77 $\pm$ 3.22	48.96 $\pm$ 3.15	98.39 $\pm$ 1.08	79.57 $\pm$ 5.36	15.52 $\pm$ 3.60	70.91
	Binoculars	91.60 $\pm$ 7.82	94.26 $\pm$ 2.49	85.36 $\pm$ 18.99	98.55 $\pm$ 1.03	62.83 $\pm$ 17.81	25.20 $\pm$ 12.61	76.3
	ChatGPT-D	60.72 $\pm$ 14.49	71.48 $\pm$ 4.55	65.36 $\pm$ 17.74	85.78 $\pm$ 7.09	23.18 $\pm$ 4.53	1.68 $\pm$ 0.89	51.37
	ChatGPT-STK	63.68 $\pm$ 15.32	66.39 $\pm$ 7.37	65.92 $\pm$ 13.67	88.92 $\pm$ 6.98	20.86 $\pm$ 10.25	1.68 $\pm$ 1.02	51.24
	OpenAI-D	79.12 $\pm$ 37.59	98.11 $\pm$ 1.12	79.76 $\pm$ 39.29	98.55 $\pm$ 1.21	70.90 $\pm$ 32.85	59.28 $\pm$ 31.57	80.96
	OpenAI-STK	95.44 $\pm$ 2.82	96.56 $\pm$ 1.90	99.76 $\pm$ 0.48	98.23 $\pm$ 1.21	70.99 $\pm$ 13.03	74.00 $\pm$ 15.92	89.16
	MPU	95.36 $\pm$ 1.94	96.97 $\pm$ 1.61	98.24 $\pm$ 0.82	98.63 $\pm$ 1.07	57.34 $\pm$ 6.19	66.72 $\pm$ 14.28	85.54
	MPU-STK	95.92 $\pm$ 1.98	95.98 $\pm$ 2.60	98.24 $\pm$ 0.90	98.63 $\pm$ 0.94	58.54 $\pm$ 5.27	65.36 $\pm$ 13.32	85.45
	RADAR	95.76 $\pm$ 1.57	94.67 $\pm$ 2.06	98.08 $\pm$ 1.39	98.96 $\pm$ 0.54	63.26 $\pm$ 4.51	62.32 $\pm$ 8.83	85.51
	RADAR-STK	95.76 $\pm$ 1.87	95.57 $\pm$ 1.59	98.32 $\pm$ 1.11	99.12 $\pm$ 0.64	64.64 $\pm$ 5.05	61.52 $\pm$ 10.33	85.82
Reuters	Log-Likelihood	77.84 $\pm$ 5.19	14.88 $\pm$ 5.98	86.08 $\pm$ 3.38	93.76 $\pm$ 2.03	11.20 $\pm$ 4.45	15.04 $\pm$ 6.86	49.80
	Log-Rank	82.40 $\pm$ 5.24	25.92 $\pm$ 7.08	90.96 $\pm$ 4.12	96.80 $\pm$ 0.88	14.00 $\pm$ 4.82	17.60 $\pm$ 8.29	54.61
	DetectGPT	2.64 $\pm$ 1.53	0.32 $\pm$ 0.30	1.20 $\pm$ 1.07	0.96 $\pm$ 0.82	0.88 $\pm$ 1.06	2.88 $\pm$ 1.85	1.48
	F-DetectGPT	48.00 $\pm$ 9.48	6.80 $\pm$ 1.88	92.96 $\pm$ 1.65	88.96 $\pm$ 4.80	0.00 $\pm$ 0.00	0.48 $\pm$ 0.39	39.53
	RepreGuard	26.16 $\pm$ 20.21	8.80 $\pm$ 6.34	3.04 $\pm$ 5.88	0.72 $\pm$ 1.06	10.32 $\pm$ 20.64	0.08 $\pm$ 0.16	8.19
	Lastde	92.64 $\pm$ 1.06	88.96 $\pm$ 1.71	87.36 $\pm$ 3.51	98.56 $\pm$ 0.65	66.00 $\pm$ 3.27	9.68 $\pm$ 3.04	73.87
	Binoculars	95.84 $\pm$ 0.93	70.88 $\pm$ 3.93	96.08 $\pm$ 0.93	99.28 $\pm$ 0.30	40.88 $\pm$ 4.94	26.16 $\pm$ 4.18	71.52
	ChatGPT-D	77.60 $\pm$ 24.67	74.48 $\pm$ 24.64	76.24 $\pm$ 28.15	76.00 $\pm$ 29.72	37.52 $\pm$ 15.02	24.72 $\pm$ 22.20	61.09
	ChatGPT-STK	93.60 $\pm$ 4.55	88.96 $\pm$ 5.73	96.64 $\pm$ 3.56	97.68 $\pm$ 1.20	39.76 $\pm$ 10.49	25.44 $\pm$ 16.75	73.68
	OpenAI-D	82.88 $\pm$ 32.04	81.12 $\pm$ 18.73	81.84 $\pm$ 34.33	88.08 $\pm$ 20.85	48.08 $\pm$ 11.99	22.24 $\pm$ 5.85	67.37
	OpenAI-STK	98.32 $\pm$ 0.59	83.84 $\pm$ 3.66	99.52 $\pm$ 0.30	98.16 $\pm$ 0.48	38.08 $\pm$ 6.02	37.36 $\pm$ 2.91	75.88
	MPU	100.00 $\pm$ 0.00	98.40 $\pm$ 0.76	100.00 $\pm$ 0.00	99.20 $\pm$ 0.36	72.72 $\pm$ 4.80	91.44 $\pm$ 2.13	93.63
	MPU-STK	100.00 $\pm$ 0.00	98.40 $\pm$ 0.80	100.00 $\pm$ 0.00	98.96 $\pm$ 0.41	70.80 $\pm$ 6.48	94.72 $\pm$ 3.76	93.81
	RADAR	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.68 $\pm$ 0.30	99.92 $\pm$ 0.16	91.36 $\pm$ 2.30	95.84 $\pm$ 1.73	97.80
	RADAR-STK	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.68 $\pm$ 0.30	99.92 $\pm$ 0.16	91.44 $\pm$ 2.31	95.84 $\pm$ 1.73	97.81
SQuAD1	Log-Likelihood	1.04 $\pm$ 0.57	5.00 $\pm$ 2.93	2.11 $\pm$ 0.29	1.85 $\pm$ 1.12	5.93 $\pm$ 3.44	21.85 $\pm$ 10.14	6.30
	Log-Rank	1.97 $\pm$ 1.19	4.88 $\pm$ 3.25	2.34 $\pm$ 0.83	1.97 $\pm$ 1.35	4.88 $\pm$ 2.72	19.77 $\pm$ 11.36	5.97
	DetectGPT	0.46 $\pm$ 0.67	0.48 $\pm$ 0.45	0.47 $\pm$ 0.44	0.58 $\pm$ 0.73	0.81 $\pm$ 0.59	0.69 $\pm$ 0.43	0.58
	F-DetectGPT	4.28 $\pm$ 1.35	6.90 $\pm$ 1.53	2.69 $\pm$ 1.41	2.77 $\pm$ 1.12	5.35 $\pm$ 2.19	28.44 $\pm$ 3.93	8.41
	RepreGuard	0.58 $\pm$ 0.52	0.00 $\pm$ 0.00	0.23 $\pm$ 0.47	0.35 $\pm$ 0.46	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.19
	Lastde	1.16 $\pm$ 0.82	0.83 $\pm$ 0.48	2.22 $\pm$ 0.57	1.50 $\pm$ 0.78	1.86 $\pm$ 0.44	2.31 $\pm$ 1.21	1.65
	Binoculars	1.39 $\pm$ 0.46	2.02 $\pm$ 1.04	2.46 $\pm$ 0.86	1.39 $\pm$ 0.28	2.67 $\pm$ 1.63	11.56 $\pm$ 2.64	3.58
	ChatGPT-D	1.27 $\pm$ 0.43	1.07 $\pm$ 0.87	1.17 $\pm$ 0.83	0.81 $\pm$ 0.59	1.16 $\pm$ 0.74	5.32 $\pm$ 3.88	1.80
	ChatGPT-STK	1.73 $\pm$ 1.75	3.81 $\pm$ 1.53	1.87 $\pm$ 0.44	1.50 $\pm$ 1.58	1.51 $\pm$ 1.31	6.59 $\pm$ 4.18	2.84
	OpenAI-D	51.21 $\pm$ 23.65	51.07 $\pm$ 22.43	32.28 $\pm$ 19.01	39.88 $\pm$ 18.98	62.67 $\pm$ 26.90	76.18 $\pm$ 12.19	52.22
	OpenAI-STK	53.41 $\pm$ 26.65	51.90 $\pm$ 20.23	32.98 $\pm$ 18.26	41.39 $\pm$ 20.40	60.58 $\pm$ 27.86	72.25 $\pm$ 14.07	52.09
	MPU	41.16 $\pm$ 29.06	17.02 $\pm$ 6.63	46.78 $\pm$ 23.16	33.06 $\pm$ 23.42	41.86 $\pm$ 25.14	51.33 $\pm$ 23.66	38.54
	MPU-STK	46.71 $\pm$ 29.55	66.43 $\pm$ 24.57	48.07 $\pm$ 22.73	37.23 $\pm$ 24.65	41.40 $\pm$ 25.34	55.84 $\pm$ 22.94	49.28
	RADAR	28.32 $\pm$ 13.89	30.48 $\pm$ 9.01	33.45 $\pm$ 10.05	21.97 $\pm$ 11.86	29.88 $\pm$ 5.56	26.94 $\pm$ 13.33	28.51
	RADAR-STK	28.90 $\pm$ 15.02	32.14 $\pm$ 9.55	36.37 $\pm$ 7.51	21.27 $\pm$ 13.05	31.63 $\pm$ 5.01	30.06 $\pm$ 13.26	30.06

TABLE VII: Performance concerning AUROC. Detectors are trained on ChatGLM texts.

Dataset	Method	ChatGPT	GPT-4	GPT-turbo	ChatGLM	Dolly	Claude/StableLM	Avg.
Essay	Log-Likelihood	98.48 $\pm$ 0.43	96.46 $\pm$ 1.05	98.85 $\pm$ 0.42	98.78 $\pm$ 0.15	90.64 $\pm$ 0.35	92.16 $\pm$ 0.32	95.90
	Log-Rank	98.64 $\pm$ 0.43	96.85 $\pm$ 0.89	98.92 $\pm$ 0.40	98.94 $\pm$ 0.16	90.41 $\pm$ 0.50	91.47 $\pm$ 0.45	95.87
	DetectGPT	96.86 $\pm$ 0.97	95.64 $\pm$ 0.45	44.40 $\pm$ 1.73	95.60 $\pm$ 0.62	92.55 $\pm$ 0.58	46.36 $\pm$ 0.71	78.57
	F-DetectGPT	83.86 $\pm$ 1.52	84.45 $\pm$ 1.58	94.12 $\pm$ 0.72	93.62 $\pm$ 0.73	55.86 $\pm$ 2.54	58.96 $\pm$ 1.52	78.48
	RepreGuard	99.81 $\pm$ 0.04	97.52 $\pm$ 0.38	41.79 $\pm$ 1.84	90.08 $\pm$ 1.84	94.78 $\pm$ 1.25	44.62 $\pm$ 2.02	78.1
	Lastde	99.68 $\pm$ 0.23	99.51 $\pm$ 0.14	93.99 $\pm$ 0.64	99.93 $\pm$ 0.04	97.99 $\pm$ 0.48	83.94 $\pm$ 0.81	95.84
	Binoculars	99.29 $\pm$ 0.50	98.73 $\pm$ 0.47	99.57 $\pm$ 0.17	99.87 $\pm$ 0.07	95.74 $\pm$ 0.56	92.18 $\pm$ 0.57	97.56
	ChatGPT-D	97.86 $\pm$ 0.63	95.40 $\pm$ 2.44	90.65 $\pm$ 1.69	99.90 $\pm$ 0.16	79.10 $\pm$ 6.61	33.03 $\pm$ 7.16	82.66
	ChatGPT-STK	98.45 $\pm$ 0.40	97.08 $\pm$ 0.99	88.03 $\pm$ 3.10	99.92 $\pm$ 0.11	85.81 $\pm$ 4.37	46.70 $\pm$ 6.20	86.00
	OpenAI-D	99.11 $\pm$ 0.53	99.77 $\pm$ 0.23	94.01 $\pm$ 1.32	99.89 $\pm$ 0.13	96.03 $\pm$ 1.37	62.51 $\pm$ 1.96	91.89
	OpenAI-STK	99.72 $\pm$ 0.14	99.67 $\pm$ 0.18	96.37 $\pm$ 0.49	99.84 $\pm$ 0.17	97.79 $\pm$ 0.43	71.36 $\pm$ 1.53	94.13
	MPU	99.86 $\pm$ 0.12	99.93 $\pm$ 0.04	96.75 $\pm$ 0.25	99.99 $\pm$ 0.01	96.72 $\pm$ 0.53	86.57 $\pm$ 1.95	96.64
	MPU-STK	99.87 $\pm$ 0.12	99.93 $\pm$ 0.04	96.64 $\pm$ 0.24	99.99 $\pm$ 0.01	96.99 $\pm$ 0.29	91.69 $\pm$ 1.39	97.52
	RADAR	99.79 $\pm$ 0.11	99.80 $\pm$ 0.09	99.75 $\pm$ 0.19	99.98 $\pm$ 0.02	97.43 $\pm$ 0.46	97.41 $\pm$ 0.39	99.03
	RADAR-STK	99.80 $\pm$ 0.11	99.76 $\pm$ 0.09	99.76 $\pm$ 0.20	99.98 $\pm$ 0.03	97.67 $\pm$ 0.36	97.36 $\pm$ 0.43	99.05
Reuters	Log-Likelihood	97.59 $\pm$ 0.37	74.85 $\pm$ 0.50	98.54 $\pm$ 0.38	99.54 $\pm$ 0.22	60.09 $\pm$ 2.13	85.66 $\pm$ 1.04	86.05
	Log-Rank	97.86 $\pm$ 0.35	79.95 $\pm$ 0.29	98.82 $\pm$ 0.37	99.65 $\pm$ 0.20	61.24 $\pm$ 2.15	84.88 $\pm$ 1.13	87.07
	DetectGPT	92.78 $\pm$ 1.34	85.71 $\pm$ 2.02	49.24 $\pm$ 1.65	91.59 $\pm$ 1.67	83.41 $\pm$ 1.73	66.59 $\pm$ 2.53	78.22
	F-DetectGPT	96.23 $\pm$ 0.19	66.02 $\pm$ 1.53	98.85 $\pm$ 0.30	98.58 $\pm$ 0.35	40.38 $\pm$ 1.31	66.16 $\pm$ 2.58	77.70
	RepreGuard	98.71 $\pm$ 0.49	96.86 $\pm$ 0.83	47.69 $\pm$ 1.79	96.99 $\pm$ 0.68	6.54 $\pm$ 1.05	21.22 $\pm$ 2.40	61.34
	Lastde	99.68 $\pm$ 0.09	99.28 $\pm$ 0.30	99.36 $\pm$ 0.19	99.48 $\pm$ 0.29	95.45 $\pm$ 0.97	86.48 $\pm$ 1.29	96.62
	Binoculars	99.37 $\pm$ 0.14	93.10 $\pm$ 0.78	99.43 $\pm$ 0.28	99.71 $\pm$ 0.19	75.70 $\pm$ 1.89	87.10 $\pm$ 1.28	92.4
	ChatGPT-D	98.80 $\pm$ 0.71	97.95 $\pm$ 0.83	98.73 $\pm$ 0.79	99.72 $\pm$ 0.31	80.34 $\pm$ 5.32	40.49 $\pm$ 6.96	86.00
	ChatGPT-STK	98.98 $\pm$ 0.60	98.56 $\pm$ 0.43	99.23 $\pm$ 0.39	99.79 $\pm$ 0.23	84.47 $\pm$ 4.73	59.35 $\pm$ 5.00	90.06
	OpenAI-D	99.71 $\pm$ 0.22	99.69 $\pm$ 0.29	99.74 $\pm$ 0.16	99.99 $\pm$ 0.01	97.83 $\pm$ 0.82	66.96 $\pm$ 2.61	93.99
	OpenAI-STK	99.77 $\pm$ 0.17	99.61 $\pm$ 0.11	99.78 $\pm$ 0.09	99.98 $\pm$ 0.01	96.91 $\pm$ 0.53	78.19 $\pm$ 2.14	95.71
	MPU	100.00 $\pm$ 0.00	99.79 $\pm$ 0.11	99.97 $\pm$ 0.04	99.99 $\pm$ 0.01	96.62 $\pm$ 0.53	94.78 $\pm$ 1.36	98.52
	MPU-STK	100.00 $\pm$ 0.00	99.75 $\pm$ 0.17	99.99 $\pm$ 0.00	99.99 $\pm$ 0.01	97.10 $\pm$ 0.52	98.00 $\pm$ 0.22	99.14
	RADAR	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.97 $\pm$ 0.03	100.00 $\pm$ 0.00	99.44 $\pm$ 0.19	99.79 $\pm$ 0.09	99.87
	RADAR-STK	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.98 $\pm$ 0.03	100.00 $\pm$ 0.00	99.46 $\pm$ 0.20	99.79 $\pm$ 0.09	99.87
SQuAD1	Log-Likelihood	69.62 $\pm$ 0.28	69.61 $\pm$ 1.06	67.18 $\pm$ 0.67	65.67 $\pm$ 0.56	67.24 $\pm$ 1.10	85.82 $\pm$ 1.96	70.86
	Log-Rank	68.78 $\pm$ 0.34	69.52 $\pm$ 0.79	66.33 $\pm$ 0.59	64.99 $\pm$ 0.65	66.83 $\pm$ 1.04	85.50 $\pm$ 1.97	70.33
	DetectGPT	52.60 $\pm$ 2.74	47.47 $\pm$ 1.42	52.05 $\pm$ 1.87	50.88 $\pm$ 2.47	50.24 $\pm$ 2.20	51.12 $\pm$ 2.44	50.73
	F-DetectGPT	67.91 $\pm$ 0.23	63.84 $\pm$ 1.73	64.67 $\pm$ 0.20	63.36 $\pm$ 0.42	66.32 $\pm$ 1.36	79.62 $\pm$ 0.63	67.62
	RepreGuard	49.58 $\pm$ 1.93	48.18 $\pm$ 2.79	50.07 $\pm$ 1.50	49.91 $\pm$ 1.53	48.51 $\pm$ 3.61	43.94 $\pm$ 8.17	48.37
	Lastde	58.18 $\pm$ 0.59	56.58 $\pm$ 1.03	58.46 $\pm$ 0.35	57.33 $\pm$ 0.39	58.30 $\pm$ 0.68	67.65 $\pm$ 0.76	59.42
	Binoculars	66.34 $\pm$ 0.21	62.58 $\pm$ 1.33	63.55 $\pm$ 0.36	62.49 $\pm$ 0.50	64.22 $\pm$ 1.56	77.28 $\pm$ 1.06	66.08
	ChatGPT-D	58.03 $\pm$ 1.19	42.38 $\pm$ 4.85	56.33 $\pm$ 1.02	55.71 $\pm$ 0.92	55.98 $\pm$ 1.41	67.19 $\pm$ 2.64	55.94
	ChatGPT-STK	57.96 $\pm$ 1.30	59.49 $\pm$ 0.74	56.25 $\pm$ 0.99	55.62 $\pm$ 1.04	55.92 $\pm$ 0.93	65.95 $\pm$ 1.72	58.53
	OpenAI-D	98.77 $\pm$ 0.34	98.99 $\pm$ 0.57	98.33 $\pm$ 0.66	95.32 $\pm$ 0.93	97.71 $\pm$ 0.39	99.25 $\pm$ 0.31	98.06
	OpenAI-STK	98.90 $\pm$ 0.16	98.98 $\pm$ 0.40	98.31 $\pm$ 0.64	95.29 $\pm$ 0.93	97.62 $\pm$ 0.51	99.34 $\pm$ 0.19	98.07
	MPU	98.60 $\pm$ 0.34	98.47 $\pm$ 1.08	98.72 $\pm$ 0.54	96.32 $\pm$ 1.34	97.56 $\pm$ 0.47	98.72 $\pm$ 0.29	98.06
	MPU-STK	98.50 $\pm$ 0.50	98.96 $\pm$ 0.88	98.62 $\pm$ 0.64	96.16 $\pm$ 1.36	97.41 $\pm$ 0.40	98.65 $\pm$ 0.41	98.05
	RADAR	81.33 $\pm$ 1.72	80.23 $\pm$ 0.80	80.91 $\pm$ 1.51	77.72 $\pm$ 0.95	77.52 $\pm$ 1.95	80.95 $\pm$ 1.89	79.78
	RADAR-STK	81.59 $\pm$ 1.86	80.69 $\pm$ 0.71	81.37 $\pm$ 1.39	77.88 $\pm$ 0.76	77.91 $\pm$ 1.90	81.03 $\pm$ 1.95	80.08

TABLE VIII: Performance concerning TPR@FPR-0.5%. Detectors are trained on ChatGLM texts.

Dataset	Method	ChatGPT	GPT-4	GPT-turbo	ChatGLM	Dolly	Claude/StableLM	Avg.
Essay	Log-Likelihood	24.08 $\pm$ 22.67	37.70 $\pm$ 30.63	23.12 $\pm$ 24.38	5.86 $\pm$ 7.02	12.45 $\pm$ 12.04	2.48 $\pm$ 3.06	17.62
	Log-Rank	28.72 $\pm$ 26.90	46.48 $\pm$ 36.20	25.04 $\pm$ 26.83	28.19 $\pm$ 27.90	17.34 $\pm$ 15.65	2.96 $\pm$ 3.73	24.79
	DetectGPT	37.04 $\pm$ 10.21	24.75 $\pm$ 10.03	5.52 $\pm$ 1.93	21.45 $\pm$ 14.25	15.45 $\pm$ 7.05	4.48 $\pm$ 2.57	18.12
	F-DetectGPT	4.24 $\pm$ 1.51	3.85 $\pm$ 2.05	31.28 $\pm$ 3.18	35.74 $\pm$ 3.97	0.00 $\pm$ 0.00	0.16 $\pm$ 0.20	12.55
	RepreGuard	92.08 $\pm$ 2.28	62.62 $\pm$ 24.10	0.32 $\pm$ 0.16	10.44 $\pm$ 2.86	20.86 $\pm$ 3.22	0.16 $\pm$ 0.20	31.08
	Lastde	94.24 $\pm$ 1.83	88.77 $\pm$ 3.22	48.96 $\pm$ 3.15	98.39 $\pm$ 1.08	79.57 $\pm$ 5.36	15.52 $\pm$ 3.60	70.91
	Binoculars	91.60 $\pm$ 7.82	94.26 $\pm$ 2.49	85.36 $\pm$ 18.99	98.55 $\pm$ 1.03	62.83 $\pm$ 17.81	25.20 $\pm$ 12.61	76.3
	ChatGPT-D	80.08 $\pm$ 8.29	79.51 $\pm$ 2.47	34.48 $\pm$ 8.86	96.71 $\pm$ 5.80	34.76 $\pm$ 8.77	0.48 $\pm$ 0.30	54.34
	ChatGPT-STK	85.36 $\pm$ 3.83	83.36 $\pm$ 4.43	39.20 $\pm$ 7.94	97.99 $\pm$ 3.27	43.43 $\pm$ 4.46	2.48 $\pm$ 1.09	58.64
	OpenAI-D	83.84 $\pm$ 23.24	85.66 $\pm$ 21.19	57.36 $\pm$ 15.97	95.74 $\pm$ 8.11	72.19 $\pm$ 19.00	5.84 $\pm$ 4.57	66.77
	OpenAI-STK	93.12 $\pm$ 1.55	94.51 $\pm$ 2.60	61.68 $\pm$ 7.31	98.63 $\pm$ 1.07	61.89 $\pm$ 10.59	9.68 $\pm$ 5.32	69.92
	MPU	96.80 $\pm$ 1.29	98.69 $\pm$ 0.66	51.52 $\pm$ 7.75	99.60 $\pm$ 0.44	70.39 $\pm$ 6.73	18.16 $\pm$ 11.12	72.53
	MPU-STK	97.28 $\pm$ 0.89	98.85 $\pm$ 0.40	52.48 $\pm$ 7.96	99.60 $\pm$ 0.44	72.88 $\pm$ 5.59	27.84 $\pm$ 12.28	74.82
	RADAR	89.92 $\pm$ 8.96	94.34 $\pm$ 2.28	86.72 $\pm$ 14.46	99.12 $\pm$ 1.20	58.28 $\pm$ 16.43	47.68 $\pm$ 19.59	79.34
	RADAR-STK	90.24 $\pm$ 8.37	94.10 $\pm$ 2.97	87.04 $\pm$ 14.19	99.04 $\pm$ 1.36	58.63 $\pm$ 16.56	49.68 $\pm$ 19.93	79.79
Reuters	Log-Likelihood	77.84 $\pm$ 5.19	14.88 $\pm$ 5.98	86.08 $\pm$ 3.38	93.76 $\pm$ 2.03	11.20 $\pm$ 4.45	15.04 $\pm$ 6.86	49.80
	Log-Rank	82.40 $\pm$ 5.24	25.92 $\pm$ 7.08	90.96 $\pm$ 4.12	96.80 $\pm$ 0.88	14.00 $\pm$ 4.82	17.60 $\pm$ 8.29	54.61
	DetectGPT	4.40 $\pm$ 2.62	0.64 $\pm$ 0.54	2.32 $\pm$ 1.87	2.56 $\pm$ 2.80	0.48 $\pm$ 0.47	3.04 $\pm$ 1.61	2.24
	F-DetectGPT	48.00 $\pm$ 9.48	6.80 $\pm$ 1.88	92.96 $\pm$ 1.65	88.96 $\pm$ 4.80	0.00 $\pm$ 0.00	0.48 $\pm$ 0.39	39.53
	RepreGuard	26.88 $\pm$ 19.31	9.04 $\pm$ 6.02	0.08 $\pm$ 0.16	0.80 $\pm$ 1.01	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	6.13
	Lastde	92.64 $\pm$ 1.06	88.96 $\pm$ 1.71	87.36 $\pm$ 3.51	98.56 $\pm$ 0.65	66.00 $\pm$ 3.27	9.68 $\pm$ 3.04	73.87
	Binoculars	95.84 $\pm$ 0.93	70.88 $\pm$ 3.93	96.08 $\pm$ 0.93	99.28 $\pm$ 0.30	40.88 $\pm$ 4.94	26.16 $\pm$ 4.18	71.52
	ChatGPT-D	91.28 $\pm$ 6.11	85.20 $\pm$ 7.05	89.12 $\pm$ 6.82	97.92 $\pm$ 1.46	41.28 $\pm$ 16.63	2.80 $\pm$ 2.83	67.93
	ChatGPT-STK	94.64 $\pm$ 2.52	90.16 $\pm$ 2.97	93.92 $\pm$ 2.57	98.88 $\pm$ 0.93	50.40 $\pm$ 11.41	8.88 $\pm$ 0.89	72.81
	OpenAI-D	96.80 $\pm$ 2.65	86.96 $\pm$ 13.16	96.24 $\pm$ 2.02	99.60 $\pm$ 0.44	60.64 $\pm$ 12.43	10.32 $\pm$ 3.91	75.09
	OpenAI-STK	97.44 $\pm$ 0.41	87.92 $\pm$ 2.92	97.36 $\pm$ 0.86	99.44 $\pm$ 0.20	50.08 $\pm$ 6.35	15.36 $\pm$ 3.96	74.60
	MPU	100.00 $\pm$ 0.00	97.12 $\pm$ 1.32	99.76 $\pm$ 0.20	99.76 $\pm$ 0.20	67.60 $\pm$ 5.76	55.92 $\pm$ 10.37	86.69
	MPU-STK	99.92 $\pm$ 0.16	96.96 $\pm$ 1.40	99.76 $\pm$ 0.32	99.60 $\pm$ 0.44	63.44 $\pm$ 7.23	69.36 $\pm$ 8.86	88.17
	RADAR	99.92 $\pm$ 0.16	99.92 $\pm$ 0.16	99.28 $\pm$ 0.85	99.92 $\pm$ 0.16	90.16 $\pm$ 2.82	93.92 $\pm$ 2.62	97.19
	RADAR-STK	99.92 $\pm$ 0.16	99.92 $\pm$ 0.16	99.28 $\pm$ 0.85	99.92 $\pm$ 0.16	89.84 $\pm$ 2.66	93.84 $\pm$ 2.62	97.12
SQuAD1	Log-Likelihood	1.04 $\pm$ 0.57	5.00 $\pm$ 2.93	2.11 $\pm$ 0.29	1.85 $\pm$ 1.12	5.93 $\pm$ 3.44	21.85 $\pm$ 10.14	6.30
	Log-Rank	1.97 $\pm$ 1.19	4.88 $\pm$ 3.25	2.34 $\pm$ 0.83	1.97 $\pm$ 1.35	4.88 $\pm$ 2.72	19.77 $\pm$ 11.36	5.97
	DetectGPT	0.35 $\pm$ 0.69	0.36 $\pm$ 0.48	0.47 $\pm$ 0.44	0.58 $\pm$ 0.73	0.58 $\pm$ 0.37	0.81 $\pm$ 0.28	0.52
	F-DetectGPT	4.28 $\pm$ 1.35	6.90 $\pm$ 1.53	2.69 $\pm$ 1.41	2.77 $\pm$ 1.12	5.35 $\pm$ 2.19	28.44 $\pm$ 3.93	8.41
	RepreGuard	0.69 $\pm$ 0.43	0.00 $\pm$ 0.00	0.23 $\pm$ 0.47	0.35 $\pm$ 0.46	0.12 $\pm$ 0.23	0.12 $\pm$ 0.23	0.25
	Lastde	1.16 $\pm$ 0.82	0.83 $\pm$ 0.48	2.22 $\pm$ 0.57	1.50 $\pm$ 0.78	1.86 $\pm$ 0.44	2.31 $\pm$ 1.21	1.65
	Binoculars	1.39 $\pm$ 0.46	2.02 $\pm$ 1.04	2.46 $\pm$ 0.86	1.39 $\pm$ 0.28	2.67 $\pm$ 1.63	11.56 $\pm$ 2.64	3.58
	ChatGPT-D	1.04 $\pm$ 0.99	1.31 $\pm$ 0.69	1.40 $\pm$ 0.95	1.16 $\pm$ 0.82	1.74 $\pm$ 1.27	6.01 $\pm$ 3.26	2.11
	ChatGPT-STK	2.66 $\pm$ 1.96	3.81 $\pm$ 1.28	1.40 $\pm$ 0.60	1.62 $\pm$ 1.69	1.05 $\pm$ 0.44	3.82 $\pm$ 1.49	2.39
	OpenAI-D	36.53 $\pm$ 17.45	41.79 $\pm$ 19.02	36.96 $\pm$ 16.77	28.67 $\pm$ 13.27	50.58 $\pm$ 24.47	63.24 $\pm$ 19.04	42.96
	OpenAI-STK	39.77 $\pm$ 15.28	47.26 $\pm$ 22.70	31.70 $\pm$ 14.38	29.94 $\pm$ 10.70	51.86 $\pm$ 26.87	65.66 $\pm$ 16.99	44.37
	MPU	41.85 $\pm$ 26.11	23.69 $\pm$ 33.19	45.03 $\pm$ 14.50	33.64 $\pm$ 20.00	54.19 $\pm$ 29.09	53.29 $\pm$ 18.57	41.95
	MPU-STK	38.73 $\pm$ 19.95	53.33 $\pm$ 24.68	42.69 $\pm$ 16.63	30.17 $\pm$ 16.73	55.81 $\pm$ 29.23	51.33 $\pm$ 14.03	45.34
	RADAR	28.09 $\pm$ 13.16	31.19 $\pm$ 11.73	35.32 $\pm$ 6.78	22.66 $\pm$ 11.61	30.93 $\pm$ 5.47	27.40 $\pm$ 10.59	29.27
	RADAR-STK	30.52 $\pm$ 11.76	31.07 $\pm$ 11.41	37.43 $\pm$ 4.62	23.82 $\pm$ 10.28	31.40 $\pm$ 5.95	26.82 $\pm$ 9.64	30.17

TABLE IX: Performance concerning AUROC. Detectors are trained on Dolly texts.

Dataset	Method	ChatGPT	GPT-4	GPT-turbo	ChatGLM	Dolly	Claude/StableLM	Avg.
Essay	Log-Likelihood	98.48 $\pm$ 0.43	96.46 $\pm$ 1.05	98.85 $\pm$ 0.42	98.78 $\pm$ 0.15	90.64 $\pm$ 0.35	92.16 $\pm$ 0.32	95.90
	Log-Rank	98.64 $\pm$ 0.43	96.85 $\pm$ 0.89	98.92 $\pm$ 0.40	98.94 $\pm$ 0.16	90.41 $\pm$ 0.50	91.47 $\pm$ 0.45	95.87
	DetectGPT	96.86 $\pm$ 0.97	95.64 $\pm$ 0.45	44.40 $\pm$ 1.73	95.60 $\pm$ 0.62	92.55 $\pm$ 0.58	46.36 $\pm$ 0.71	78.57
	F-DetectGPT	83.86 $\pm$ 1.52	84.45 $\pm$ 1.58	94.12 $\pm$ 0.72	93.62 $\pm$ 0.73	55.86 $\pm$ 2.54	58.96 $\pm$ 1.52	78.48
	RepreGuard	99.81 $\pm$ 0.04	97.52 $\pm$ 0.38	41.79 $\pm$ 1.84	94.46 $\pm$ 1.02	92.47 $\pm$ 1.56	44.62 $\pm$ 2.02	78.44
	Lastde	99.68 $\pm$ 0.23	99.51 $\pm$ 0.14	93.99 $\pm$ 0.64	99.93 $\pm$ 0.04	97.99 $\pm$ 0.48	83.94 $\pm$ 0.81	95.84
	Binoculars	99.29 $\pm$ 0.50	98.73 $\pm$ 0.47	99.57 $\pm$ 0.17	99.87 $\pm$ 0.07	95.74 $\pm$ 0.56	92.18 $\pm$ 0.57	97.56
	ChatGPT-D	99.08 $\pm$ 0.84	98.90 $\pm$ 1.45	89.54 $\pm$ 2.35	99.86 $\pm$ 0.22	93.15 $\pm$ 6.08	62.74 $\pm$ 13.69	90.54
	ChatGPT-STK	99.08 $\pm$ 0.45	98.89 $\pm$ 1.07	90.05 $\pm$ 2.28	99.89 $\pm$ 0.11	93.32 $\pm$ 4.06	65.73 $\pm$ 12.96	91.16
	OpenAI-D	99.81 $\pm$ 0.21	99.86 $\pm$ 0.15	99.21 $\pm$ 0.40	99.98 $\pm$ 0.03	99.48 $\pm$ 0.46	93.16 $\pm$ 2.83	98.58
	OpenAI-STK	99.90 $\pm$ 0.14	99.95 $\pm$ 0.04	99.22 $\pm$ 0.20	99.99 $\pm$ 0.01	99.68 $\pm$ 0.25	92.71 $\pm$ 0.80	98.58
	MPU	100.00 $\pm$ 0.00	99.98 $\pm$ 0.03	97.88 $\pm$ 0.43	100.00 $\pm$ 0.00	99.86 $\pm$ 0.04	98.74 $\pm$ 0.46	99.41
	MPU-STK	100.00 $\pm$ 0.00	99.98 $\pm$ 0.03	98.16 $\pm$ 0.39	100.00 $\pm$ 0.00	99.86 $\pm$ 0.04	98.80 $\pm$ 0.32	99.47
	RADAR	99.69 $\pm$ 0.10	99.89 $\pm$ 0.07	99.39 $\pm$ 0.23	99.96 $\pm$ 0.02	99.36 $\pm$ 0.13	99.69 $\pm$ 0.05	99.66
	RADAR-STK	99.70 $\pm$ 0.11	99.89 $\pm$ 0.07	99.39 $\pm$ 0.31	99.97 $\pm$ 0.02	99.35 $\pm$ 0.16	99.68 $\pm$ 0.08	99.66
Reuters	Log-Likelihood	21.47 $\pm$ 38.08	35.27 $\pm$ 20.02	20.75 $\pm$ 38.74	20.35 $\pm$ 39.69	44.75 $\pm$ 8.87	28.28 $\pm$ 28.30	28.48
	Log-Rank	2.14 $\pm$ 0.35	20.05 $\pm$ 0.29	1.18 $\pm$ 0.37	0.35 $\pm$ 0.20	38.76 $\pm$ 2.15	15.12 $\pm$ 1.13	12.93
	DetectGPT	92.78 $\pm$ 1.34	85.71 $\pm$ 2.02	49.24 $\pm$ 1.65	91.59 $\pm$ 1.67	83.41 $\pm$ 1.73	66.59 $\pm$ 2.53	78.22
	F-DetectGPT	96.23 $\pm$ 0.19	66.02 $\pm$ 1.53	98.85 $\pm$ 0.30	98.58 $\pm$ 0.35	40.38 $\pm$ 1.31	66.16 $\pm$ 2.58	77.70
	RepreGuard	99.81 $\pm$ 0.04	97.52 $\pm$ 0.38	41.79 $\pm$ 1.84	94.46 $\pm$ 1.02	92.47 $\pm$ 1.56	44.62 $\pm$ 2.02	78.44
	Lastde	99.68 $\pm$ 0.23	99.51 $\pm$ 0.14	93.99 $\pm$ 0.64	99.93 $\pm$ 0.04	97.99 $\pm$ 0.48	83.94 $\pm$ 0.81	95.84
	Binoculars	99.29 $\pm$ 0.50	98.73 $\pm$ 0.47	99.57 $\pm$ 0.17	99.87 $\pm$ 0.07	95.74 $\pm$ 0.56	92.18 $\pm$ 0.57	97.56
	ChatGPT-D	99.67 $\pm$ 0.16	99.26 $\pm$ 0.68	99.28 $\pm$ 0.50	99.13 $\pm$ 0.97	94.75 $\pm$ 2.38	78.91 $\pm$ 2.78	95.16
	ChatGPT-STK	99.53 $\pm$ 0.50	98.96 $\pm$ 0.89	99.36 $\pm$ 0.65	98.84 $\pm$ 1.48	94.58 $\pm$ 1.73	80.78 $\pm$ 8.15	95.34
	OpenAI-D	99.09 $\pm$ 1.02	99.17 $\pm$ 0.96	99.08 $\pm$ 1.00	99.04 $\pm$ 1.03	98.61 $\pm$ 0.81	88.19 $\pm$ 1.27	97.20
	OpenAI-STK	99.45 $\pm$ 0.60	99.55 $\pm$ 0.52	99.44 $\pm$ 0.58	99.43 $\pm$ 0.61	99.23 $\pm$ 0.49	89.64 $\pm$ 1.88	97.79
	MPU	100.00 $\pm$ 0.00	99.99 $\pm$ 0.01	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.95 $\pm$ 0.04	99.55 $\pm$ 0.11	99.91
	MPU-STK	100.00 $\pm$ 0.00	99.98 $\pm$ 0.02	99.99 $\pm$ 0.01	100.00 $\pm$ 0.00	99.93 $\pm$ 0.07	99.62 $\pm$ 0.26	99.92
	RADAR	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.96 $\pm$ 0.03	99.99 $\pm$ 0.01	99.86 $\pm$ 0.04	99.82 $\pm$ 0.09	99.94
	RADAR-STK	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.96 $\pm$ 0.03	99.99 $\pm$ 0.01	99.85 $\pm$ 0.04	99.82 $\pm$ 0.09	99.94
SQuAD1	Log-Likelihood	69.62 $\pm$ 0.28	69.61 $\pm$ 1.06	67.18 $\pm$ 0.67	65.67 $\pm$ 0.56	67.24 $\pm$ 1.10	85.82 $\pm$ 1.96	70.86
	Log-Rank	68.78 $\pm$ 0.34	69.52 $\pm$ 0.79	66.33 $\pm$ 0.59	64.99 $\pm$ 0.65	66.83 $\pm$ 1.04	85.50 $\pm$ 1.97	70.33
	DetectGPT	52.60 $\pm$ 2.74	47.47 $\pm$ 1.42	52.05 $\pm$ 1.87	50.88 $\pm$ 2.47	50.24 $\pm$ 2.20	51.12 $\pm$ 2.44	50.73
	F-DetectGPT	67.91 $\pm$ 0.23	63.84 $\pm$ 1.73	64.67 $\pm$ 0.20	63.36 $\pm$ 0.42	66.32 $\pm$ 1.36	79.62 $\pm$ 0.63	67.62
	RepreGuard	99.81 $\pm$ 0.04	97.52 $\pm$ 0.38	41.79 $\pm$ 1.84	94.46 $\pm$ 1.02	92.47 $\pm$ 1.56	44.62 $\pm$ 2.02	78.44
	Lastde	99.68 $\pm$ 0.23	99.51 $\pm$ 0.14	93.99 $\pm$ 0.64	99.93 $\pm$ 0.04	97.99 $\pm$ 0.48	83.94 $\pm$ 0.81	95.84
	Binoculars	99.29 $\pm$ 0.50	98.73 $\pm$ 0.47	99.57 $\pm$ 0.17	99.87 $\pm$ 0.07	95.74 $\pm$ 0.56	92.18 $\pm$ 0.57	97.56
	ChatGPT-D	57.10 $\pm$ 1.04	42.02 $\pm$ 4.36	55.48 $\pm$ 1.28	54.90 $\pm$ 1.09	54.82 $\pm$ 0.91	65.11 $\pm$ 2.34	54.90
	ChatGPT-STK	57.87 $\pm$ 1.35	59.11 $\pm$ 1.82	55.87 $\pm$ 1.31	55.28 $\pm$ 1.11	55.70 $\pm$ 1.53	66.15 $\pm$ 3.96	58.33
	OpenAI-D	99.34 $\pm$ 0.45	99.39 $\pm$ 0.34	98.57 $\pm$ 0.81	95.94 $\pm$ 1.16	97.81 $\pm$ 0.51	99.58 $\pm$ 0.27	98.44
	OpenAI-STK	99.39 $\pm$ 0.41	99.48 $\pm$ 0.30	98.61 $\pm$ 0.83	96.13 $\pm$ 1.05	97.88 $\pm$ 0.51	99.63 $\pm$ 0.22	98.52
	MPU	98.73 $\pm$ 0.43	99.00 $\pm$ 0.92	98.46 $\pm$ 0.77	95.68 $\pm$ 0.92	97.43 $\pm$ 0.39	98.97 $\pm$ 0.48	98.04
	MPU-STK	98.80 $\pm$ 0.55	99.01 $\pm$ 0.98	98.41 $\pm$ 0.77	95.51 $\pm$ 0.96	97.16 $\pm$ 0.39	99.00 $\pm$ 0.53	97.98
	RADAR	80.32 $\pm$ 2.05	79.27 $\pm$ 1.43	79.72 $\pm$ 1.69	77.01 $\pm$ 0.57	77.44 $\pm$ 0.79	80.57 $\pm$ 2.72	79.05
	RADAR-STK	81.23 $\pm$ 2.56	80.21 $\pm$ 2.10	80.53 $\pm$ 2.46	77.56 $\pm$ 1.28	78.69 $\pm$ 1.27	81.17 $\pm$ 3.01	79.90

TABLE X: Performance concerning TPR@FPR-0.5%. Detectors are trained on Dolly texts.

Dataset	Method	ChatGPT	GPT-4	GPT-turbo	ChatGLM	Dolly	Claude/StableLM	Avg.
Essay	Log-Likelihood	24.08 $\pm$ 22.67	37.70 $\pm$ 30.63	23.12 $\pm$ 24.38	5.86 $\pm$ 7.02	12.45 $\pm$ 12.04	2.48 $\pm$ 3.06	17.62
	Log-Rank	28.72 $\pm$ 26.90	46.48 $\pm$ 36.20	25.04 $\pm$ 26.83	28.19 $\pm$ 27.90	17.34 $\pm$ 15.65	2.96 $\pm$ 3.73	24.79
	DetectGPT	37.04 $\pm$ 10.21	24.75 $\pm$ 10.03	5.52 $\pm$ 1.93	21.45 $\pm$ 14.25	15.45 $\pm$ 7.05	4.48 $\pm$ 2.57	18.12
	F-DetectGPT	4.24 $\pm$ 1.51	3.85 $\pm$ 2.05	31.28 $\pm$ 3.18	35.74 $\pm$ 3.97	0.00 $\pm$ 0.00	0.16 $\pm$ 0.20	12.55
	RepreGuard	92.08 $\pm$ 2.28	62.62 $\pm$ 24.10	0.32 $\pm$ 0.16	31.89 $\pm$ 2.22	9.53 $\pm$ 2.24	0.16 $\pm$ 0.20	32.77
	Lastde	94.24 $\pm$ 1.83	88.77 $\pm$ 3.22	48.96 $\pm$ 3.15	98.39 $\pm$ 1.08	79.57 $\pm$ 5.36	15.52 $\pm$ 3.60	70.91
	Binoculars	91.60 $\pm$ 7.82	94.26 $\pm$ 2.49	85.36 $\pm$ 18.99	98.55 $\pm$ 1.03	62.83 $\pm$ 17.81	25.20 $\pm$ 12.61	76.3
	ChatGPT-D	82.08 $\pm$ 5.30	83.28 $\pm$ 14.28	32.00 $\pm$ 6.21	97.75 $\pm$ 2.99	46.87 $\pm$ 10.12	1.92 $\pm$ 1.32	57.32
	ChatGPT-STK	69.44 $\pm$ 15.68	83.52 $\pm$ 8.05	26.00 $\pm$ 9.13	95.58 $\pm$ 5.06	39.74 $\pm$ 14.81	3.52 $\pm$ 4.97	52.97
	OpenAI-D	92.88 $\pm$ 13.04	98.61 $\pm$ 1.46	79.36 $\pm$ 16.76	99.44 $\pm$ 0.60	85.41 $\pm$ 16.99	34.40 $\pm$ 18.17	81.68
	OpenAI-STK	98.88 $\pm$ 0.82	98.52 $\pm$ 1.20	79.76 $\pm$ 8.15	99.60 $\pm$ 0.51	92.02 $\pm$ 4.04	29.76 $\pm$ 11.57	83.09
	MPU	99.84 $\pm$ 0.20	99.67 $\pm$ 0.31	42.88 $\pm$ 12.55	100.00 $\pm$ 0.00	95.02 $\pm$ 1.40	51.36 $\pm$ 8.25	81.46
	MPU-STK	99.84 $\pm$ 0.20	99.67 $\pm$ 0.31	46.48 $\pm$ 15.65	100.00 $\pm$ 0.00	95.28 $\pm$ 2.38	56.88 $\pm$ 7.73	83.03
	RADAR	87.20 $\pm$ 5.92	96.31 $\pm$ 3.07	79.20 $\pm$ 9.10	98.15 $\pm$ 1.36	78.71 $\pm$ 6.05	84.48 $\pm$ 7.31	87.34
	RADAR-STK	87.92 $\pm$ 5.68	95.57 $\pm$ 3.04	80.88 $\pm$ 9.28	98.15 $\pm$ 1.41	78.54 $\pm$ 6.01	84.72 $\pm$ 7.49	87.63
Reuters	Log-Likelihood	15.76 $\pm$ 31.52	4.16 $\pm$ 6.53	16.96 $\pm$ 33.92	18.88 $\pm$ 37.76	8.24 $\pm$ 1.94	2.80 $\pm$ 5.60	11.13
	Log-Rank	0.00 $\pm$ 0.00	0.80 $\pm$ 0.36	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	7.04 $\pm$ 2.31	0.00 $\pm$ 0.00	1.31
	DetectGPT	4.40 $\pm$ 2.62	0.64 $\pm$ 0.54	2.32 $\pm$ 1.87	2.56 $\pm$ 2.80	0.48 $\pm$ 0.47	3.04 $\pm$ 1.61	2.24
	F-DetectGPT	48.00 $\pm$ 9.48	6.80 $\pm$ 1.88	92.96 $\pm$ 1.65	88.96 $\pm$ 4.80	0.00 $\pm$ 0.00	0.48 $\pm$ 0.39	39.53
	RepreGuard	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	16.88 $\pm$ 1.92	0.00 $\pm$ 0.00	46.96 $\pm$ 6.08	0.00 $\pm$ 0.00	10.64
	Lastde	92.64 $\pm$ 1.06	88.96 $\pm$ 1.71	87.36 $\pm$ 3.51	98.56 $\pm$ 0.65	66.00 $\pm$ 3.27	9.68 $\pm$ 3.04	73.87
	Binoculars	95.84 $\pm$ 0.93	70.88 $\pm$ 3.93	96.08 $\pm$ 0.93	99.28 $\pm$ 0.30	40.88 $\pm$ 4.94	26.16 $\pm$ 4.18	71.52
	ChatGPT-D	94.72 $\pm$ 3.88	92.00 $\pm$ 10.07	91.52 $\pm$ 3.28	92.08 $\pm$ 13.06	73.12 $\pm$ 3.95	17.52 $\pm$ 9.95	76.83
	ChatGPT-STK	91.04 $\pm$ 10.00	88.72 $\pm$ 13.24	90.72 $\pm$ 8.98	88.00 $\pm$ 18.04	70.88 $\pm$ 16.85	22.32 $\pm$ 13.71	75.28
	OpenAI-D	59.84 $\pm$ 48.53	61.60 $\pm$ 43.63	59.44 $\pm$ 47.89	59.44 $\pm$ 48.04	51.36 $\pm$ 32.85	18.64 $\pm$ 10.58	51.72
	OpenAI-STK	60.00 $\pm$ 48.50	65.52 $\pm$ 41.11	59.92 $\pm$ 47.62	59.60 $\pm$ 48.17	58.56 $\pm$ 33.66	24.72 $\pm$ 14.98	54.72
	MPU	100.00 $\pm$ 0.00	99.68 $\pm$ 0.16	99.84 $\pm$ 0.20	99.84 $\pm$ 0.20	98.32 $\pm$ 1.25	90.72 $\pm$ 1.72	98.07
	MPU-STK	100.00 $\pm$ 0.00	99.28 $\pm$ 0.64	99.76 $\pm$ 0.20	99.84 $\pm$ 0.20	97.76 $\pm$ 1.38	93.04 $\pm$ 2.43	98.28
	RADAR	99.92 $\pm$ 0.16	100.00 $\pm$ 0.00	99.36 $\pm$ 0.48	99.92 $\pm$ 0.16	96.32 $\pm$ 0.93	96.80 $\pm$ 1.75	98.72
	RADAR-STK	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.60 $\pm$ 0.44	99.92 $\pm$ 0.16	96.24 $\pm$ 0.86	97.04 $\pm$ 1.65	98.80
SQuAD1	Log-Likelihood	1.04 $\pm$ 0.57	5.00 $\pm$ 2.93	2.11 $\pm$ 0.29	1.85 $\pm$ 1.12	5.93 $\pm$ 3.44	21.85 $\pm$ 10.14	6.30
	Log-Rank	1.97 $\pm$ 1.19	4.88 $\pm$ 3.25	2.34 $\pm$ 0.83	1.97 $\pm$ 1.35	4.88 $\pm$ 2.72	19.77 $\pm$ 11.36	5.97
	DetectGPT	0.35 $\pm$ 0.69	0.36 $\pm$ 0.48	0.47 $\pm$ 0.44	0.58 $\pm$ 0.73	0.58 $\pm$ 0.37	0.81 $\pm$ 0.28	0.52
	F-DetectGPT	4.28 $\pm$ 1.35	6.90 $\pm$ 1.53	2.69 $\pm$ 1.41	2.77 $\pm$ 1.12	5.35 $\pm$ 2.19	28.44 $\pm$ 3.93	8.41
	RepreGuard	0.58 $\pm$ 0.37	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.35 $\pm$ 0.46	0.12 $\pm$ 0.23	0.12 $\pm$ 0.23	0.19
	Lastde	1.16 $\pm$ 0.82	0.83 $\pm$ 0.48	2.22 $\pm$ 0.57	1.50 $\pm$ 0.78	1.86 $\pm$ 0.44	2.31 $\pm$ 1.21	1.65
	Binoculars	1.39 $\pm$ 0.46	2.02 $\pm$ 1.04	2.46 $\pm$ 0.86	1.39 $\pm$ 0.28	2.67 $\pm$ 1.63	11.56 $\pm$ 2.64	3.58
	ChatGPT-D	1.73 $\pm$ 0.97	1.55 $\pm$ 0.97	1.29 $\pm$ 0.68	0.46 $\pm$ 0.23	1.05 $\pm$ 0.57	4.16 $\pm$ 3.17	1.71
	ChatGPT-STK	1.16 $\pm$ 0.97	2.98 $\pm$ 1.36	1.17 $\pm$ 0.83	0.58 $\pm$ 0.37	1.05 $\pm$ 0.85	4.16 $\pm$ 2.60	1.85
	OpenAI-D	53.87 $\pm$ 26.50	60.36 $\pm$ 10.30	43.27 $\pm$ 23.38	40.58 $\pm$ 20.72	54.19 $\pm$ 25.62	76.42 $\pm$ 14.26	54.78
	OpenAI-STK	53.64 $\pm$ 26.69	70.95 $\pm$ 5.64	46.08 $\pm$ 24.36	40.69 $\pm$ 21.89	51.74 $\pm$ 22.89	76.07 $\pm$ 16.06	56.53
	MPU	43.12 $\pm$ 23.37	54.17 $\pm$ 31.77	37.89 $\pm$ 22.45	32.14 $\pm$ 19.43	47.33 $\pm$ 28.64	68.09 $\pm$ 12.98	47.12
	MPU-STK	44.16 $\pm$ 30.43	59.05 $\pm$ 27.99	40.23 $\pm$ 23.59	32.49 $\pm$ 24.94	42.21 $\pm$ 27.72	67.98 $\pm$ 14.68	47.69
	RADAR	34.45 $\pm$ 7.07	31.07 $\pm$ 7.41	33.10 $\pm$ 3.96	27.17 $\pm$ 6.57	27.91 $\pm$ 4.92	35.72 $\pm$ 8.20	31.57
	RADAR-STK	36.18 $\pm$ 6.63	32.74 $\pm$ 7.63	35.32 $\pm$ 3.89	28.90 $\pm$ 6.89	29.53 $\pm$ 4.00	35.14 $\pm$ 8.28	32.97

TABLE XI: Performance concerning AUROC. Detectors are trained on Claude/StableLM texts.

Dataset	Method	ChatGPT	GPT-4	GPT-turbo	ChatGLM	Dolly	Claude/StableLM	Avg.
Essay	Log-Likelihood	98.48 $\pm$ 0.43	96.46 $\pm$ 1.05	98.85 $\pm$ 0.42	98.78 $\pm$ 0.15	90.64 $\pm$ 0.35	92.16 $\pm$ 0.32	95.90
	Log-Rank	98.64 $\pm$ 0.43	96.85 $\pm$ 0.89	98.92 $\pm$ 0.40	98.94 $\pm$ 0.16	90.41 $\pm$ 0.50	91.47 $\pm$ 0.45	95.87
	DetectGPT	96.86 $\pm$ 0.97	95.64 $\pm$ 0.45	44.40 $\pm$ 1.73	95.60 $\pm$ 0.62	92.55 $\pm$ 0.58	46.36 $\pm$ 0.71	78.57
	F-DetectGPT	83.86 $\pm$ 1.52	84.45 $\pm$ 1.58	94.12 $\pm$ 0.72	93.62 $\pm$ 0.73	55.86 $\pm$ 2.54	58.96 $\pm$ 1.52	78.48
	RepreGuard	0.19 $\pm$ 0.04	2.48 $\pm$ 0.38	58.21 $\pm$ 1.84	5.54 $\pm$ 1.02	5.22 $\pm$ 1.25	51.21 $\pm$ 1.94	20.48
	Lastde	99.68 $\pm$ 0.23	99.51 $\pm$ 0.14	93.99 $\pm$ 0.64	99.93 $\pm$ 0.04	97.99 $\pm$ 0.48	83.94 $\pm$ 0.81	95.84
	Binoculars	99.29 $\pm$ 0.50	98.73 $\pm$ 0.47	99.57 $\pm$ 0.17	99.87 $\pm$ 0.07	95.74 $\pm$ 0.56	92.18 $\pm$ 0.57	97.56
	ChatGPT-D	95.91 $\pm$ 2.81	96.62 $\pm$ 2.16	93.96 $\pm$ 4.28	98.72 $\pm$ 1.11	86.14 $\pm$ 7.72	81.87 $\pm$ 14.51	92.20
	ChatGPT-STK	96.89 $\pm$ 2.07	97.78 $\pm$ 1.79	95.41 $\pm$ 4.00	99.49 $\pm$ 0.42	89.16 $\pm$ 3.99	88.89 $\pm$ 9.25	94.60
	OpenAI-D	97.02 $\pm$ 3.00	97.11 $\pm$ 3.91	99.89 $\pm$ 0.13	98.99 $\pm$ 1.10	91.43 $\pm$ 9.06	99.77 $\pm$ 0.19	97.37
	OpenAI-STK	98.77 $\pm$ 0.50	99.18 $\pm$ 0.35	99.93 $\pm$ 0.08	99.17 $\pm$ 0.38	97.74 $\pm$ 0.69	99.93 $\pm$ 0.07	99.12
	MPU	99.92 $\pm$ 0.09	99.84 $\pm$ 0.11	97.68 $\pm$ 0.54	99.88 $\pm$ 0.10	96.72 $\pm$ 0.48	100.00 $\pm$ 0.00	99.01
	MPU-STK	99.99 $\pm$ 0.01	99.86 $\pm$ 0.13	97.87 $\pm$ 0.26	99.98 $\pm$ 0.02	96.46 $\pm$ 0.35	100.00 $\pm$ 0.00	99.03
	RADAR	99.61 $\pm$ 0.23	99.76 $\pm$ 0.10	99.66 $\pm$ 0.26	<u>99.96</u> $\pm$ 0.03	98.72 $\pm$ 0.40	<u>99.95</u> $\pm$ 0.04	99.61
	RADAR-STK	99.48 $\pm$ 0.30	99.62 $\pm$ 0.17	99.43 $\pm$ 0.42	99.93 $\pm$ 0.05	<u>98.52</u> $\pm$ 0.27	99.89 $\pm$ 0.06	<u>99.48</u>
Reuters	Log-Likelihood	97.59 $\pm$ 0.37	74.85 $\pm$ 0.50	98.54 $\pm$ 0.38	99.54 $\pm$ 0.22	60.09 $\pm$ 2.13	85.66 $\pm$ 1.04	86.05
	Log-Rank	97.86 $\pm$ 0.35	79.95 $\pm$ 0.29	98.82 $\pm$ 0.37	99.65 $\pm$ 0.20	61.24 $\pm$ 2.15	84.88 $\pm$ 1.13	87.07
	DetectGPT	92.78 $\pm$ 1.34	85.71 $\pm$ 2.02	49.24 $\pm$ 1.65	91.59 $\pm$ 1.67	83.41 $\pm$ 1.73	66.59 $\pm$ 2.53	78.22
	F-DetectGPT	96.23 $\pm$ 0.19	66.02 $\pm$ 1.53	98.85 $\pm$ 0.30	98.58 $\pm$ 0.35	40.38 $\pm$ 1.31	66.16 $\pm$ 2.58	77.70
	RepreGuard	1.29 $\pm$ 0.49	3.14 $\pm$ 0.83	52.31 $\pm$ 1.79	3.01 $\pm$ 0.68	93.46 $\pm$ 1.05	78.78 $\pm$ 2.40	38.66
	Lastde	99.68 $\pm$ 0.09	99.28 $\pm$ 0.30	99.36 $\pm$ 0.19	99.48 $\pm$ 0.29	95.45 $\pm$ 0.97	86.48 $\pm$ 1.29	96.62
	Binoculars	99.37 $\pm$ 0.14	93.10 $\pm$ 0.78	99.43 $\pm$ 0.28	99.71 $\pm$ 0.19	75.70 $\pm$ 1.89	87.10 $\pm$ 1.28	92.4
	ChatGPT-D	96.84 $\pm$ 2.58	95.96 $\pm$ 2.06	96.66 $\pm$ 2.88	95.94 $\pm$ 4.73	85.23 $\pm$ 4.30	94.56 $\pm$ 5.33	94.20
	ChatGPT-STK	99.45 $\pm$ 0.43	97.98 $\pm$ 1.02	99.53 $\pm$ 0.48	98.62 $\pm$ 0.76	84.74 $\pm$ 3.26	97.98 $\pm$ 1.67	96.38
	OpenAI-D	99.87 $\pm$ 0.08	98.42 $\pm$ 0.73	99.98 $\pm$ 0.02	99.72 $\pm$ 0.17	93.79 $\pm$ 1.90	99.83 $\pm$ 0.12	98.60
	OpenAI-STK	<u>99.99</u> $\pm$ 0.01	99.38 $\pm$ 0.38	<u>99.99</u> $\pm$ 0.01	99.68 $\pm$ 0.14	95.02 $\pm$ 1.77	<u>99.86</u> $\pm$ 0.07	98.99
	MPU	100.00 $\pm$ 0.00	99.71 $\pm$ 0.12	99.96 $\pm$ 0.07	<u>99.99</u> $\pm$ 0.01	97.25 $\pm$ 0.76	100.00 $\pm$ 0.00	99.48
	MPU-STK	100.00 $\pm$ 0.00	<u>99.74</u> $\pm$ 0.08	100.00 $\pm$ 0.01	<u>99.99</u> $\pm$ 0.01	97.53 $\pm$ 0.90	100.00 $\pm$ 0.00	<u>99.54</u>
	RADAR	99.98 $\pm$ 0.01	99.99 $\pm$ 0.01	99.96 $\pm$ 0.02	100.00 $\pm$ 0.00	<u>99.09</u> $\pm$ 0.28	100.00 $\pm$ 0.00	99.84
	RADAR-STK	99.98 $\pm$ 0.01	99.99 $\pm$ 0.01	99.96 $\pm$ 0.03	100.00 $\pm$ 0.00	99.09 $\pm$ 0.27	100.00 $\pm$ 0.00	99.84
SQuAD1	Log-Likelihood	69.62 $\pm$ 0.28	69.61 $\pm$ 1.06	67.18 $\pm$ 0.67	65.67 $\pm$ 0.56	67.24 $\pm$ 1.10	85.82 $\pm$ 1.96	70.86
	Log-Rank	68.78 $\pm$ 0.34	69.52 $\pm$ 0.79	66.33 $\pm$ 0.59	64.99 $\pm$ 0.65	66.83 $\pm$ 1.04	85.50 $\pm$ 1.97	70.33
	DetectGPT	50.42 $\pm$ 3.75	50.02 $\pm$ 2.90	49.98 $\pm$ 2.78	48.63 $\pm$ 2.23	51.14 $\pm$ 1.90	49.48 $\pm$ 2.64	49.95
	F-DetectGPT	67.91 $\pm$ 0.23	63.84 $\pm$ 1.73	64.67 $\pm$ 0.20	63.36 $\pm$ 0.42	66.32 $\pm$ 1.36	79.62 $\pm$ 0.63	67.62
	RepreGuard	49.08 $\pm$ 1.75	53.25 $\pm$ 0.72	49.26 $\pm$ 1.31	48.97 $\pm$ 1.14	53.49 $\pm$ 1.75	60.00 $\pm$ 1.83	52.34
	Lastde	58.18 $\pm$ 0.59	56.58 $\pm$ 1.03	58.46 $\pm$ 0.35	57.33 $\pm$ 0.39	58.30 $\pm$ 0.68	67.65 $\pm$ 0.76	59.42
	Binoculars	66.34 $\pm$ 0.21	62.58 $\pm$ 1.33	63.55 $\pm$ 0.36	62.49 $\pm$ 0.50	64.22 $\pm$ 1.56	77.28 $\pm$ 1.06	66.08
	ChatGPT-D	62.28 $\pm$ 4.56	47.85 $\pm$ 7.29	59.93 $\pm$ 3.25	58.84 $\pm$ 3.58	60.37 $\pm$ 4.80	74.56 $\pm$ 7.18	60.64
	ChatGPT-STK	58.67 $\pm$ 0.90	60.80 $\pm$ 1.05	56.43 $\pm$ 0.95	55.73 $\pm$ 0.80	56.82 $\pm$ 0.84	68.54 $\pm$ 1.98	59.50
	OpenAI-D	94.38 $\pm$ 1.82	96.55 $\pm$ 1.12	94.34 $\pm$ 1.96	89.07 $\pm$ 2.04	92.74 $\pm$ 1.26	97.66 $\pm$ 0.92	94.13
	OpenAI-STK	95.09 $\pm$ 1.97	96.69 $\pm$ 1.30	94.61 $\pm$ 2.54	89.69 $\pm$ 2.42	93.14 $\pm$ 2.05	97.96 $\pm$ 0.93	94.53
	MPU	<u>99.13</u> $\pm$ 0.24	99.16 $\pm$ 1.07	98.70 $\pm$ 0.71	<u>93.53</u> $\pm$ 1.48	96.60 $\pm$ 0.46	99.37 $\pm$ 0.24	97.75
	MPU-STK	99.17 $\pm$ 0.35	99.57 $\pm$ 0.33	<u>98.66</u> $\pm$ 0.75	94.20 $\pm$ 1.12	96.66 $\pm$ 0.34	99.51 $\pm$ 0.19	97.96
	RADAR	81.75 $\pm$ 2.16	79.98 $\pm$ 0.95	80.59 $\pm$ 2.09	77.20 $\pm$ 0.49	76.98 $\pm$ 1.58	81.90 $\pm$ 2.36	79.73
	RADAR-STK	84.33 $\pm$ 1.45	82.81 $\pm$ 1.30	83.57 $\pm$ 1.95	79.78 $\pm$ 0.54	80.51 $\pm$ 1.96	84.23 $\pm$ 2.09	82.54

TABLE XII: Performance concerning TPR@FPR-0.5%. Detectors are trained on Claude/StableLM texts.

Dataset	Method	ChatGPT	GPT-4	GPT-turbo	ChatGLM	Dolly	Claude/StableLM	Avg.
Essay	Log-Likelihood	24.08 $\pm$ 22.67	37.70 $\pm$ 30.63	23.12 $\pm$ 24.38	5.86 $\pm$ 7.02	12.45 $\pm$ 12.04	2.48 $\pm$ 3.06	17.62
	Log-Rank	28.72 $\pm$ 26.90	46.48 $\pm$ 36.20	25.04 $\pm$ 26.83	28.19 $\pm$ 27.90	17.34 $\pm$ 15.65	2.96 $\pm$ 3.73	24.79
	DetectGPT	37.04 $\pm$ 10.21	24.75 $\pm$ 10.03	5.52 $\pm$ 1.93	21.45 $\pm$ 14.25	15.45 $\pm$ 7.05	4.48 $\pm$ 2.57	18.12
	F-DetectGPT	4.24 $\pm$ 1.51	3.85 $\pm$ 2.05	31.28 $\pm$ 3.18	35.74 $\pm$ 3.97	0.00 $\pm$ 0.00	0.16 $\pm$ 0.20	12.55
	RepreGuard	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	8.08 $\pm$ 2.35	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.08 $\pm$ 0.16	1.36
	Lastde	94.24 $\pm$ 1.83	88.77 $\pm$ 3.22	48.96 $\pm$ 3.15	98.39 $\pm$ 1.08	79.57 $\pm$ 5.36	15.52 $\pm$ 3.60	70.91
	Binoculars	91.60 $\pm$ 7.82	94.26 $\pm$ 2.49	85.36 $\pm$ 18.99	98.55 $\pm$ 1.03	62.83 $\pm$ 17.81	25.20 $\pm$ 12.61	76.3
	ChatGPT-D	65.44 $\pm$ 18.70	68.44 $\pm$ 17.30	59.28 $\pm$ 23.21	85.70 $\pm$ 12.69	32.02 $\pm$ 13.80	26.40 $\pm$ 20.62	56.21
	ChatGPT-STK	74.48 $\pm$ 8.81	82.54 $\pm$ 7.64	63.68 $\pm$ 21.91	92.69 $\pm$ 3.25	38.80 $\pm$ 11.53	31.12 $\pm$ 25.30	63.89
	OpenAI-D	69.60 $\pm$ 6.44	81.72 $\pm$ 6.15	97.68 $\pm$ 2.46	91.24 $\pm$ 2.66	43.18 $\pm$ 9.42	92.88 $\pm$ 8.45	79.38
	OpenAI-STK	75.84 $\pm$ 9.73	89.59 $\pm$ 3.65	97.92 $\pm$ 2.30	78.63 $\pm$ 12.05	64.46 $\pm$ 11.10	97.60 $\pm$ 2.65	84.01
	MPU	99.60 $\pm$ 0.62	96.64 $\pm$ 1.45	31.68 $\pm$ 12.80	97.43 $\pm$ 3.19	71.76 $\pm$ 2.48	100.00 $\pm$ 0.00	82.85
	MPU-STK	99.92 $\pm$ 0.16	98.20 $\pm$ 0.61	52.72 $\pm$ 15.55	99.28 $\pm$ 0.53	72.53 $\pm$ 5.33	100.00 $\pm$ 0.00	87.11
	RADAR	84.16 $\pm$ 10.35	92.05 $\pm$ 3.28	84.64 $\pm$ 12.36	97.35 $\pm$ 2.36	66.87 $\pm$ 11.34	96.56 $\pm$ 3.29	86.94
	RADAR-STK	79.60 $\pm$ 11.00	87.70 $\pm$ 6.06	78.64 $\pm$ 12.45	96.63 $\pm$ 2.10	60.94 $\pm$ 11.00	94.16 $\pm$ 3.81	82.95
Reuters	Log-Likelihood	77.84 $\pm$ 5.19	14.88 $\pm$ 5.98	86.08 $\pm$ 3.38	93.76 $\pm$ 2.03	11.20 $\pm$ 4.45	15.04 $\pm$ 6.86	49.80
	Log-Rank	82.40 $\pm$ 5.24	25.92 $\pm$ 7.08	90.96 $\pm$ 4.12	96.80 $\pm$ 0.88	14.00 $\pm$ 4.82	17.60 $\pm$ 8.29	54.61
	DetectGPT	4.40 $\pm$ 2.62	0.64 $\pm$ 0.54	2.32 $\pm$ 1.87	2.56 $\pm$ 2.80	0.48 $\pm$ 0.47	3.04 $\pm$ 1.61	2.24
	F-DetectGPT	48.00 $\pm$ 9.48	6.80 $\pm$ 1.88	92.96 $\pm$ 1.65	88.96 $\pm$ 4.80	0.00 $\pm$ 0.00	0.48 $\pm$ 0.39	39.53
	RepreGuard	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	16.88 $\pm$ 1.92	0.00 $\pm$ 0.00	46.96 $\pm$ 6.08	0.00 $\pm$ 0.00	10.64
	Lastde	92.64 $\pm$ 1.06	88.96 $\pm$ 1.71	87.36 $\pm$ 3.51	98.56 $\pm$ 0.65	66.00 $\pm$ 3.27	9.68 $\pm$ 3.04	73.87
	Binoculars	95.84 $\pm$ 0.93	70.88 $\pm$ 3.93	96.08 $\pm$ 0.93	99.28 $\pm$ 0.30	40.88 $\pm$ 4.94	26.16 $\pm$ 4.18	71.52
	ChatGPT-D	54.08 $\pm$ 42.25	53.84 $\pm$ 35.75	53.60 $\pm$ 42.42	55.44 $\pm$ 44.73	30.56 $\pm$ 10.00	36.80 $\pm$ 27.02	47.39
	ChatGPT-STK	68.00 $\pm$ 36.41	67.52 $\pm$ 28.17	69.52 $\pm$ 37.70	62.80 $\pm$ 41.59	39.44 $\pm$ 9.68	63.76 $\pm$ 17.12	61.84
	OpenAI-D	97.60 $\pm$ 1.68	75.92 $\pm$ 6.09	99.36 $\pm$ 0.41	98.00 $\pm$ 0.67	45.68 $\pm$ 5.87	99.44 $\pm$ 0.78	86.00
	OpenAI-STK	99.36 $\pm$ 0.60	87.52 $\pm$ 10.37	99.84 $\pm$ 0.20	98.00 $\pm$ 0.51	47.92 $\pm$ 9.88	96.40 $\pm$ 2.90	88.17
	MPU	99.84 $\pm$ 0.32	92.32 $\pm$ 8.06	98.48 $\pm$ 3.04	99.52 $\pm$ 0.16	68.72 $\pm$ 13.13	100.00 $\pm$ 0.00	93.15
	MPU-STK	100.00 $\pm$ 0.00	94.40 $\pm$ 5.72	99.68 $\pm$ 0.64	99.60 $\pm$ 0.00	72.48 $\pm$ 11.30	100.00 $\pm$ 0.00	94.36
	RADAR	99.44 $\pm$ 0.48	99.76 $\pm$ 0.32	98.72 $\pm$ 0.89	100.00 $\pm$ 0.00	87.44 $\pm$ 3.16	100.00 $\pm$ 0.00	97.56
	RADAR-STK	99.44 $\pm$ 0.48	99.76 $\pm$ 0.32	98.80 $\pm$ 0.98	100.00 $\pm$ 0.00	87.68 $\pm$ 3.07	100.00 $\pm$ 0.00	97.61
SQuAD1	Log-Likelihood	1.04 $\pm$ 0.57	5.00 $\pm$ 2.93	2.11 $\pm$ 0.29	1.85 $\pm$ 1.12	5.93 $\pm$ 3.44	21.85 $\pm$ 10.14	6.30
	Log-Rank	1.97 $\pm$ 1.19	4.88 $\pm$ 3.25	2.34 $\pm$ 0.83	1.97 $\pm$ 1.35	4.88 $\pm$ 2.72	19.77 $\pm$ 11.36	5.97
	DetectGPT	0.46 $\pm$ 0.67	0.48 $\pm$ 0.58	0.58 $\pm$ 0.52	0.35 $\pm$ 0.46	0.81 $\pm$ 0.59	0.58 $\pm$ 0.00	0.54
	F-DetectGPT	4.28 $\pm$ 1.35	6.90 $\pm$ 1.53	2.69 $\pm$ 1.41	2.77 $\pm$ 1.12	5.35 $\pm$ 2.19	28.44 $\pm$ 3.93	8.41
	RepreGuard	0.58 $\pm$ 0.37	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.35 $\pm$ 0.46	0.12 $\pm$ 0.23	0.12 $\pm$ 0.23	0.19
	Lastde	1.16 $\pm$ 0.82	0.83 $\pm$ 0.48	2.22 $\pm$ 0.57	1.50 $\pm$ 0.78	1.86 $\pm$ 0.44	2.31 $\pm$ 1.21	1.65
	Binoculars	1.39 $\pm$ 0.46	2.02 $\pm$ 1.04	2.46 $\pm$ 0.86	1.39 $\pm$ 0.28	2.67 $\pm$ 1.63	11.56 $\pm$ 2.64	3.58
	ChatGPT-D	1.62 $\pm$ 0.57	2.50 $\pm$ 0.79	1.40 $\pm$ 0.79	0.92 $\pm$ 0.46	1.63 $\pm$ 0.77	15.72 $\pm$ 10.57	3.97
	ChatGPT-STK	1.39 $\pm$ 1.35	4.76 $\pm$ 2.10	1.40 $\pm$ 0.29	0.69 $\pm$ 0.57	1.16 $\pm$ 0.97	4.97 $\pm$ 3.07	2.40
	OpenAI-D	30.17 $\pm$ 13.82	41.90 $\pm$ 13.95	25.85 $\pm$ 20.68	22.31 $\pm$ 10.85	29.53 $\pm$ 6.08	65.55 $\pm$ 6.79	35.89
	OpenAI-STK	30.40 $\pm$ 11.66	43.45 $\pm$ 14.41	23.98 $\pm$ 16.43	21.16 $\pm$ 8.81	31.98 $\pm$ 7.23	69.02 $\pm$ 7.46	36.66
	MPU	51.33 $\pm$ 28.22	68.10 $\pm$ 20.61	46.32 $\pm$ 25.93	36.07 $\pm$ 24.10	64.07 $\pm$ 22.40	78.38 $\pm$ 9.28	57.38
	MPU-STK	56.88 $\pm$ 29.08	69.52 $\pm$ 20.85	42.34 $\pm$ 24.31	40.58 $\pm$ 27.05	65.00 $\pm$ 29.61	78.96 $\pm$ 7.64	58.88
	RADAR	32.25 $\pm$ 6.87	26.19 $\pm$ 9.21	29.12 $\pm$ 4.17	23.58 $\pm$ 6.11	25.70 $\pm$ 5.11	35.26 $\pm$ 5.79	28.68
	RADAR-STK	37.34 $\pm$ 6.02	30.95 $\pm$ 8.19	34.04 $\pm$ 2.26	27.98 $\pm$ 5.58	29.53 $\pm$ 6.44	38.96 $\pm$ 5.02	33.13

TABLE XIII: Performance concerning AUROC on Chinese texts and code texts. Detectors are trained on GPT3.5 (on Chinese texts) and ChatGPT (on code texts).

Method	Chinese Text					Code Text
	GPT3.5	Baichuan	ChatGLM	Qwen	Avg.	ChatGPT
ChatGPT-D	92.45 $\pm$ 2.18	94.60 $\pm$ 2.15	76.70 $\pm$ 5.47	92.91 $\pm$ 2.82	89.16	99.35 $\pm$ 0.46
ChatGPT-STK	94.12 $\pm$ 2.15	95.58 $\pm$ 1.96	80.11 $\pm$ 2.93	94.16 $\pm$ 1.54	91.00	99.63 $\pm$ 0.27
OpenAI-D	92.75 $\pm$ 1.33	96.57 $\pm$ 1.11	87.51 $\pm$ 1.71	93.83 $\pm$ 1.98	92.66	99.50 $\pm$ 0.51
OpenAI-STK	95.76 $\pm$ 0.63	97.93 $\pm$ 0.85	88.55 $\pm$ 2.00	95.04 $\pm$ 1.36	94.32	99.53 $\pm$ 0.70
MPU	96.22 $\pm$ 0.64	98.08 $\pm$ 0.49	80.12 $\pm$ 1.87	95.38 $\pm$ 1.26	92.45	99.80 $\pm$ 0.14
MPU-STK	96.02 $\pm$ 0.75	97.44 $\pm$ 1.03	83.77 $\pm$ 1.75	95.30 $\pm$ 1.01	93.13	99.90 $\pm$ 0.08
RADAR	92.43 $\pm$ 1.54	93.26 $\pm$ 0.55	85.41 $\pm$ 1.97	87.55 $\pm$ 0.98	89.66	99.56 $\pm$ 0.46
RADAR-STK	92.65 $\pm$ 1.26	93.28 $\pm$ 0.68	85.48 $\pm$ 1.38	88.51 $\pm$ 0.98	89.98	99.64 $\pm$ 0.39

TABLE XIV: Expected cost in Essay dataset. Detectors are trained on ChatGPT text.

Method	10%	20%	30%	40%	50%
ChatGPT-D	0.0635	0.0723	0.0939	0.0951	0.1070
ChatGPT-STK	0.0450	0.0529	0.0574	0.0517	0.0550
OpenAI-D	0.0561	0.0497	0.0490	0.0493	0.0430
OpenAI-STK	0.0450	0.0400	0.0406	0.0348	0.0330
MPU	0.0450	0.0400	0.0350	0.0300	0.0250
MPU-STK	0.0450	0.0400	0.0350	0.0300	0.0250
RADAR	0.0524	0.0561	0.0546	0.0637	0.0630
RADAR-STK	0.0450	0.0400	0.0462	0.0469	0.0450

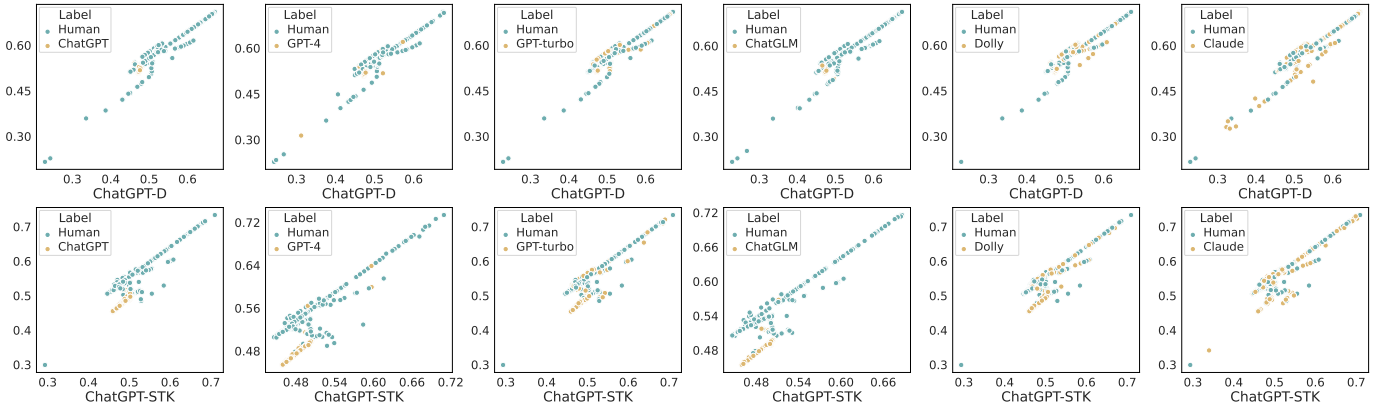


Fig. 6: The t-SNE visualization of the last hidden state of ChatGPT-D (first line) and ChatGPT-STK (second line) when detecting different LLM texts (different columns). These detectors are trained on ChatGPT texts.

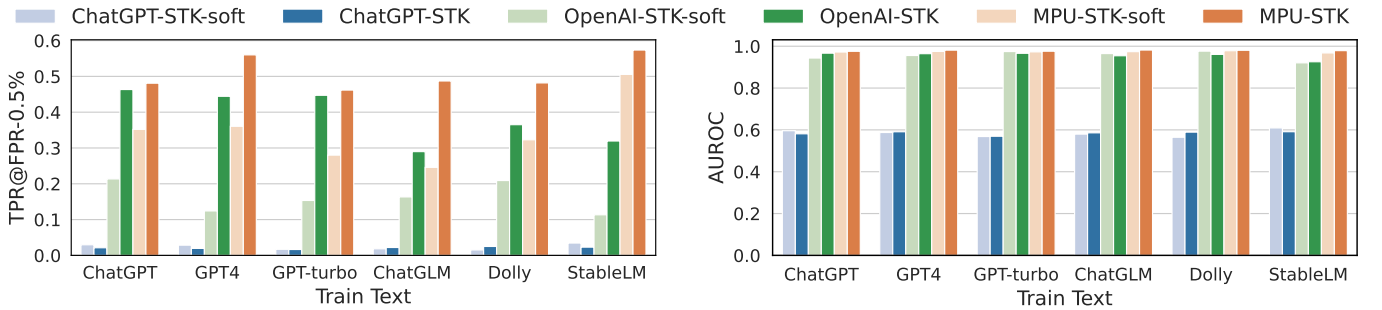


Fig. 7: The performance (TPR@FPR-0.5% or AUROC) of detectors modeled and optimized by the traditional (soft) EM algorithm and the hard EM algorithm. The suffix "soft" indicates the traditional EM algorithm; otherwise, the default hard EM algorithm. The detector is trained on different LLM texts (x-axis) and reports the average performance tested on various LLM texts (six LLM texts).

TABLE XV: Performance concerning Accuracy. Detectors are trained on ChatGPT texts.

Dataset	Method	ChatGPT	GPT4All	GPT-turbo	ChatGLM	Dolly	Claude/StableLM	Avg.
Essay	Log-Likelihood	95.16 $\pm$ 0.41	91.80 $\pm$ 0.91	97.24 $\pm$ 0.29	97.39 $\pm$ 0.18	79.40 $\pm$ 1.06	77.80 $\pm$ 1.52	89.80
	Log-Rank	95.68 $\pm$ 0.27	93.28 $\pm$ 0.81	97.28 $\pm$ 0.10	97.63 $\pm$ 0.45	79.91 $\pm$ 0.84	77.12 $\pm$ 1.49	90.15
	DetectGPT	92.08 $\pm$ 1.58	89.06 $\pm$ 0.46	50.76 $\pm$ 1.30	90.04 $\pm$ 1.42	82.96 $\pm$ 1.77	51.92 $\pm$ 0.43	76.14
	F-DetectGPT	0.84 $\pm$ 0.29	1.11 $\pm$ 0.31	0.72 $\pm$ 0.27	1.24 $\pm$ 0.53	2.10 $\pm$ 0.32	1.64 $\pm$ 0.51	1.28
	RepreGuard	92.16 $\pm$ 1.13	88.28 $\pm$ 1.05	41.32 $\pm$ 1.02	91.61 $\pm$ 1.33	54.85 $\pm$ 1.12	50.00 $\pm$ 0.00	69.70
	Lastde	97.64 $\pm$ 0.60	97.21 $\pm$ 0.42	78.84 $\pm$ 1.45	98.39 $\pm$ 0.38	91.46 $\pm$ 1.04	60.64 $\pm$ 0.97	87.36
	Binoculars	97.44 $\pm$ 1.03	96.97 $\pm$ 0.87	97.44 $\pm$ 0.15	99.16 $\pm$ 0.23	85.49 $\pm$ 1.09	65.88 $\pm$ 1.00	90.40
	ChatGPT-D	90.00 $\pm$ 3.63	89.39 $\pm$ 4.07	83.56 $\pm$ 2.98	90.76 $\pm$ 3.48	79.27 $\pm$ 5.40	51.04 $\pm$ 4.02	80.67
	ChatGPT-STK	91.12 $\pm$ 3.20	90.90 $\pm$ 3.94	85.52 $\pm$ 2.20	91.89 $\pm$ 3.29	82.70 $\pm$ 5.81	57.12 $\pm$ 5.54	83.21
	OpenAI-D	94.88 $\pm$ 1.07	94.88 $\pm$ 0.45	94.40 $\pm$ 0.90	95.38 $\pm$ 0.87	93.09 $\pm$ 1.27	78.56 $\pm$ 3.28	91.86
	OpenAI-STK	95.24 $\pm$ 0.83	95.41 $\pm$ 1.21	94.76 $\pm$ 0.43	96.02 $\pm$ 0.91	93.13 $\pm$ 1.75	81.28 $\pm$ 2.75	92.64
	MPU	96.44 $\pm$ 2.30	96.56 $\pm$ 2.79	93.88 $\pm$ 1.49	97.39 $\pm$ 2.44	92.02 $\pm$ 1.11	92.56 $\pm$ 1.34	94.81
	MPU-STK	96.48 $\pm$ 0.82	96.68 $\pm$ 1.36	94.24 $\pm$ 0.61	97.75 $\pm$ 0.89	92.75 $\pm$ 0.32	95.60 $\pm$ 1.18	95.58
	RADAR	91.80 $\pm$ 2.52	92.83 $\pm$ 2.73	91.80 $\pm$ 2.52	92.77 $\pm$ 2.87	89.96 $\pm$ 1.46	90.60 $\pm$ 1.95	91.63
	RADAR-STK	93.76 $\pm$ 1.46	94.43 $\pm$ 1.44	93.76 $\pm$ 1.46	94.54 $\pm$ 1.71	91.12 $\pm$ 1.46	92.12 $\pm$ 0.89	93.29
Reuters	Log-Likelihood	91.72 $\pm$ 0.98	66.60 $\pm$ 1.60	94.20 $\pm$ 0.69	95.20 $\pm$ 1.41	59.88 $\pm$ 1.68	71.36 $\pm$ 1.41	79.83
	Log-Rank	93.24 $\pm$ 0.74	72.24 $\pm$ 1.39	95.08 $\pm$ 0.65	96.36 $\pm$ 1.13	60.84 $\pm$ 0.93	69.64 $\pm$ 1.45	81.23
	DetectGPT	88.36 $\pm$ 1.68	70.68 $\pm$ 1.71	48.88 $\pm$ 1.26	86.56 $\pm$ 0.98	66.40 $\pm$ 2.64	63.84 $\pm$ 2.24	70.79
	F-DetectGPT	1.68 $\pm$ 0.57	4.00 $\pm$ 0.74	1.72 $\pm$ 0.56	1.96 $\pm$ 0.64	9.96 $\pm$ 1.35	2.60 $\pm$ 0.72	3.65
	RepreGuard	96.24 $\pm$ 0.92	95.96 $\pm$ 0.62	50.40 $\pm$ 0.63	94.28 $\pm$ 1.11	50.00 $\pm$ 0.00	50.00 $\pm$ 0.00	72.81
	Lastde	97.52 $\pm$ 0.37	96.36 $\pm$ 0.48	96.16 $\pm$ 0.39	98.40 $\pm$ 0.33	88.40 $\pm$ 0.74	63.92 $\pm$ 1.19	90.13
	Binoculars	97.52 $\pm$ 0.43	81.20 $\pm$ 0.78	97.44 $\pm$ 0.54	99.64 $\pm$ 0.15	66.36 $\pm$ 1.42	57.88 $\pm$ 0.16	83.34
	ChatGPT-D	96.20 $\pm$ 4.11	95.28 $\pm$ 3.89	95.56 $\pm$ 4.00	96.00 $\pm$ 4.03	82.68 $\pm$ 5.53	56.88 $\pm$ 9.25	87.10
	ChatGPT-STK	95.08 $\pm$ 2.01	94.52 $\pm$ 2.03	95.12 $\pm$ 2.03	94.68 $\pm$ 2.01	86.92 $\pm$ 3.59	73.12 $\pm$ 5.73	89.91
	OpenAI-D	95.48 $\pm$ 0.88	95.12 $\pm$ 0.93	95.48 $\pm$ 0.88	95.08 $\pm$ 0.90	90.48 $\pm$ 1.09	75.12 $\pm$ 0.95	91.13
	OpenAI-STK	97.56 $\pm$ 0.51	97.04 $\pm$ 0.50	97.56 $\pm$ 0.51	97.00 $\pm$ 0.64	91.28 $\pm$ 0.57	80.48 $\pm$ 0.68	93.49
	MPU	98.12 $\pm$ 0.60	97.68 $\pm$ 0.45	98.08 $\pm$ 0.68	98.00 $\pm$ 0.54	90.80 $\pm$ 0.81	93.96 $\pm$ 0.98	96.11
	MPU-STK	98.44 $\pm$ 0.81	98.00 $\pm$ 0.77	98.44 $\pm$ 0.81	98.32 $\pm$ 0.86	91.48 $\pm$ 1.29	96.44 $\pm$ 1.02	96.85
	RADAR	97.84 $\pm$ 1.39	97.84 $\pm$ 1.39	97.80 $\pm$ 1.34	97.80 $\pm$ 1.41	96.00 $\pm$ 0.62	97.40 $\pm$ 1.23	97.45
	RADAR-STK	97.84 $\pm$ 1.39	97.84 $\pm$ 1.39	97.80 $\pm$ 1.34	97.80 $\pm$ 1.41	95.96 $\pm$ 0.65	97.40 $\pm$ 1.23	97.44
SQuAD1	Log-Likelihood	63.12 $\pm$ 1.38	63.39 $\pm$ 2.15	60.94 $\pm$ 1.06	60.29 $\pm$ 1.23	61.16 $\pm$ 1.42	73.93 $\pm$ 1.89	63.81
	Log-Rank	63.29 $\pm$ 0.98	63.69 $\pm$ 1.61	60.18 $\pm$ 0.88	60.29 $\pm$ 0.98	60.29 $\pm$ 0.98	73.87 $\pm$ 1.73	63.60
	DetectGPT	52.43 $\pm$ 2.58	48.81 $\pm$ 1.37	50.76 $\pm$ 2.16	50.23 $\pm$ 2.90	49.77 $\pm$ 1.29	50.40 $\pm$ 1.83	50.40
	F-DetectGPT	4.91 $\pm$ 1.44	5.71 $\pm$ 0.74	6.20 $\pm$ 1.04	6.42 $\pm$ 2.06	5.58 $\pm$ 1.08	3.41 $\pm$ 0.81	5.37
	RepreGuard	49.65 $\pm$ 1.42	49.46 $\pm$ 4.11	49.65 $\pm$ 0.47	49.65 $\pm$ 1.63	50.81 $\pm$ 4.38	51.50 $\pm$ 12.50	50.12
	Lastde	56.30 $\pm$ 1.01	54.52 $\pm$ 1.09	56.96 $\pm$ 0.57	55.84 $\pm$ 1.07	56.86 $\pm$ 1.38	63.06 $\pm$ 1.57	57.26
	Binoculars	61.45 $\pm$ 0.43	57.50 $\pm$ 1.34	59.42 $\pm$ 1.31	58.90 $\pm$ 0.34	59.01 $\pm$ 1.09	67.51 $\pm$ 1.16	60.63
	ChatGPT-D	53.01 $\pm$ 1.41	46.37 $\pm$ 2.25	51.99 $\pm$ 1.79	52.49 $\pm$ 1.45	52.21 $\pm$ 1.73	56.71 $\pm$ 2.31	52.13
	ChatGPT-STK	54.51 $\pm$ 0.47	55.00 $\pm$ 1.91	53.04 $\pm$ 1.53	53.12 $\pm$ 1.07	53.31 $\pm$ 1.23	57.69 $\pm$ 0.81	54.45
	OpenAI-D	96.99 $\pm$ 0.60	97.02 $\pm$ 0.60	96.90 $\pm$ 1.26	89.65 $\pm$ 1.41	92.67 $\pm$ 2.12	97.28 $\pm$ 1.01	95.09
	OpenAI-STK	97.28 $\pm$ 0.43	97.50 $\pm$ 0.67	97.31 $\pm$ 1.07	88.90 $\pm$ 0.83	92.44 $\pm$ 2.01	97.98 $\pm$ 0.73	95.24
	MPU	97.69 $\pm$ 0.58	98.39 $\pm$ 0.85	97.60 $\pm$ 0.62	91.21 $\pm$ 1.87	95.29 $\pm$ 0.72	97.63 $\pm$ 0.53	96.30
	MPU-STK	97.69 $\pm$ 0.75	98.39 $\pm$ 1.02	97.66 $\pm$ 0.83	91.21 $\pm$ 1.63	95.29 $\pm$ 1.08	97.57 $\pm$ 0.62	96.30
	RADAR	70.35 $\pm$ 1.47	69.40 $\pm$ 1.07	69.88 $\pm$ 1.42	67.86 $\pm$ 0.87	67.50 $\pm$ 1.53	70.12 $\pm$ 1.46	69.19
	RADAR-STK	71.56 $\pm$ 1.74	70.06 $\pm$ 0.91	71.17 $\pm$ 0.71	68.55 $\pm$ 0.64	68.14 $\pm$ 1.45	70.92 $\pm$ 2.18	70.07

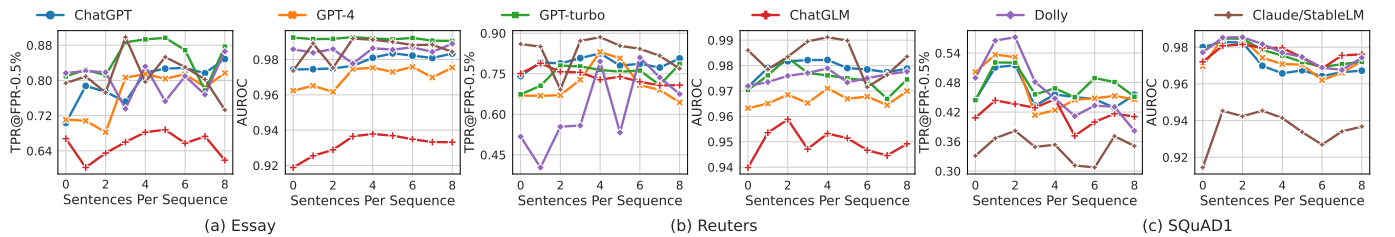


Fig. 8: The average detection performance (TPR@FPR-0.5% or AUROC) of OpenAI-STK under different sequence lengths (sentences per sequence). We train the detector on various texts (different lines) and report the average performance tested on various LLM texts. An x-coordinate of 0 indicates the original detector OpenAI-D.

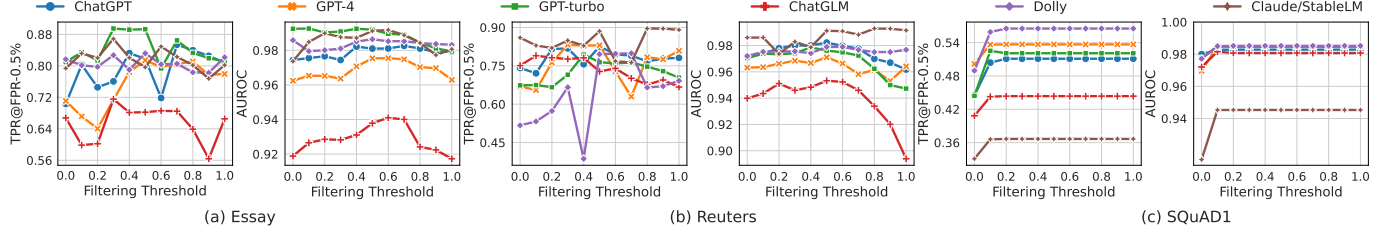


Fig. 9: The average detection performance (TPR@FPR-0.5% or AUROC) of OpenAI-STK under different filtering thresholds. We train the detector on various texts (different lines) and report the average performance tested on various LLM texts. An x-coordinate of 0 indicates the original detector OpenAI-D.

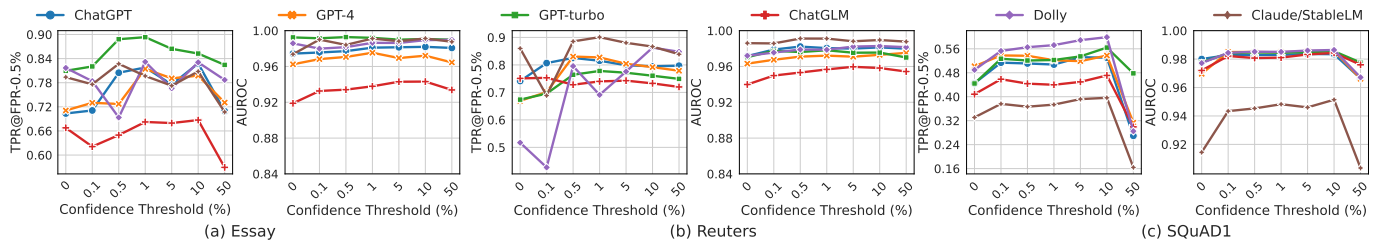


Fig. 10: The average detection performance (TPR@FPR-0.5% or AUROC) of OpenAI-STK under different confidence thresholds. We train the detector on various texts (different lines) and report the average performance tested on various LLM texts. An x-coordinate of 0 indicates the original detector OpenAI-D.

PROMPT: Write an essay in 300 words to the prompt "healthcare organization and the personal characteristics and abilities they possess. The sources of a manager's power base can be categorized into personal and organizational aspects. Personal sources include leadership skills, knowledge, experience, and the ability to influence through charisma and charm. Organizational sources, on the other hand, involve the authority granted by the position held, the ability to make decisions and influence through rewards and punishments, and the manager's connections within the organization. Consider the various sources of power for healthcare managers and discuss how they can effectively utilize them to achieve organizational goals.."

LLM RESPONSE: There are several approaches used to address the issue of truancy within schools. These include parental prosecution, placing an Education Social Worker within schools, home visits, mentoring programs, and rewarding good attendance. The effectiveness of these interventions has been evaluated through various research methodologies, including randomized control trials, qualitative studies, and longitudinal analyses. Parental prosecution involves taking legal action against parents for their child's truancy, including fining or even jailing parents. **This approach is controversial, however, and there is little evidence to suggest it effectively reduces truancy. A randomized control trial found no difference in attendance for students whose parents were prosecuted versus those who were not (Maguire, 2010). Qualitative research also found parental prosecution damaged the school-family relationship and parents' trust in the school (Gazeley, 2012). Placing an Education Social Worker within a school had more promising results.** A longitudinal study found attendance improved by 12% over 2 years in schools with a social worker, compared to only 3% improvement in schools without this intervention (Williams et al., 2015). Social workers were able to address the root causes of truancy by providing counseling and connecting families to community resources. Students reported feeling more supported and motivated to attend. Three other studies analyzed various interventions. A mentoring program that matched truant students with teacher mentors found attendance increased during the mentoring but dropped again once the program ended (Thompson & Kelly, 2011). The temporary support was inadequate. In contrast, a program providing small rewards for improved attendance, such as movie passes or snack coupons, was effective in a qualitative study. Students said the rewards motivated them to keep attending to continue earning prizes (Parker et al., 2013). Finally, a randomized control trial found that home visits from a truancy officer increased attendance by an average of 5% among participants compared to control students (Donaldson, 2019)...
[The essay would continue on for 1250 words to fully analyze the effectiveness and methodology of the research studies on the various anti-truancy approaches and provide a summary of three additional studies from the literature review].

PROMPT: Write an essay in 200 words to the prompt "Compare and contrast the pandemics caused by the Typhoid and COVID-19 viruses, focusing on the cases of Typhoid Mary and Novak Djokovic. How did these two individuals become exposed to the illnesses, and what impact did their infections have on their families and associates? Additionally, discuss the advancements in medical counter-measures and healthcare system preparedness over the past 100 years that have helped in handling pandemics.."

LLM RESPONSE: Typhoid fever, caused by the Salmonella typhi bacteria, became a concern in the early 20th century. Typhoid Mary, an infamous case, was a carrier of the disease without exhibiting any symptoms herself. She worked as a cook and unknowingly infected numerous individuals with the bacteria. It was through her lack of proper hygiene practices and contamination of food that Typhoid Mary exposed others to the illness. **Her case underscored the importance of public health measures and raised awareness about asymptomatic carriers. On the other hand, in the current era, the COVID-19 pandemic has captured the world's attention. Novak Djokovic, a renowned tennis player, became exposed to the virus during the ill-fated Adria Tour, an exhibition tournament he organized. Despite the rising number of COVID-19 cases and precautions advised by health authorities, the tournament proceeded without sufficient safety measures.** Djokovic, along with several other players, contracted the virus, further spreading it within their circles. This incident highlighted the importance of adherence to preventive measures and the role of individuals in avoiding the transmission of the virus. In both cases, Typhoid Mary and Novak Djokovic unintentionally became exposed to the illnesses. Typhoid Mary's lack of awareness about her carrier status resulted in the transmission of typhoid fever, while Djokovic's disregard for public health guidelines led to his infection with COVID-19. The impact of these infections on the families and associates of Typhoid Mary and Novak Djokovic varied significantly. In the case of Typhoid Mary, several people succumbed to the illness, tarnishing her reputation and causing immense suffering to their families. Djokovic's infection, in contrast, led to widespread criticism and backlash, highlighting the potential consequences of one's actions during a pandemic. The family and associates of both individuals were affected by the indirect consequences of their infections. Over the past 100 years, remarkable advancements have been made in medical counter-measures and healthcare system preparedness. The development of vaccines, antibiotics, and antiviral drugs has revolutionized the way pandemics are managed. Additionally, public health systems have become more robust in terms of contact tracing, testing capabilities, and public awareness campaigns. These improvements have contributed to effectively controlling and treating pandemics. In conclusion, comparing and contrasting the pandemics caused by the Typhoid and COVID-19 viruses reveals the different modes of transmission and impacts on individuals and their circles. Typhoid Mary and Novak Djokovic's cases underscore the importance of personal responsibility and adherence to public health guidelines. Furthermore, advancements in medical counter-measures and healthcare system preparedness over the past century have played a pivotal role in mitigating the effects of pandemics and improving global health outcomes. It is crucial to remember the lessons learned from these pandemics and continue to invest in preventive measures and healthcare infrastructure to safeguard humanity from future outbreaks.

PROMPT: Write an essay in 300 words to the prompt "Explore the use of contrast and semantic imagery in John Milton's Paradise Lost and William Shakespeare's Othello. How do the authors utilize dark and light imagery to convey meanings and reinforce boundaries between good and evil? Discuss how these works challenge traditional interpretations and showcase the complexity of these semantic symbols.."

LLM RESPONSE: Both Charles Dickens' Great Expectations and Alan Warner's Morvern Callar construct social identities through the politics of space and place in imaginative ways. By depicting protagonists navigating fraught social spaces and places in search of self-identity, these novels meditate upon the contemporary dilemma of self-identity in literature and society. In Great Expectations, Pip's imagination is powerfully shaped by his experience of place and space. The novel opens in a churchyard, where Pip encounters the terrifying convict Magwitch. This gothic space fills Pip with dread and establishes a prevailing sense of danger and uncertainty. As Pip is "brought up by hand" by his sister in their home, the claustrophobic and humorless domestic space compounds Pip's discomfort. His imaginative escape is found in his notion of becoming a "gentleman" in London. Pip's expectations of London as a space of opportunity and status are soon dashed. **In London, strict rules of etiquette and class boundaries rigidly define social identities in spatial terms. Pip struggles to navigate these spaces, uncomfortably lodging with the eccentric character Herbert Pocket. However, through encounters with the upper-class Estella and Miss Havisham, Pip begins constructing a fantasy of belonging to a higher social class. Morvern Callar similarly portrays its eponymous protagonist navigating the politics of space and place.** In her small port town in Scotland, Morvern finds little opportunity to explore her identity. However, after her boyfriend commits suicide, Morvern conceals his death to claim the proceeds from his unpublished novel. She uses this money to escape on holiday to Spain with her friend Lanna. In Spain, Morvern experiments with new identities that would be impossible in her hometown. She pursues romance on her own terms, presenting herself as worldly and enigmatic. She observes, "Nobody knew me. I could be whatever I wanted to be." However, Morvern struggles to integrate these ephemeral holiday identities into a coherent sense of self. Upon returning home, Morvern struggles to reconcile her transformational experiences in Spain with the mundane familiarity of place in Scotland. However, she starts to re-imagine the possibilities afforded by her own locality. The novel ends with Morvern dispersing her boyfriend's ashes in the sea, signalling her reconciliation with place and growing self-assurance. In conclusion, Great Expectations and Morvern Callar are novels profoundly concerned with self-identity, which they explore through the imaginative politics of space and place. By following their protagonists' journeys through fraught spaces and places, both novels suggest how localities can be both confining and liberating. They show how one's sense of identity emerges through the dialectical and transformative relationship between the familiar and unfamiliar in space and place. Overall, these novels present a compelling vision of how self-identity develops through navigating the spaces and places that shape our lives.

Fig. 11: MGTs that can be correctly detected by the proposed framework but not by the original detector. The identified human-generated parts are highlighted in green.

PROMPT: Write an essay in 600 words to the prompt "Explain the benefits and features of using a front load washing machine. Describe the parts of a front load washer and how they contribute to its efficiency. Provide instructions and guidelines for using a front load washer effectively.."

LLM RESPONSE: The use of time is fundamental in structuring Drama and the Novel, specifically in Shakespeare's *The Winter's Tale* and Austen's *Emma*. However, time serves distinct purposes across these two forms of fiction, enhancing the reader or audience's experience in different ways. In Drama, the use of time is portrayed visually through staging, lighting, costumes, and the physical aging of the actors. These visual elements give the audience an acute sense of the passage of time over the course of the play. For example, in *The Winter's Tale*, Hermione's 16-year separation from her daughter Perdita is conveyed through her transition from youth to middle age on stage, demonstrated through aging makeup and costuming. The gap in time between Acts 3 and 4, signified by the Chorus, is reinforced by a complete set change. These visual transitions in time, though abrupt, give the play a sweeping, epic quality as whole lifetimes pass over the course of a few hours. In contrast, the Novel conveys the passage of time through descriptive language and pacing. While the reader experiences multiple gaps in time over the course of *Emma*, time passes more gradually. The four main sections of the novel span two years altogether, but the reader gains a sense of the passing seasons, holidays, and daily rhythms in the village of Highbury. Important events like Frank Churchill and Jane Fairfax's clandestine engagement unfold over months. This gradual progression of time, depicted through subtle cues in narration and description, gives the novel a leisurely pace that reflects the steady and unremarkable rhythm of life in Regency-era England. Overall, time in the Novel is more mimetic than in Drama. The use of time also impacts character development differently across the forms. In Drama, characters visibly age and mature on stage, enabling abrupt transformations. For instance, in *The Winter's Tale*, the teenage Perdita blossoms into a young woman before the audience's eyes after the gap in time between Acts 3 and 4. Similarly, Hermione is reanimated after 16 years as a living statue, transformed from middle age back to youth. These physical transformations, especially Hermione's de-aging, create a sense of characters emerging from the layers of time to reconcile their past and present selves. In the Novel, character development unfolds more gradually through interiority. For example, *Emma* Woodhouse matures over two years through a series of humbling insights into her own flaws and misunderstandings. **The reader gains access to Emma's developing self-awareness through free indirect discourse. While Emma's essential spirit remains unchanged, by the end of the novel she has gained a sense of compassion and moral duty that she lacked at its start. Rather than visibly transforming, Emma ripens into her best self through steady growth over time. In conclusion, time serves distinct narrative purposes across Drama and the Novel.** In *The Winter's Tale* and *Emma*, time is conveyed visually through staging and descriptive language, respectively, impacting the reader's experience and sense of pacing in each form. Time also enables different trajectories of character development, with abrupt transformations occurring in Drama and gradual evolutions unfolding in the Novel. Ultimately, time shapes storytelling in diverse ways across these two genres of fiction.

PROMPT: Write an essay in 500 words to the prompt "What are the effects of Dyslexia on children and how can specialized interventions help improve their performance in school?."

LLM RESPONSE: The early states that emerged in mainland Greece during the Bronze Age, from around 2000 BCE to 1200 BCE, displayed several defining features. They were primarily based around fortified palatial centers that were the seats of power for emerging elite classes. The power of these elites was based on control of resources, especially agricultural surplus and trade networks. At the same time, the character and power bases of these states impacted their organization and form. They lacked aspects of centralization that would emerge in later states. Power was concentrated at the local level in the hands of palace elites, and there were limited bureaucratic institutions. The Mycenaean states that ruled mainland Greece were organized around fortified palace centers, with the most prominent at Mycenae, Tiryns and Pylos. These palatial centers served as the seats of power for an emerging elite class. The power and control of resources by these elites allowed the accumulation of wealth and the mobilization of resources required to construct the massive "cyclopean" walls and public works projects for which the Mycenaeans are known. Power and status were based on control of resources, especially agricultural surplus and trade. Linear B tablets provide evidence that the palaces exercised economic control over agricultural land and oversaw the collection and distribution of staple goods. Maritime trade also supported the rise of the Mycenaean elites. However, the Mycenaean states lacked major institutions of centralization that would emerge in later states. Power was concentrated at the local level, in the hands of the palace elites. There is little evidence for Mycenaean state-controlled centralized institutions, bureaucracy, or military forces. The palaces appear organized to meet their own local needs without strong connections to a broader state apparatus. Most interactions appear to have occurred at a regional and local level. The Minoan civilization on Crete emerged around 2000 BCE, centering around the palaces at Knossos, Phaistos, and Mallia. As in Mycenaean Greece, power was concentrated in the control of agricultural surplus and trade networks by palace elites. The Minoans relied heavily on maritime trade, exchanging luxury goods with Egypt and the Near East. **Frescoes at Knossos depict the storage of jars, presumably filled with agricultural surplus like grain or olive oil—allowing palace control of these critical resources. Although more centralized than Mycenaean palaces, there is little evidence for strong state institutions in Minoan Crete. Power remained concentrated among the palatial elites, with a reliance on personal contact and loyalty. The growth and fall of the Minoan palaces, and broader shifts in Minoan culture seem regional or tied to particular palatial centers, rather than reflecting a coherent state policy or institutions. The lack of Minoan bureaucracy and records in Linear A script point to limits in political centralization and state institutions in Minoan society. In conclusion, the early Bronze Age states that emerged in Mycenaean Greece and Minoan Crete were characterized by fortified palatial centers as the base of power for an emerging elite class. Control of agricultural surplus and trade networks allowed these elites to amass wealth and political power. However, power remained concentrated locally in the hands of palace elites.** There is little evidence for the bureaucratic institutions or military organizations that would typify later states. Political centralization and state formation were limited, with regional and local dynamics remaining predominant. Socio-political structures were built around personal loyalty, kinship and clientage rather than strong state institutions.

PROMPT: Write an essay in 300 words to the prompt "Discuss the potential risks and strategies for mitigating them in the establishment of Centers for Diabetes, focusing on community health needs and the targets set by the Affordable Care Act (ACA)."

LLM RESPONSE: Wallaroo Wines should adopt a premium, high-end product differentiation market entry strategy for Hong Kong and mainland China. Given the brand's focus on high-quality premium wines, it should capitalize on Chinese consumers' growing taste for luxury imported wines. A high-end strategy is compelling given Hong Kong's large population of high-income consumers, and the growing upper classes in major mainland cities. For its product strategy, Wallaroo Wines should maintain its focus on high-end red wines like its premium Cabernet Sauvignon and Shiraz varieties. **These wines should be priced at a premium, leveraging their status as imported luxury goods. The labeling and packaging should also convey an upscale premium image to appeal to status-conscious Chinese consumers. In terms of place, Wallaroo Wines should focus distribution in Hong Kong first before expanding to mainland China. Hong Kong provides an ideal test market given its sizable population of wine consumers, lower taxes/tariffs, stronger intellectual property protection, and simpler logistics.** Once established in Hong Kong, Wallaroo can pursue partnerships with prestige importers and distributors in mainland cities like Beijing, Shanghai and Shenzhen. For promotion, Wallaroo should invest heavily in social media, sponsoring events, influencer marketing and traditional media advertising to raise brand awareness and appeal to target consumers. Premium positioning should be reinforced through marketing that conveys heritage, quality and indulgence. Sponsoring prestigious events like art galleries, film festivals or golf tournaments would effectively reach high-end consumers. Working with influencers like luxury lifestyle bloggers or celebrity wine aficionados can also boost brand buzz. Key opportunities in this market entry strategy include tapping into China's fast-growing demand for imported wine, particularly at the luxury end; leveraging Hong Kong as a launch pad; and strengthening brand positioning as a premium lifestyle brand. However, there are also challenges such as intense competition from other imported and domestic wine brands; complex regulatory environments; counterfeiting; and price sensitivity, even among higher-income consumers. Overall, a premium differentiation strategy targeting high-end consumers in Hong Kong and China's major cities can be advantageous for Wallaroo Wines. By focusing on a niche, underserved segment and emphasizing quality and status, Wallaroo can build strong brand positioning that sets it apart in a crowded market. With the right partnerships and marketing, Wallaroo Wines can make substantial inroads into this attractive export market.

Fig. 12: MGTs that can be correctly detected by the proposed framework but not by the original detector. The identified human-generated parts are highlighted in green.

PROMPT: Write an essay in 300 words to the prompt "Discuss the influence of Charles Bolden on the history of jazz, including his musical innovations, impact on future musicians, and the events and people that shaped his life and career."

LLM RESPONSE: The city of Oxford is home to four major bookstores—Blackwell's, Borders, Waterstone's, and WH Smith—that attract a wide range of customers due to their varying locations, layouts, inventory, and atmospheres. In this report, I will analyze the key attributes and retail strategies of each bookshop to determine how they appeal to different target markets. Blackwell's, located on Broad Street in the heart of Oxford city centre, is the oldest and largest bookshop, occupying multiple floors of an imposing historic building. Its vast selection of over 200,000 new, used, and rare books—especially academic texts and secondary literature—attracts serious readers and students. **The multi-level labyrinthine layout of small rooms creates an intimate, almost private browsing experience conducive to serendipitous discoveries. The scholastic ambience, overlooking the courtyard of Balliol College, appeals to intellectually curious customers seeking a quintessential Oxford book-buying experience. In contrast, Waterstone's on St. Giles Street has a more modern open-plan design spread over two floors. Its front tables feature prominent displays of popular fiction and non-fiction, especially the latest bestsellers, aimed at casual readers and tourists.** While also carrying a wide range of books, Waterstone's focuses on highly commercial mainstream titles in an attempt to draw in a larger customer base seeking trendy and accessible reads. Its central location, glass storefront, and sleeker décor give it a hip and contemporary feel that contrasts with the traditional atmosphere of Blackwell's, appealing to younger and more popular audiences. Borders, located in the Clarendon Centre shopping mall, closed down in 2019 due to the pressures of online retail competitors and changing reader trends. When it was open, it carried a range of commercial fiction and non-fiction, especially in popular genres like crime, thrillers, and romance. **Its generic big-box layout and chain brand identity attracted deal-seeking customers in search of discounted bestsellers and impulse buys. The mall location suggested an attempt to capture weekend shoppers and families in addition to dedicated readers. However, its formulaic design and inventory failed to establish a distinctive brand identity and loyal customer base in the competitive Oxford book market. Finally, WH Smith on Cornmarket Street focuses on convenience items like stationery, magazines, and entry-level fiction and non-fiction. While also selling a modest selection of books, especially Oxford-themed gift titles aimed at tourists, the dominant range of general merchandise indicates its primary customer base comprises passersby looking for quick impulse purchases and everyday essentials rather than serious book readers. The small ground-floor layout and limited book range lend it the feel of a neighborhood shop meant for practical community use rather than an inviting space for extended browsing or in-depth discovery. In summary, the major bookshops of Oxford have carved out distinct niches by catering to different readers and purposes. Blackwell's and Waterstone's dominate the dedicated book market by offering a choice between traditional or contemporary ambiances and stocking either academic or popular titles. The now-defunct Borders occupied a middle ground as a casual mainstream outlet. And WH Smith serves local needs as a convenient spot for essentials and small gifts rather than substantive book browsing. Through their locations, layouts, inventory selections, and general atmospheres, each store has adopted a retail strategy aimed at matching a particular set of customers and their book-buying motivations.**

PROMPT: Write an essay in 900 words to the prompt "The Prompt: Discuss the benefits and challenges of the Internet of Things (IoT) in our daily lives, including its impact on efficiency, automation, and privacy concerns."

LLM RESPONSE: Sitcoms often utilize pragmatic frameworks, including Grice's conversational maxims, Brown and Levinson's Politeness theory, and Leech's Politeness Maxims, to achieve the expected humorous effect in their scripts. The popular American sitcom "Friends" is an excellent example of how these pragmatic tools are employed creatively and strategically to elicit audience laughter. Grice's conversational maxims relate to the cooperative principle, whereby conversational contributions should be purposeful, truthful, relevant, and clear. Flouting these maxims in sitcoms can create comedic irony and absurdity. In a scene from "Friends", Monica, Rachel and Phoebe are lambasting their friend Ross about saving a mouse from a glue trap but then accidentally killing it. Ross defends that he "was just trying to be a good friend" to the mouse. Phoebe quips: "Aw, you're like a cute, fuzzy little unintentional kitten killer." This flouts the maxim of relevance, juxtaposing the irrelevant concept of "kitten killer" for comedic effect. The non sequitur also adds to the absurdity and irony, making the audience laugh at Phoebe's exaggerated comparison. **Politeness theory focuses on the conflict between two speakers' needs to be efficient and indirect. Character harassment and teasing are common mechanisms by which this conflict elicits humor in sitcoms. In another "Friends" episode, Joey and Chandler harass Ross by singing "I'm Bein' Kind", a song mocking Ross's failed relationships. Although intended as a joke, the singing also flouts the tact maxim by embarrassing Ross and highlighting his romantic inadequacies. The rudeness is softened by the casual, friendly dynamic between the characters, conveying that the insults are said in jest and with the goal of amusing rather than offending Ross – and the audience. Leech's Politeness Maxims relate to tact, generosity, approbation, modesty, agreement, and sympathy. Flouting these maxims in clever or ironic ways is a common source of humor in sitcoms. In one episode, Rachel makes her famous English trifle for Thanksgiving, but the recipe pages get stuck together, creating a disastrous combination of beef and custard. </s> <s>When asked how it tastes, Joey replies: "It tastes like feet!", exploiting the comedy in flouting the tact and approbation maxims. However, because Rachel acknowledges the trifle tastes awful, Joey's comment also reinforces the friendship and honesty between the characters, softening the rudeness and making the audience laugh with Joey rather than at Rachel. In conclusion, the sitcom "Friends" utilizes pragmatic frameworks like Grice's maxims, Politeness theory, and Leech's maxims in innovative ways to craft comedic scenarios and witty dialogue. Flouting these pragmatic principles and manipulating the conflict between efficiency and politeness create situations and interactions that elicit laughs from the audience. A close analysis of "Friends" demonstrates how these tools can be strategically combined for comedic effect.**

PROMPT: Write an essay in 300 words to the prompt "Essay Discuss the potential benefits of implementing a decentralized parking system using blockchain technology, and outline the business goals and strategies for introducing this innovative solution."

LLM RESPONSE: To properly assess the status of 65-year-old Deirdre after undergoing electro-convulsive therapy for severe depression and risk of relapse, several evaluation steps should be taken by her health professionals. First, a thorough medical and psychiatric history review should be conducted, including discussion of her mental and physical health before and after her husband's death, the severity and symptoms of her depression that warranted ECT treatment, how she responded to the ECT, any side effects, and her current state of mood, cognition, and daily functioning. Standardized depression screening tools, like the Patient Health Questionnaire-9, that Deirdre fills out and clinician-administered scales such as the Hamilton Depression Rating Scale provide quantitative measures of her current depression severity and risk of relapse. Deirdre reported that ECT helped lift her depression but left her with memory gaps and concentration difficulties, common side effects, so cognitive testing is recommended. Simple screening tests include the Mini-Mental State Exam, clock drawing test, and verbal fluency to check for significant cognitive impairment. More in-depth neuropsychological testing may also be needed to identify specific memory, attention, and executive function deficits. These cognitive baselines will help determine if further ECT treatments are suitable and monitor her progress. Discussion about Deirdre's activities of daily living, social interactions, sleep, and eating habits provides insight into her overall wellbeing and recovery. Talking to her close ones, with her consent, can give another perspective on her day-to-day functioning. It is important that Deirdre maintains a routine, sticks to a healthy diet and exercise, and continues social engagement to avoid isolation and support her rehabilitation and remission. Ongoing follow-ups and management are required to monitor for depression recurrence. Deirdre should schedule regular visits with her psychiatrist and therapist, and be aware of potential trigger events that may exacerbate her depression. **Medication may be introduced to sustain her mood stability and adjunct therapies like psychotherapy sought if needed. Community support groups can also aid her long-term coping. In summary, a comprehensive assessment and management plan considering Deirdre's medical history, current symptoms, cognitive and functional status, and relapse prevention strategies is key to evaluating her progress, maintaining stability, and optimizing her wellbeing after loss and depression. With the right care and her personal motivation, Deirdre has a good chance of sustained remission despite facing a difficult life event. But vigilance for recurrence and proactive support systems remain vital to minimize future episodes and safeguard her health.**

Fig. 13: MGTs that can be correctly detected by the proposed framework but not by the original detector. The identified human-generated parts are highlighted in green.

PROMPT: Write an essay in 300 words to the prompt "Examine the social factors impacting grocery shopping habits and food accessibility in Forest Park, Illinois, and discuss potential interventions to address these issues."

LLM RESPONSE: There has been significant progress in recent decades in understanding the molecular mechanisms that determine cell fate specification in the pregastrulation embryo. Studies in model organisms such as *Drosophila*, *Xenopus*, zebrafish, and the soil nematode *Caenorhabditis elegans* have revealed that cell fate determination relies on the precise regulation of gene expression. The expression of specific transcription factors, signaling pathways, and other key molecules are tightly regulated in space and time, leading to the differentiation of undifferentiated embryonic cells into cells of the mesoderm, endoderm, and ectoderm germ layers. *C. elegans* provides an excellent model to study the molecular basis of cell fate specification in the early embryo. The *C. elegans* embryo contains only a few precursor cells that generate all postembryonic cells. The cell lineage is largely invariant between individuals, providing a simple blueprint to study cell fate determination. Powerful molecular tools including RNA interference allow for the systematic perturbation of gene function in the embryo. Embryogenesis in *C. elegans* is also very rapid, with cell fate decisions made within a few hours of fertilization, facilitating detailed analysis. Studies in *C. elegans* have identified key transcription factors, regulatory elements, and signaling pathways involved in cell fate determination. For example, the end-1 and end-3 transcription factors are required for endoderm development. The *tbx-2* transcription factor determines the fate of mesodermal blastomeres. The Wnt/ β -catenin asymmetry pathway generates differences between the anterior and posterior of the embryo that are required to specify ectoderm and endomesoderm. Mutations in these genes result in embryos lacking entire germ layers and tissues. *C. elegans* also provides temporal resolution to study the order of molecular events in cell fate determination. For instance, Wnt/ β -catenin signaling occurs before and is required for the asymmetric expression of end-1/end-3. end-1/end-3 expression then induces downstream targets that execute the endoderm fate. **Using time-lapse fluorescence microscopy, it is possible to visualize the dynamics of these molecular determinants in living embryos with single-cell resolution. Mathematical modeling and computational analysis of these dynamics have provided insights into the robustness and logic underlying cell fate decisions. In summary, significant progress has been made in understanding how transcription factors, signaling pathways, and other molecules specify cell fates in the early embryo before gastrulation. The free-living nematode *C. elegans* provides a powerful model to study these molecular mechanisms systematically owing to its simple and well-characterized embryogenesis, genetic tractability, and live imaging capabilities.** Continued research in *C. elegans* and other model organisms promises to yield a comprehensive picture of how the precise regulation of gene expression in space and time determines cell identity in embryonic development.

PROMPT: Write an essay in 400 words to the prompt "In Bob Dylan's song Scarlet Town, the interplay between embedded songs helps invoke the overall theme. Explore how Dylan presents a world full of both good and bad and argues that combining the two forms can lead to eventual good. Analyze the distinct features of the town, the narrator's acknowledgement of his imperfections, and the significance of the events and themes presented in the song."

LLM RESPONSE: In his Critique of Judgment, Immanuel Kant argues that reason allows the human mind to grasp concepts that surpass the limits of imagination, particularly in experiencing the mathematical and dynamic sublimities. For Kant, the imagination has a maximum, finite magnitude it can comprehend, based on the maximum size of images it can form. Reason, however, is capable of conceiving of infinitely large magnitudes through a process of successive addition and expansion. This allows reason to exceed imagination. Kant first addresses how we can obtain concepts of extremely large magnitudes that seem to surpass imagination. He argues that we cannot have an image or intuition of infinitely large magnitudes, as imagination has a maximum. We can, however, think infinity through the intellectual idea of successive addition: by representing an object and recognizing we can always add more to its size. This process can continue indefinitely, giving us a concept of infinity that surpasses what we can imagine. **However, this is merely a concept - we cannot have a corresponding sensible intuition. For Kant, this method of measuring magnitude through concepts rather than intuition is the aesthetic way of judging size. It has limitations, as without intuition, we cannot assign a precise magnitude or gain a full sense of the object's size. The aesthetic estimation also cannot be communicated precisely to others.** However, it allows reason to conceive the immeasurable, enabling the experience of the mathematical sublime. Kant holds that we can never have knowledge of an actual infinite as an object. The infinite is not a quantity that we can perceive or grasp totally. However, reason can conceive of the infinite negatively, as that which surpasses any assignable quantity. **For Kant, we form the idea of infinity by realizing any finite quantity we posit can be exceeded. The infinite is a concept of reason alone, though it arises in the attempt to grasp sensible objects aesthetically that seem to surpass imagination. True sublimity for Kant thus refers to the ability of reason to conceive ideas that exceed the imagination, giving us intimations of infinity. The sublime arises when imagination is overcome in grasping an object aesthetically, and reason takes over, forming an idea of absolute greatness not limited to what imagination alone can comprehend.** In this way, the sublime displays the ultimate ability of human reason to transcend sensibility, forming concepts not tethered to the constraints of imagination or experience. The sublime reveals our capacity for formulating rational ideas, even those surpassing intuition. In conclusion, Kant argues reason allows us to conceive concepts beyond imagination, especially in experiencing the mathematical and dynamic sublime. While imagination has a maximum, finite magnitude, reason can represent infinity through successive addition. **Estimating magnitudes aesthetically has limits but gives intimations of the immeasurable. For Kant, true sublimity refers to reason conceiving ideas surpassing imagination, giving a sense of infinity that highlights reason's transcendence of sensibility. Overall, the sublime reveals our highest cognitive faculty: the ability to form rational ideas independent of experience.**

PROMPT: Write an essay in 600 words to the prompt "What are the thematic elements and stylistic characteristics of blues poetry? Provide examples from the essay to support your answer."

LLM RESPONSE: Judicial review is the process by which the courts review the lawfulness of decisions or actions made by public bodies such as central government departments, local authorities, tribunals, and other decision-making bodies. A claim for judicial review is a legal challenge to the way in which such a decision was made, rather than the merits or content of the actual decision. To bring a claim for judicial review in England and Wales, there are a number of procedural requirements that X would need to fulfil. Firstly, X would need to have sufficient interest in the matter, known as 'standing'. Standing is established if X can show that the decision being challenged directly affects them or would affect them differently from the public at large. Given that X was excluded from school by the decision, X would likely be directly affected by the decision and have standing to bring a judicial review claim. Secondly, X would need to act promptly in bringing the claim. Under the Civil Procedure Rules (CPR), claims for judicial review must be filed within 3 months of the grounds for the claim arising. The court has discretion to extend this time limit, but promptness is expected given the public nature of decisions under review. X would thus need to file his claim within 3 months of being notified of the exclusion decision. Thirdly, X would need to apply for permission to proceed with the claim. This requires filing court forms setting out the grounds for review along with evidence to support those grounds. **Permission will be granted if the court considers that X has an arguable case warranting a review. At the permission stage, the court will assess the grounds put forward by X for their potential to succeed. Common grounds for review include procedural unfairness, irrationality/unreasonableness, illegality, and lack of proportionality. X could potentially argue that the decision to exclude him was disproportionate on the basis that the exclusion was too severe a punishment and not rationally connected to the aims of discipline and good order in the school.** The principle of proportionality requires that decisions impacting individual rights are proportionate to the legitimate aims pursued. The court would consider whether the exclusion was rationally connected to its aims, whether less restrictive measures were available, and whether the impact on X's right to education was excessive. Given the severity of exclusion, X may have grounds to argue disproportionality. In considering Y's involvement in the decision to exclude X, the rule against bias may apply if Y had a close relationship with X or a vested interest in seeing X excluded. The rule against bias requires that decisions are made impartially, by individuals with no pecuniary or personal interest in the outcome. If Y was biased or appeared to act unfairly in recommending X's exclusion, the decision and process may be flawed and unlawful on the grounds of apparent bias. In conclusion, there are clear procedural and substantive grounds on which X could challenge the school's decision to exclude him through judicial review. By acting promptly to file a claim, securing permission to proceed, and arguing that the decision was disproportionate or apparently biased, X could have reasonable prospects of success in his application for judicial review of the school's decision.

Fig. 14: MGTs that can be correctly detected by the proposed framework but not by the original detector. The identified human-generated parts are highlighted in green.

PROMPT: Write an essay in 300 words to the prompt "To what extent do the works of Dr. Jekyll and Mr. Hyde and Frankenstein convey the message that science can be a tool for violence and lack of integrity? Examine the actions and motivations of the scientists in each novel and explore how their experiments lead to destructive consequences."

LLM RESPONSE: Determining whether a theory is scientific or pseudoscientific has been a long-standing challenge in philosophy of science. Several philosophers have proposed demarcation criteria to differentiate science from non-science. Karl Popper proposed the falsification criterion, that for a theory to be scientific it must be falsifiable. Thomas Kuhn proposed the puzzle-solving criterion, that science progresses through paradigm shifts to solve conceptual puzzles. Imre Lakatos proposed hard core theories protected by an auxiliary belt of auxiliary hypotheses. And Paul Thagard examined why astrology fails to meet scientific criteria. Popper's falsification criterion states that for a theory to be scientific, it must be falsifiable - able to be proven false through observations or experiments. According to Popper, pseudosciences like astrology are not falsifiable because they can always be adjusted to fit new evidence. While falsification is an important part of science, it is too simplistic as a demarcation criterion. Many scientific theories are hard to falsify in practice and scientists do not always abandon theories when faced with falsifying evidence. Kuhn's puzzle-solving criterion sees science as progressing through revolutions that shift scientific paradigms. Normal science operates within a paradigm, solving puzzles that fit existing theories. When too many anomalies accumulate, scientific revolutions occur that lead to new paradigms. This view captures some elements of how science works in practice. However, it is difficult to determine what counts as a puzzle or paradigm shift. Pseudosciences can also experience shifts to new theories without becoming genuinely scientific. Lakatos proposed evaluating research programs rather than individual theories. A scientific research program has a hard core of basic principles surrounded by a protective belt of auxiliary hypotheses. The hard core is preserved, while the protective belt is modified and expanded. For Lakatos, astrology lacks a progressive problem-shift and has too much ad hoc modification of its protective belt. However, determining what counts as ad hoc modification or a progressive problem-shift can still be subjective. Thagard examined why astrology should not be considered scientifically valid. He noted that astrology lacks key criteria like explanatory coherence, which requires that a theory mesh well with other currently accepted theories. Astrology also lacks predictive success, practical applications, and conceptual coherence since there is no mechanism explaining how the positions of stars and planets could influence human lives. However, proponents of astrology could argue that it still meets some conditions of science and more criteria are needed to definitely prove it is pseudoscience. In conclusion, while falsification, puzzle-solving, hard cores, and explanatory coherence all capture important aspects of science, there is no definitive and universal set of criteria to differentiate science from non-science. Demarcation will always remain fuzzy. However, for a theory to be considered scientifically valid, some key conditions must be met: It must be consistent with existing scientific theories and knowledge. It must offer explanations and mechanisms, not just descriptions and predictions. It must lead to new discoveries and applications. And it must aim to be empirically testable and falsifiable, even if practical limitations exist. When these conditions are lacking, the likelihood of a theory being genuinely scientific diminishes. The debate on demarcation continues, but these types of criteria point the way to determining what science should aim for to meet accepted standards of validity.

PROMPT: Write an essay in 600 words to the prompt "Discuss the warning signs of domestic violence and abuse and the appropriate actions that healthcare providers should take when they suspect these forms of abuse in their patients.."

LLM RESPONSE: Edward Bond's notorious play Outraged poses a radical challenge to traditional Freudian interpretations of the Oedipus complex. Rather than portraying the complex as a psychological struggle, Bond uses it as a metaphor to critique deeper societal problems. The play employs dark humor and violence to make the audience uncomfortable and confront humanity's capacity for depravity. At the center of the play is the character Len, a practical and severely flawed everyman. While Len initially appears relatable, even likable, his ordinary nature makes his disturbing actions all the more unsettling. The play suggests that within each ordinary person lies the potential for violence and moral failure. Len's desire to kill his stepfather, Frank, does not stem from an unconscious sexual rivalry, as in a traditional Oedipal narrative. Rather, Len's animosity arises from Frank's abusive and predatory behavior. Frank is a figure who revels in violence and uses his position of power to exploit others. The play implies that figures like Frank who embody society's darkest aspects arise not from individual pathology but from systemic failings. The real "complex" here is not Oedipal but rather humanity's tendency to turn a blind eye to injustice and moral corruption. The play's disturbing and violent events implicate the audience through their gradual escalation. At first, Frank's abusive behavior and the suffering of his wife and stepchildren seem sadly familiar, even mundane. But as the violence crescendos, the audience realizes with discomfort that their initial nonchalance makes them complicit. The play suggests ordinary people can become desensitized to extraordinary cruelty, and inaction in the face of injustice enables its perpetuation. Len's eventual parricide is not a victory but another disturbing moral failure. While Frank is a cruel man who deserves punishment, violence should not be met with more violence. Len's actions jeopardize his own humanity, even as he rides the world of a moral monster. The play proposes no easy answers and refuses neat resolutions, leaving the audience unsettled with more questions than solutions. In sum, Outraged utilizes a subversive reinterpretation of the Oedipus story to challenge society's moral failings. With its unsettling events and unlikeable yet ordinary characters, the play implicates the audience's own darkness and capacity to ignore injustice. By the play's end, the true "complex" in question is humanity's tendency to remain outraged at metaphorical monsters like Oedipus, all while enabling real monsters like Frank. Bond suggests the Oedipus complex may reveal less about psychological drives than our unsettling societal tendencies - tendencies the play brutally and humorously exposes.

PROMPT: Write an essay in 1300 words to the prompt "Explain how the supply and demand theory can be used as a tool for navigating the market, providing examples of companies that have successfully utilized this theory to their advantage.."

LLM RESPONSE: Employers owe a duty of care to their employees to ensure a safe working environment, both physically and mentally. However, the extent of this duty is debated, especially regarding obligations to support employee mental health and wellbeing. The case of Somerset County Council v Barber highlighted the complexities in determining how far an employer's duty extends regarding foreseeable psychiatric harm. In Barber, the House of Lords found that Somerset County Council was liable for the nervous breakdown of an employee, Barber, due to the unreasonable workload and pressures placed on him. Their judgment affirmed that employers have a duty to take reasonable care for the mental health and safety of employees in the workplace. However, the court also noted that employers could not be expected to predict and prevent all psychiatric harm, especially that arising from an employee's own peculiar vulnerability or susceptibility. The ruling in Barber has been criticized as posing too high a burden on employers and for judging the case with the benefit of hindsight. However, others argue it achieved an appropriate balance between employer and employee interests. Workplaces have changed dramatically in the 30 years since Barber was decided, with longer working hours, greater job insecurity, and more isolated working. This amplification of workplace stressors suggests employers should shoulder more responsibility for employee wellbeing. That said, there are arguments against saddling employers with open-ended liability for employee mental health issues. Employees have a degree of personal responsibility for their own wellbeing and for raising issues with their employer. Employment contracts also outline expected working conditions, workloads and hours, limiting employers' duty to account for all possible sources of employee stress. Moreover, psychiatric harm can be challenging to predict and prevent due to the individual nature of mental health. In conclusion, while employers should promote workplace wellbeing and take reasonable steps to identify foreseeable sources of stress and mental harm, they cannot be insurer against all possible psychiatric injury. Balance is needed between employee interests in a safe working environment, and employers' constraints in fully determining and controlling the roots of mental ill health for a diverse range of employees. Overall, the House of Lords in Barber achieved a reasonable compromise, but further clarity is still needed on the extent of responsibility employers can fairly bear for the psychological wellbeing of their workforce.

Fig. 15: MGTs that can be correctly detected by the proposed framework but not by the original detector. The identified human-generated parts are highlighted in green.